Optimal Rényi and Logarithmic Additive Functional Entropy Estimation

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Abstract

We characterize the minimax sample complexity of estimating Rényi entropy from independent samples on an alphabet of size at most k, when the known order $\alpha > 1$ may vary with k.[1] For every fixed additive accuracy $\varepsilon \in (0, 1/2)$, we obtain matching bounds with constants independent of both k and α. Below order $\log k$, the sample complexity is $\Theta_\varepsilon(\alpha k^{1-1/\alpha} X_\alpha(k, \varepsilon))$, where a monotone scalar inequality determines the noninteger multiplier. This characterization resolves the transitions as the order approaches one or an integer, or grows with the alphabet. Integer orders have complexity $\Theta_\varepsilon(\alpha k^{1-1/\alpha})$, while orders at least $\log k$ and min-entropy have complexity $\Theta_\varepsilon(k \log k)$.

We also develop a framework for estimating logarithms of positive additive functionals. Exact factorial-count variance identities and weighted approximation duality retain the multiplicative thresholds required for logarithmic accuracy. For every fixed continuous function f satisfying $f(0) = 0$, $f(x) > 0$ for $x > 0$, and $f(x) \sim ax^r$ as $x \downarrow 0$, with $a > 0$ and $r > 1$, we show that estimating $-\beta^{-1} \log \sum_i f(p_i)$, for fixed $\beta > 0$, has sample complexity $\Theta(k^{1-1/r})$ when r is an integer and $\Theta(k/\log k)$ otherwise. The proof uses a replication reduction that preserves the sampling experiment. Smooth flat functions demonstrate that regularity alone does not determine these rates.

The Rényi upper bound combines polynomial estimates for small probabilities, Gamma-function count statistics for larger probabilities, and an independent pilot. A global affine correction preserves the known first moment near order one. The lower bound constructs positive mean-one priors by finite interpolation and controls both statistical indistinguishability and entropy separation after normalization.

[1] Lean verification files of our proofs are available at https://github.com/adibiarman/Optimal-Renyi-and-Logarithmic-Additive-Functional-Entropy-Estimation.

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1 Introduction

Estimating a distribution and estimating a functional of that distribution are different statistical tasks. A sample may contain too little information to recover most small probabilities, while still determining their aggregate contribution to entropy. This distinction is central to large-alphabet statistics and distribution property estimation. It also creates a precise algorithmic question: which nonlinear summaries can be estimated from substantially fewer observations than the alphabet size?

Rényi entropy [Rén61] provides a family in which the answer depends sharply on a real parameter. For a distribution $p = (p_1, \dots, p_k)$ and order $\alpha > 1$, it is $H_\alpha(p) = -(\alpha - 1)^{-1} \log \sum_i p_i^\alpha$. At integer orders, collisions provide unbiased power-sum estimators. At noninteger orders, the behavior of x^α near zero creates an approximation problem. The distinction becomes more delicate when α changes with k: a fixed-order constant can conceal a transition near an integer, near one, or near the high-order regime.

We resolve these transitions for all regimes of α through a uniform sample-complexity characterization. We also ask which parts of our arguments extend beyond powers. For a known positive function f, the logarithmic target $-\beta^{-1} \log \sum_i f(p_i)$ requires relative accuracy in its additive sum. Its statistical difficulty can therefore differ from that of estimating the same sum to fixed absolute accuracy. We develop the count and approximation identities for this loss and prove a transfer theorem for every fixed function with a finite leading power at zero.

Contributions. Our first result gives matching bounds for every real $\alpha > 1$, with the accuracy fixed before the alphabet size and order. We define a scalar root that retains the dependence on the distance to adjacent integers and on the requested accuracy. The result includes the collision scale $\alpha k^{1-1/\alpha}$ at integer orders and the saturation scale $k \log k$ at large orders. At every fixed noninteger order it gives $k/\log k$.

Our second result separates the general functional framework from its quantitative evaluation for powers. We give exact count-variance and likelihood identities, an approximation duality with the exact logarithmic threshold, and explicit conditions needed to turn these identities into statistical bounds. Our third result shows that, for fixed function $f(x) \sim ax^r$, the leading order determines the alphabet dependence of the logarithmic additive estimation. The reduction assumes only continuity and positivity, and retains its accuracy loss explicitly.

1.1 Prior work and the remaining uniformity question

Entropy and Rényi estimation. Valiant and Valiant [VV11] established the optimal $\Theta(k/\log k)$ sample complexity for constant-accuracy Shannon entropy estimation. Acharya, Orlitsky, Suresh, and Tyagi [AOST17] identified the integer/noninteger distinction for Rényi entropy: fixed integer orders admit the collision scale $k^{1-1/\alpha}$, whereas noninteger orders have near-linear alphabet dependence. Their polynomial estimator gives an $O(k/\log k)$ upper bound for fixed noninteger orders above one. Obremski and Skorski [OS17] refined Rényi bounds, including entropy-dependent integer-order estimates and falling-factorial variance calculations. Adibi and Krysta [AK26] obtained matching bounds for growing integer orders and min-entropy. Those results supply important parts of the present picture; they do not give the noninteger transition uniformly as the order approaches one or an integer.

Polynomial methods and additive functionals. Jiao, Venkat, Han, and Weissman [JVHW15] developed a general method that estimates polynomial approximations in nonsmooth regions and uses bias correction in smooth regions. Wu and Yang [WY16] proved sharp entropy risk bounds

through best polynomial approximation. Their later work [WY19, WY20] develops the associated connections between approximation, moment matching, and estimation of unseen probabilities. Fukuchi and Sakuma [FS18] study general additive functionals under squared-error loss, with rates governed by derivative behavior. Our functional conclusions concern logarithmic accuracy and require no differentiability under the finite-leading-power assumption. The underlying use of polynomial estimates and moment matching is an established methodology.

Approximation and statistical duality. Polyanskiy and Wu [PW19] connect functional estimation to convex duality and optimized testing bounds. We use a weighted form of approximation duality that records an exact multiplicative ratio. The identity alone does not evaluate a minimax rate: uniform variance, likelihood tails, concentration of the realized target, and the sampling reduction must also be controlled. Our main technical work here is the order-uniform evaluation of these quantities for powers at the original accuracy. In particular, a constant-factor reduction in entropy accuracy cannot be absorbed into a sample-size constant when the order varies.

Organization. Section 2 states our two main statistical results. Section 3 explains our common functional framework. Section 4 gives the i.i.d. proof and its principal quantitative estimates; Section 5 explains our functional reduction. The appendices give the full versions of our count identities, polynomial constructions, sampling bounds, lower-bounds, transfer proofs, and approximation dualities.

2 Models and main results

All logarithms are natural. Write $\Delta_k = \{p \in [0,1]^k : \sum_i p_i = 1\}$; zero probabilities are allowed. The observations are independent samples from an unknown $p \in \Delta_k$, and the alphabet bound k is known. For any target T, define

$$S_T(k,\varepsilon,\zeta) = \min\left\{ n \in \mathbb{N}_0 : \exists \widehat{T} \quad \sup_{p\in\Delta_k} \mathbb{P}_p\{|\widehat{T} - T(p)| > \varepsilon\} \le \zeta \right\}. \tag{2.1}$$

Estimators may use independent randomization. For real $\alpha > 1$, set

$$P_\alpha(p) = \sum_i p_i^\alpha, \qquad H_\alpha(p) = -\frac{\log P_\alpha(p)}{\alpha - 1}, \qquad S_\alpha = S_{H_\alpha}, \tag{2.2}$$

where H_α is the Rényi entropy of order α. Jensen's inequality gives

$$k^{1-\alpha} \le P_\alpha(p) \le 1, \qquad 0 \le H_\alpha(p) \le \log k. \tag{2.3}$$

Min-entropy is $H_\infty(p) = -\log \max_i p_i$, with complexity S_∞ defined by the same loss. We suppress the failure parameter when it equals $1/3$.

2.1 A rate uniform over all real orders above one

Fix $0 < \varepsilon < 1/2$. For noninteger $1 < \alpha < \log k$, put

$$\begin{aligned} &\beta = \alpha - 1, \quad && m = \lfloor \alpha \rfloor, && \gamma = \alpha - m, \quad && \delta = 1 - \gamma, && h = \gamma\delta, \\ &E = 2\varepsilon\beta, && \Psi = \gamma^{1+1/\delta}\delta^{1+1/\gamma}, && && \mathcal{Q}^2 = \frac{\Psi}{\alpha^2\beta^2}\exp\!\left(\frac{\log k}{\alpha} - \frac{E}{h}\right). \end{aligned} \tag{2.4}$$

Table 1: Consequences at fixed additive accuracy and failure probability 1/3. Constants are uniform in α only in the first, second, and fourth rows.

Order range	Sample complexity
$1 < \alpha \le 3/2$	$\Theta_\varepsilon(k/\log k)$
Integer $2 \le \alpha < \log k$	$\Theta_\varepsilon(\alpha k^{1-1/\alpha})$
Fixed noninteger $\alpha > 1$	$\Theta_{\alpha,\varepsilon}(k/\log k)$
$\alpha \ge \log k$, or $\alpha = \infty$	$\Theta_\varepsilon(k \log k)$

The letters γ, δ denote distances to the adjacent integers, not failure probabilities. For $[t]_+ = \max\{t, 0\}$, define

$$X_\alpha^{\mathrm{root}} = \inf\left\{x \ge 1 : \frac{\alpha h}{\beta} x^\alpha \left[1 + \log \frac{x(1+\log x)}{\mathcal{Q}^2}\right]_+ \ge 1\right\}. \tag{2.5}$$

The defining function is continuous and nondecreasing, and strictly increasing wherever positive. Thus the infimum is attained and, unless it equals one, is the unique crossing of level one. It can be computed by monotone search in $\log x$.

For $\log k > 2$, extend this multiplier to all orders by

$$X_\alpha(k,\varepsilon) = \begin{cases} X_\alpha^{\mathrm{root}}, & 1 < \alpha < \log k,\ \alpha \notin \mathbb{N}, \\ \max\{1, (2\varepsilon r)^{-1/r}\}, & \alpha = r \in \mathbb{N},\ 2 \le r < \log k, \\ (\log k) k^{1/\alpha}/\alpha, & \alpha \ge \log k, \end{cases} \tag{2.6}$$

and set

$$R_\alpha(k,\varepsilon) = \alpha k^{1-1/\alpha} X_\alpha(k,\varepsilon). \tag{2.7}$$

The integer multiplier is bounded above and below by positive accuracy-dependent constants. The high-order branch gives $R_\alpha = k \log k$ exactly; its multiplier need not be at least one.

Theorem 2.1 (Uniform Rényi sample complexity)**.** *For every $0 < \varepsilon < 1/2$ there are constants $c_\varepsilon, C_\varepsilon > 0$ and $K_\varepsilon \in \mathbb{N}$, depending only on ε, such that for every $k \ge K_\varepsilon$ and every real $\alpha > 1$,*

$$c_\varepsilon R_\alpha(k,\varepsilon) \le S_\alpha(k,\varepsilon,1/3) \le C_\varepsilon R_\alpha(k,\varepsilon).$$

Moreover, $S_\infty(k,\varepsilon,1/3) = \Theta_\varepsilon(k \log k)$.

The constants are chosen before both k and α. The estimator may depend on the known order; different ranges use different auxiliary parameters. Table 1 records useful consequences. The root itself is needed for moving noninteger orders outside the first row. Its exponential term retains the original accuracy: replacing ε by a constant fraction can change the rate.

2.2 Logarithmic additive functionals

Let $f \in C([0,1])$ satisfy $f(0) = 0$ and $f(x) > 0$ for $x > 0$. For known $\beta_f > 0$, define

$$F_f(p) = \sum_i f(p_i), \qquad H_{f,\beta_f}(p) = -\beta_f^{-1} \log F_f(p). \tag{2.8}$$

The scale β_f is independent of any leading exponent of f.

Theorem 2.2 (Finite leading power)**.** *Suppose $f(x)/(ax^r) \to 1$ as $x \downarrow 0$, for fixed $a > 0$ and $r > 1$. For every fixed $\beta_f, \varepsilon > 0$ and $0 < \zeta < 1/2$, as $k \to \infty$,*

$$S_{H_{f,\beta_f}}(k, \varepsilon, \zeta) = \begin{cases} \Theta_{f,r,\beta_f,\varepsilon,\zeta}(k^{1-1/r}), & r \in \{2, 3, \ldots\}, \\ \Theta_{f,r,\beta_f,\varepsilon,\zeta}(k/\log k), & r > 1 \text{ noninteger.} \end{cases}$$

For example, $x^2 + x^{5/2}$, $x^2 e^x$, and $e^x - 1 - x$ have complexity $\Theta(\sqrt{k})$, whereas $x^{3/2}(1 + x)$ has complexity $\Theta(k/\log k)$. No differentiability is required. The function, its leading exponent, and the accuracy are fixed here; the theorem does not claim uniform constants over changing functions or orders. The reduction proving it is stated in Theorem H.2 independently of any formula for the Rényi rate. It incurs an explicit deterministic accuracy loss while preserving the sampling experiment exactly.

3 The general functional framework

Three identities explain the common structure of the proofs. They separate deterministic approximation, sampling noise, and statistical indistinguishability. Their quantitative evaluation for powers is the additional step needed for Theorem 2.1.

Polynomial approximation and exact count noise. In the Poisson experiment, counts are independent with $N_i \sim \mathrm{Poi}(np_i)$. For a polynomial $Q(x) = \sum_{j=0}^{L} c_j x^j$, factorial substitution gives

$$Z_Q = \sum_i \sum_{j=0}^{L} c_j (N_i)_j / n^j, \qquad (N)_j = N(N-1)\cdots(N-j+1).$$

Its expectation is $\sum_i Q(p_i)$, and

$$\operatorname{Var} Z_Q = \sum_i \sum_{\ell=1}^{L} \frac{p_i^\ell}{\ell! n^\ell} \{Q^{(\ell)}(p_i)\}^2. \tag{3.1}$$

Lemma A.1 proves the identity, including all cross terms. Thus approximation error controls the bias, while derivatives control the noise. If the polynomial has a linear term, its aggregate is known: $\sum_i c_1 p_i = c_1$. Replacing that term globally by its deterministic value is particularly important near order one.

Exact logarithmic accuracy. For a positive functional F and scale $\beta > 0$, set $\eta = \tanh(\beta\varepsilon)$. If $|\mathbb{E}Z - F| \le \eta F/2$ and $\operatorname{Var} Z \le \zeta\eta^2 F^2/4$, Chebyshev's inequality gives $Z/F \in [1 - \eta, 1 + \eta]$ with failure at most ζ. The centered output

$$-\frac{\log Z}{\beta} + \frac{\log(1 - \eta^2)}{2\beta}$$

then has error at most ε. The logarithmic half-width is exactly $\operatorname{arctanh}(\eta) = \beta\varepsilon$. Proposition A.2 gives the full certificate.

Approximation duality at the same threshold. Let G be positive and continuous on a compact parameter space Θ, with $\min G > 0$, and let $V \subset C(\Theta)$ be a finite-dimensional space containing constants. The weighted approximation error satisfies

$$\inf_{v \in V} \left\| \frac{G - v}{G} \right\|_\infty = \max_{\substack{\int v \, d\omega = 0 \ (v \in V) \\ \int G \, d|\omega| \le 1}} \int G \, d\omega. \tag{3.2}$$

Theorem I.1 proves attainment and shows that, when $d > 0$, the normalized Jordan parts of an optimizer have matching V-moments and target-expectation ratio $(1 + d)/(1 - d)$, where d is the displayed value. Therefore the threshold $d = \tanh(\beta\varepsilon)$ corresponds exactly to a logarithmic separation 2ε. The regularized version in Theorem I.3 permits controlled moment defects.

The sampling counterpart is equally explicit. For independent $N_e \sim \mathrm{Poi}(\nu_e \theta_e)$ and a positive reference θ^*, the likelihood ratios obey

$$\mathbb{E}_{\theta^*} L_\theta L_\vartheta = \exp\left\{ \sum_e \frac{\nu_e (\theta_e - \theta_e^*)(\vartheta_e - \theta_e^*)}{\theta_e^*} \right\}. \tag{3.3}$$

An orthonormal Poisson polynomial basis turns coefficient energy into reference variance and moment-defect energy into likelihood overlap. Appendix I derives both identities and the truncation tail. Reference variance is not a uniform variance bound, and separated prior expectations do not ensure concentration of the realized target. These points are verified separately in the i.i.d. proof.

4 Proof of the uniform i.i.d. theorem

We describe the construction and its quantitative obligations before giving the complete proofs in the appendices. Write $\beta = \alpha - 1$ and $P = P_\alpha(p)$.

4.1 Upper bound: approximation with a sampling margin

The noninteger estimator targets the exact interval

$$e^{-2\varepsilon\beta} P \le \widehat{P} \le P. \tag{4.1}$$

On this event the output

$$\widehat{H} = -\beta^{-1} \log \widehat{P} - \varepsilon \tag{4.2}$$

has error at most ε. The construction must leave positive margins at both ends of this interval, so that random fluctuations do not consume the deterministic approximation tolerance.

Orders bounded away from one. For noninteger $3/2 \le \alpha < \log k$, use main counts $N_i \sim \mathrm{Poi}(np_i)$ and independent pilot counts $Y_i \sim \mathrm{Poi}(c_p np_i)$, with $n = D\alpha k^{1-1/\alpha} X$ and $X = X_\alpha^{\mathrm{root}}$. Set $\chi = \min\{\beta, 1\}$ and choose

$$L \asymp_\varepsilon \alpha(1 + \log X), \quad b = c_b L/n, \quad v = (\beta/\alpha) X^{-\alpha}, \quad u = \min\{v + h, u_0\}.$$

The exact two-arm envelope of Lemma D.1 chooses a lower cutoff a and a deterministic tolerance $E_0 = E - \min\{v + C_h h, E/2\}$. The root calibration ensures $a \le c_\varepsilon \beta/(\alpha k)$.

Appendix D.4 constructs an explicit polynomial $Q(x) = \sum_{j=0}^d q_j x^j$, of degree $d = O(L)$, from two endpoint profiles. Reversing one profile makes its fractional power cancel against p^α, so both

profiles yield ordinary polynomials in p. For the larger probabilities use the analytic count transform $\mathsf{H}_\alpha(j) = \Gamma(j+1)/\Gamma(j+1-\alpha)$, interpreted with the entire reciprocal Gamma function. It can be signed at small counts; those values must be included in the error analysis. With $I_i = \mathbf{1}\{Y_i \le c_p B\alpha\}$, define

$$\widehat{P} = 2E_a P_* + q_1 + \sum_i \left[I_i(\widehat{Q}_i - q_1 N_i/n) + (1 - I_i)(\widehat{G}_i - q_1 N_i/n) \right], \tag{4.3}$$

where $P_* = k^{1-\alpha}$, $\widehat{Q}_i = \sum_j q_j (N_i)_j / n^j$, $\widehat{G}_i = \omega n^{-\alpha} \mathsf{H}_\alpha(N_i)$, and $E_a = \chi d_\varepsilon C_0^m (ka)^\beta$. The known coefficients, damping factor ω, and constants are specified in Appendix D. The deterministic q_1 in (4.3) replaces the random global linear total.

Lemma 4.1 (Upper-bound budgets)**.** *The design constants can be chosen in terms of ε alone so that (4.3) satisfies (4.1) with probability at least $3/4$, uniformly over $p \in \Delta_k$ and noninteger $3/2 \le \alpha < \log k$, for all sufficiently large k.*

Proof structure. The deterministic construction and the correctly selected heavy aggregate leave margins comparable to

$$\mathcal{M} = E_a P_* + e^{-E} u P + \chi e^{-E} \sum_i p_i^\alpha \min\{Lnp_i, 1\}.$$

The derivative energy (3.1) bounds the conditional polynomial noise by a small multiple of $\mathcal{M}^2$. The global affine replacement obeys the same variance budget. Pilot misclassification and the lower cutoff contribute expected absolute error at most a small multiple of $\mathcal{M}$. The heavy-count norm estimate supplies the remaining concentration event. Assigning failure $1/16$ to each of these four contributions proves the assertion by Chebyshev, Markov, and a union bound. The constructions and all four quantitative estimates are proved in Appendices D and E; Theorem E.13 assembles them. □

Orders approaching one. For $1 < \alpha \le 3/2$, the root gives, uniformly,

$$X \asymp_\varepsilon k^{1/\alpha}/\log k, \qquad R_\alpha \asymp_\varepsilon k/\log k. \tag{4.4}$$

We use separate fixed choices of the auxiliary parameters in this range. The localized polynomial is anchored to the tangent of $p^{1+\beta}$ and has a factor β in every nonlinear coefficient. After the global linear replacement, the complete variance, including the pilot and signed low-count corrections, is $o(\beta^2 P^2)$. Its bias is an arbitrarily small multiple of βP, and its center is at distance at least $c_\varepsilon \beta P$ from both ends of (4.1). Chebyshev proves the upper bound. Appendix F gives the construction and proves the exact affine cancellation.

Integer and high orders. For integers $2 \le r < \log k$, use $Z_r = n^{-r} \sum_i (N_i)_r$. Its mean is P_r and

$$\frac{\operatorname{Var} Z_r}{P_r^2} \le \sum_{j=1}^{r} \binom{r}{j}^2 j! (nP_r^{1/r})^{-j}.$$

At $n = C_\varepsilon r k^{1-1/r}$, the correct target is

$$e^{-\varepsilon(r-1)} P_r \le Z_r \le e^{\varepsilon(r-1)} P_r, \tag{4.5}$$

rather than a fixed relative-variance bound. The proof splits coordinates according to whether np_i exceeds a fixed multiple of r, combines count concentration with factorial variance, and controls

both tails (Appendix B). For $\alpha \geq \log k$, simultaneous count bounds at $n = C_\varepsilon k \log k$ give a relative-plus-$1/k$ approximation to each p_i. A nonnegative capped-order Gamma transform and an ℓ_α norm bound then estimate H_α; the maximum coordinate gives min-entropy. Proposition C.1 proves both guarantees.

All these count estimators can be implemented from a fixed sample pool. Draw a Poisson total of mean $(1 + c_p)n$, split it into main and pilot counts, and use a pool of twice that size, with a fallback on overflow. The overflow probability is exponentially small. The ideal and implemented experiments agree off overflow, so no conditioning destroys independence. Appendix A gives the coupling.

4.2 Lower bound: positive priors and observable counts

The collision obstruction. For $\alpha \geq 3/2$, a uniformly located heavier coordinate gives $\Omega_\varepsilon(\alpha k^{1-1/\alpha})$ below saturation and $\Omega_\varepsilon(k \log k)$ for high orders and min-entropy. The Poisson likelihood identity (3.3) controls the mixture distance. This suffices when X is bounded; the unbounded-root case requires a stronger approximation obstruction.

Constructing positive priors. Finite interpolation produces two signed factors, with exponents γ and $1 - \gamma$. Splitting their tensor product by parity of signs, applying a change of variables, and completing at zero gives genuine laws $U_0, U_1 \in \{0\} \cup [1, B]$ with $\mathbb{E}U_0 = \mathbb{E}U_1 = 1$. If $A_i = \mathbb{E}U_i^\alpha$, their exact ratio is

$$\frac{A_1}{A_0} = \frac{1 + R_- R_+}{R_- + R_+}, \tag{4.6}$$

where R_-, R_+ are the negative masses of the two normalized factors. Integer moments cancel through the selected degree, except possibly at the integer nearest α; the first moment always matches exactly. Appendix G.1 gives the nodes, weights, positivity, and moment identities.

Normalization is part of the lower bound. Draw $Q \sim \mathrm{Poi}(b)$ weights with $b = k/2$, and put $S = b^{-1} \sum_j U_j$, $V = b^{-1} \sum_j U_j^\alpha$, and $W = V - A_i S$. On $S > 0$ and $Q \leq k$, normalize to $p_j = U_j/(bS)$. Its entropy center is $h_i = \log b - (\log A_i)/\beta$, and

$$\beta(H_\alpha(p) - h_i) = \beta \log S - \log\left(1 + \frac{W}{A_i S}\right). \tag{4.7}$$

The residual $U_i^\alpha - A_i U_i$ contains a factor β, which preserves concentration near order one.

Give each weight an independent count of mean θU_i. The numbers observed exactly j times are independent Poisson variables of means bq_{ij}, where $q_{ij} = \mathbb{E}[e^{-\theta U_i}(\theta U_i)^j/j!]$ for $j \geq 1$. Their total variation is at most $\sqrt{bD}$, with

$$D = \sum_{j \geq 1} \frac{(q_{1j} - q_{0j})^2}{q_{1j} + q_{0j}}.$$

A common random labeling and ordering kernel recovers the labeled sample experiment. Conditional on the weights and observation length, the ordered labels are i.i.d. from the normalized vector. The transfer accounts separately for alphabet overflow and insufficient observations.

Calibration at the original accuracy. For X above a fixed threshold, put $T = \alpha \log X$, $v = (\beta/\alpha)e^{-T}$, and $w = h + v$. Choose $L \asymp_\varepsilon (T + \alpha)$, $\theta = c_\varepsilon \alpha k^{-1/\alpha} X$, and $B = c_m L/\theta$. The root equality gives the exact balance

$$\frac{\log k}{\alpha} = \frac{E - v}{h} + \log X + \log(1 + \log X) + 1 - \log \Psi + 2 \log \alpha + 2 \log \beta.$$

It permits separation $\log(A_1/A_0) \geq E + w$ while keeping kD small. Matched moments cancel in the mixed-Poisson expansion; actual atoms in the priors bound the denominator for the one possible moment defect. Normalization fluctuations fit inside the extra margin w, uniformly also in the degree-zero near-one case. Theorem G.7 proves these estimates and the testing contradiction at $n \asymp_\varepsilon R_\alpha$.

4.3 Completion for general real order

Proof of Theorem 2.1. Lemma 4.1, Theorem F.3, Theorem B.1, and Proposition C.1 give the four upper-bound ranges. Theorem G.7 and the hidden-coordinate construction give the corresponding lower bounds. Take the minimum of their positive lower constants, the maximum of their upper constants, and the maximum of their alphabet cutoffs. Every choice depends only on ε, giving the asserted uniform quantifiers. □

For fixed noninteger α, write $\mathcal{Q}^2 = c_{\alpha,\varepsilon} k^{1/\alpha}$ and let $y(1 + \log y) = \mathcal{Q}^2/e$. The root criterion is zero at y and exceeds one at $2y$ for large k. Thus $X \asymp_{\alpha,\varepsilon} k^{1/\alpha}/\log k$ and $R_\alpha \asymp_{\alpha,\varepsilon} k/\log k$.

5 Why the leading power determines the functional rate

The transfer argument is simpler than the uniform-order calculation. Fix $0 < \xi < \beta_f \varepsilon$ and choose $\tau > 0$ so that

$$e^{-\xi} a x^r \leq f(x) \leq e^{\xi} a x^r \qquad (0 \leq x \leq \tau).$$

For the upper bound, first estimate F_f to a fixed absolute tolerance using a number of samples independent of k. Lemma H.1 does this by approximating the continuous function $f(x)/x$ by a polynomial. If the estimated sum is substantial, this absolute estimate already gives logarithmic accuracy. Otherwise positivity implies that every coordinate is below τ, and the leading-power band reduces the task to estimating H_r with accuracy $(\beta_f \varepsilon - \xi)/(r-1)$.

For the lower bound, choose a fixed integer $m \geq 1/\tau$ and replace each symbol of $p \in \Delta_{\lfloor k/m \rfloor}$ by m copies, each of probability p_i/m. A sample from this replicated distribution is exactly a sample from p together with an independent uniform label. Discarding that label reverses the operation. Every probability now lies below τ, and

$$\left| \log F_f(q) - \log(a m^{1-r} P_r(p)) \right| \leq \xi.$$

Consequently an estimator of the logarithmic functional would estimate H_r with accuracy $(\beta_f \varepsilon + \xi)/(r-1)$ and the same sample size. Theorem H.2 proves both reductions with explicit failure accounting.

Proof of Theorem 2.2. Apply those reductions with, for example, $\xi = \beta_f \varepsilon/2$. Use Theorem G.10 for noninteger r and the direct integer bounds for integer r. These bounds cover every fixed positive accuracy and confidence, as required by both reductions. A fixed replication changes neither $k^{1-1/r}$ nor $k/\log k$ by more than a constant factor, and the auxiliary sample cost is constant. □

The finite-leading-power condition has substantive content. For every $s > 0$, a smooth, convex, increasing, unit-Lipschitz function proportional to e^{-A/x^s} has logarithmic estimation complexity $\Omega(k^{2s})$, although all its derivatives vanish at zero. Appendix H proves this and a single smooth example with an exponential lower bound. Thus continuity, smoothness, or convexity alone cannot replace the leading-power hypothesis.

A The counting identities used throughout the proof

The upper and lower bounds use the same observation model: independent Poisson counts. This section derives the count identities before they are used. There are three distinct operations to keep in mind. A polynomial in a probability becomes an unbiased statistic by replacing powers with falling factorials. A real power can also be estimated by a Gamma ratio, with a small, explicit bias. Finally, the likelihood ratios of Poisson counts have an exact overlap formula, which will make hard distributions statistically indistinguishable.

A.1 Poissonization and factorial counts

Suppose that the number of observations is a random variable $M \sim \mathrm{Poi}(n)$, independent of an iid sequence with distribution $p = (p_1, \dots, p_k)$. Write N_i for the number of occurrences of symbol i among the first M observations. Then

$$N_1, \dots, N_k \text{ are independent}, \qquad N_i \sim \mathrm{Poi}(np_i).$$

One can see independence directly from their joint probability-generating function. For $z_1, \dots, z_k \geq 0$,

$$\mathbb{E} \prod_{i=1}^{k} z_i^{N_i} = \mathbb{E} \left(\sum_i p_i z_i \right)^M = \exp\left(n \sum_i p_i (z_i - 1) \right) = \prod_i \exp(np_i(z_i - 1)).$$

The final expression is a product of Poisson generating functions. Here n is a sample-size *mean*; it need not be an integer. Later, a Poisson tail bound converts the construction into an estimator using a fixed number of observations. An independent pilot sample is obtained by independent thinning, and therefore does not change these identities.

Fixed and Poisson sample sizes have different denominators. If the sample size is a fixed integer N, then counting ordered choices of j distinct positions gives $\mathbb{E}(N_i)_j = (N)_j p_i^j$. The unbiased denominator is $(N)_j$. In the Poisson experiment it is n^j, because n is the Poisson mean. The count formulas below use the Poisson convention unless a fixed sample size is explicitly stated.

How to implement Poissonization from a fixed sample pool. With $\lceil 2n \rceil$ available observations, independently draw $M \sim \mathrm{Poi}(n)$ and use the first M observations if available. Couple this procedure to an infinite i.i.d. sequence. The ideal Poisson estimator and the implemented estimator agree unless $M > \lceil 2n \rceil$. Since $\mathbb{E}2^M = e^n$, Markov's inequality yields

$$\mathbb{P}\{M > \lceil 2n \rceil\} \leq e^{-(2\log 2 - 1)n}.$$

An ideal failure probability ζ therefore becomes at most $\zeta + e^{-(2\log 2-1)n}$. Independent main and pilot lengths of means n and $c_p n$ have a total of mean $(1 + c_p)n$; allocate successive disjoint blocks and use the same argument for their sum. Independence is asserted in the ideal experiment. We handle the finite-pool overflow by coupling, rather than conditioning the counts on no overflow.

For integers $j \geq 0$, define

$$(N)_j = N(N-1)\cdots(N-j+1), \qquad (N)_0 = 1.$$

The product vanishes when $j > N$. If $N \sim \mathrm{Poi}(\lambda)$, then

$$\mathbb{E}(N)_j = \sum_{q=j}^{\infty} \frac{q!}{(q-j)!} e^{-\lambda} \frac{\lambda^q}{q!} = \lambda^j \sum_{v=0}^{\infty} e^{-\lambda} \frac{\lambda^v}{v!} = \lambda^j. \tag{A.1}$$

Consequently $(N_i)_j / n^j$ estimates p_i^j without bias. Ordinary powers do not have this property: $\mathbb{E}(N_i/n)^2 = p_i^2 + p_i/n$.

A.2 From a polynomial approximation to a count estimator

Let $Q(x) = \sum_{j=0}^{L} c_j x^j$ approximate x^α on the probability range of interest. Replacing each x^j by $(N_i)_j/n^j$ produces an estimator whose mean is exactly $Q(p_i)$. Thus its bias is a *deterministic approximation error*. Its variance is also explicit; the next identity is why derivatives of the approximation polynomial appear in the later proof. In the following lemma, $\mathsf{G}_j(N) = (N)_j$ for integer j, with $\mathsf{G}_0(N) = 1$.

Lemma A.1 (Factorial derivative energy)**.** *Let $N_i \sim \mathrm{Poi}(np_i)$ independently, and let $Q(x) = \sum_{j=0}^{L} c_j x^j$. The statistic $Z_Q = \sum_i \sum_{j=0}^{L} c_j \mathsf{G}_j(N_i)/n^j$, with $\mathsf{G}_0 = 1$, satisfies*

$$\mathbb{E}Z_Q = \sum_i Q(p_i), \qquad \mathrm{Var}\, Z_Q = \sum_i \sum_{\ell=1}^{L} \frac{p_i^\ell}{\ell! n^\ell} \{Q^{(\ell)}(p_i)\}^2.$$

Proof. For integers $r, s \geq 0$, counting pairs of ordered injective selections by the number of shared indices gives the polynomial identity

$$(N)_r (N)_s = \sum_{\ell=0}^{\min(r,s)} \binom{r}{\ell}\binom{s}{\ell} \ell! (N)_{r+s-\ell}.$$

Taking the Poisson expectation and subtracting the $\ell = 0$ term gives the covariance. Expanding Q and collecting the terms with the same ℓ proves the formula. Independent coordinates add their variances. For a single coordinate, the term with a fixed ℓ is

$$\sum_{r,s \geq \ell} c_r c_s \binom{r}{\ell}\binom{s}{\ell} \ell! \frac{p^{r+s-\ell}}{n^\ell} = \frac{p^\ell}{\ell! n^\ell} \left(\sum_{r \geq \ell} c_r (r)_\ell p^{r-\ell} \right)^2.$$

The expression in parentheses is $Q^{(\ell)}(p)$. This proves the claimed derivative formula, including all cross terms between the polynomial coefficients. □

The order-two specialization. Putting $Q(x) = x^2$ in the derivative-energy identity recovers (A.16): the first derivative contributes $4P_3/n$ and the second contributes $2P_2/n^2$. Thus the elementary collision calculation is a special case of the general polynomial formula.

A.3 Why power-sum accuracy must be calibrated before taking a logarithm

For Rényi entropy the target is $H_\alpha(p) = -\beta^{-1} \log P_\alpha(p)$, where $\beta = \alpha - 1$. A power-sum error of relative size $O(\beta\varepsilon)$ is sufficient near order one, but this informal statement loses constants that matter when the order varies. We will instead use an exact interval. If

$$e^{-2\varepsilon\beta} P_\alpha(p) \leq \widehat{P} \leq P_\alpha(p),$$

then $-\beta^{-1} \log \widehat{P} - \varepsilon$ differs from $H_\alpha(p)$ by at most ε. The following symmetric version is also useful. The following statement applies to every positive additive sum $F(p) = \sum_i f(p_i)$ and every scale $\beta > 0$. The letter δ in this proposition denotes a failure probability, not a distance to an integer.

Proposition A.2 (Exact logarithmic accuracy bound)**.** *Let $F(p) = \sum_i f(p_i) > 0$, $\beta > 0$, and $\eta = \tanh(\beta\varepsilon)$. If*

$$\left| \sum_i Q(p_i) - F(p) \right| \leq \frac{\eta}{2} F(p), \qquad \mathrm{Var}\, Z_Q \leq \frac{\delta \eta^2}{4} F(p)^2$$

uniformly in the parameter class, then

$$\widehat{H} = -\frac{\log Z_Q}{\beta} + \frac{\log(1-\eta^2)}{2\beta}$$

estimates $-\beta^{-1}\log F$ *within* ε *with failure at most* δ*; define it arbitrarily when* $Z_Q \le 0$*. The same conclusion holds for any real statistic with the stated bias and variance, including a statistic defined using an independent pilot.*

Proof. Chebyshev gives $Z_Q/F \in [1-\eta, 1+\eta]$ except with probability δ. This interval is positive. The midpoint of its logarithms is $\frac{1}{2}\log(1-\eta^2)$, and its logarithmic half-width is $\operatorname{artanh}(\eta) = \beta\varepsilon$. For a pilot, apply this argument conditionally and account separately for the pilot's failure. No replacement of ε by a smaller accuracy occurs. □

A.4 Extending factorial counts to real powers: the nonnegative transform

A noninteger power x^s cannot be represented by a finite polynomial exactly. The Gamma function gives a second extension of factorial counts. Recall that $\Gamma(j+1) = j!$ for nonnegative integers j and $\Gamma(x+1) = x\Gamma(x)$. For a real $s \ge 1$ and an integer $j \ge 0$, define

$$\mathsf{G}_s(j) = \begin{cases} \Gamma(j+1)/\Gamma(j+1-s), & j \ge \lceil s \rceil, \\ 0, & j < \lceil s \rceil, \end{cases} \qquad \mathsf{g}_s(j) = \mathsf{G}_s(j)^{1/s}. \tag{A.2}$$

At an integer r, $\mathsf{G}_r(j) = (j)_r$ is the falling factorial.

Lemma A.3 (Mean and deterministic approximation)**.** *For every* $s \ge 1$ *and integer* $j \ge 0$*,*

$$(j-s)_+ \le \mathsf{g}_s(j) \le j, \qquad |\mathsf{g}_s(j) - j| \le s. \tag{A.3}$$

If $N \sim \mathrm{Poi}(\lambda)$*, then* $\mathbb{E}\mathsf{G}_r(N) = \lambda^r$ *for integer* r*. For noninteger* $s = m+\gamma$*,* $0 < \gamma < 1$*, and* $\delta = 1-\gamma$*,*

$$\mathbb{E}\mathsf{G}_s(N) = \frac{\lambda^s}{\Gamma(\delta)} \int_0^\lambda t^{\delta-1} e^{-t}\, dt. \tag{A.4}$$

Proof. For $j < \lceil s \rceil$, the sandwich is immediate. Otherwise, write the logarithm of the Gamma ratio as the integral of the increasing function $\psi = (\log\Gamma)'$ over $[j+1-s, j+1]$. The average of an increasing function over a terminal interval decreases when that interval is lengthened. Since $s \ge 1$, this average is at most the average over $[j, j+1]$, which equals $\log j$. Convexity of $\log\Gamma$ also gives $\psi(t) \ge \log(t-1)$ for $t > 1$ by comparison with the preceding unit secant. If $j > s$, the integral average is consequently at least $\log(j-s)$. The boundary case $j = s$ has a zero lower bound. Exponentiating proves (A.3).

For integers, factorial unbiasedness follows by shifting the Poisson series. For nonintegers, cancellation of $j!$ gives

$$\mathbb{E}\mathsf{G}_s(N) = e^{-\lambda}\lambda^{m+1} \sum_{\ell \ge 0} \frac{\lambda^\ell}{\Gamma(\ell+1+\delta)}.$$

The function $e^{-\lambda}\lambda^\delta \sum_{\ell\ge 0} \lambda^\ell/\Gamma(\ell+1+\delta)$ vanishes at zero and has derivative $e^{-\lambda}\lambda^{\delta-1}/\Gamma(\delta)$. Termwise differentiation is justified by local uniform convergence; adjacent terms cancel using the Gamma recurrence. This proves the mean. □

For a noninteger s, the integral in (A.4), divided by $\Gamma(\delta)$, is a number in $[0,1]$ that approaches one as λ grows. Thus $\mathsf{G}_s(N)$ is downward biased for λ^s, but the bias becomes small for sufficiently frequent symbols. The pointwise sandwich (A.3) will be particularly useful at high orders: taking an sth root turns a nonlinear statistic into a quantity differing from the count itself by at most s.

A.5 An analytic count extension with exact order-one cancellation

The nonnegative count family G_α has a cutoff jump when α crosses an integer. A signed extension removes that jump and retains the exact factorial members. This section supplies a common heavy-count construction. The corresponding light-coordinate approximation and the complete statistical bound are proved in Sections D.4 and E.3.

The distinction between the next transform and G_a is important. The transform G_a truncates low counts to zero; the analytic transform below retains them, even when they have a negative sign. This makes its dependence on the order continuous. For every real a and integer $j \geq 0$, define

$$\mathsf{H}_a(j) = \frac{\Gamma(j+1)}{\Gamma(j+1-a)}, \qquad \Delta v(j) = v(j+1) - v(j).$$

The reciprocal function $1/\Gamma(z)$ is entire. In particular its value is zero at every nonpositive integer, so the ratio is well-defined even where a literal denominator $\Gamma(j+1-a)$ has a pole. In particular $\mathsf{H}_r(j) = (j)_r$ for every nonnegative integer r.

Proposition A.4 (Exact analytic count identities)**.** *For $\lambda > 0$, let $m_a(\lambda) = \mathbb{E}\mathsf{H}_a(N)$ with $N \sim \mathrm{Poi}(\lambda)$. For every real a,*

$$\Delta^\ell \mathsf{H}_a(j) = (a)_\ell \mathsf{H}_{a-\ell}(j), \qquad \ell \geq 0, \tag{A.5}$$

$$m_a(\lambda) = \lambda^a - \frac{1}{\Gamma(-a)} \int_1^\infty e^{-\lambda t} t^{-a-1}\, dt, \tag{A.6}$$

$$\operatorname{Var} \mathsf{H}_a(N) = \sum_{\ell \geq 1} \frac{\lambda^\ell}{\ell!} (a)_\ell^2 m_{a-\ell}(\lambda)^2. \tag{A.7}$$

The variance series converges. At $a = r \in \mathbb{N}$ it terminates at r, and the mean is exactly λ^r. For $a > 0$,

$$|m_a(\lambda) - \lambda^a| \leq \frac{\Gamma(a+1)|\sin(\pi a)|}{\pi} e^{-\lambda} \min\{a^{-1}, \lambda^{-1}\}. \tag{A.8}$$

Proof. The Gamma recurrence gives $\Delta \mathsf{H}_a = a \mathsf{H}_{a-1}$; induction proves (A.5), including the removable zeros. For $a < 0$ the beta integral gives

$$\mathsf{H}_a(j) = \frac{1}{\Gamma(-a)} \int_0^1 t^j (1-t)^{-a-1}\, dt.$$

Take the Poisson expectation and substitute $u = 1 - t$. Subtracting the tail from the complete Gamma integral proves (A.6) for $a < 0$. For fixed $\lambda > 0$ both sides are entire in complex a. Indeed, the Gamma ratio grows at most polynomially in j on each compact set of a, so its Poisson expectation series is locally uniformly convergent; the integral on $[1, \infty)$ is also locally uniformly convergent. Analytic continuation proves the formula for all a. Its value at $\lambda = 0$ is instead read directly as $m_a(0) = 1/\Gamma(1-a)$. Bounding the integral both by $e^{-\lambda}/a$ and by $e^{-\lambda}/\lambda$, and using reflection, proves (A.8).

It remains to prove the variance identity. We use an orthogonal polynomial basis for the Poisson distribution, in the same way that Fourier coefficients decompose a squared norm. No approximation is involved in this step. Define the Charlier polynomials by

$$e^{-\lambda t}(1+t)^N = \sum_{\ell \geq 0} C_\ell(N; \lambda) t^\ell / \ell!.$$

The joint generating function gives $\mathbb{E}C_\ell C_q = \mathbf{1}\{\ell = q\}\ell!\lambda^\ell$. Repeated Poisson size bias gives

$$\mathbb{E}[v(N)C_\ell(N;\lambda)] = \lambda^\ell \mathbb{E}\Delta^\ell v(N).$$

These polynomials are complete in $L^2(\mathrm{Poi}(\lambda))$: if v is orthogonal to all of them, then the entire function $e^{-\lambda(z-1)}\mathbb{E}[v(N)z^N]$ has every derivative zero at $z = 1$. It therefore vanishes, and its coefficients imply $v(j) = 0$ for all j. Entirety follows by Cauchy–Schwarz and Poisson exponential moments. Thus Parseval gives, whenever $v(N)$ is square integrable,

$$\operatorname{Var} v(N) = \sum_{\ell \geq 1} \frac{\lambda^\ell}{\ell!} (\mathbb{E}\Delta^\ell v(N))^2. \tag{A.9}$$

The polynomial growth of $\mathsf{H}_a(j)$ ensures square integrability. Substitution of (A.5) proves (A.7). □

Lemma A.5 (Poisson Poincaré from the same count expansion)**.** *If $N \sim \mathrm{Poi}(\lambda)$ and $v(N)$ is square integrable, then*

$$\operatorname{Var} v(N) \leq \lambda \mathbb{E}\{v(N+1) - v(N)\}^2.$$

Proof. If the right side is infinite the assertion is immediate. Otherwise apply the complete Charlier expansion to Δv. Its squared norm, multiplied by λ, equals

$$\sum_{\ell \geq 1} \ell \frac{\lambda^\ell}{\ell!} (\mathbb{E}\Delta^\ell v(N))^2,$$

which dominates (A.9) term by term. □

Removing noise from a known linear term. Because $\sum_i p_i = 1$, the aggregate of a linear function cp_i is the known constant c. It is wasteful to estimate that constant by $c\sum_i N_i/n$. The following identity describes exactly how removing an affine term changes the variance. Only the first term of the Charlier expansion is affected. For constants c, d, the exact residual formula is

$$\begin{aligned} \operatorname{Var}\{\mathsf{H}_a(N) - cN - d\} &= \lambda\{a m_{a-1}(\lambda) - c\}^2 \\ &\quad + \sum_{\ell \geq 2} \frac{\lambda^\ell}{\ell!} (a)_\ell^2 m_{a-\ell}(\lambda)^2. \end{aligned} \tag{A.10}$$

Here the affine term must be removed globally, or compensated in the other part of a pilot partition: a random subset of the identity $\sum_i p_i = 1$ is not known.

Why this matters near $\alpha = 1$. Write $\alpha = 1 + \beta$. As β decreases to zero, $\mathsf{H}_{1+\beta}(N)$ approaches N, whose aggregate mean is already known. After removing a suitable linear term, the remaining fluctuation should be proportional to β. The next proposition shows that its variance is proportional to β^2, uniformly in small positive β. This matches the shrinking power-sum accuracy required before division by β in the entropy formula.

Proposition A.6 (Uniform divided-difference noise at order one)**.** *Let $0 < \beta \leq 1/2$, $N \sim \mathrm{Poi}(\lambda)$, $\lambda \geq 1$, and $B > 0$. Then, for an absolute constant C,*

$$\operatorname{Var}\left\{\frac{\mathsf{H}_{1+\beta}(N) - B^\beta N}{\beta}\right\} \leq C\lambda(\lambda^{2\beta} + B^{2\beta})\{1 + \log^2(\lambda/B)\}. \tag{A.11}$$

The bias of its numerator relative to $\lambda^{1+\beta} - B^\beta\lambda$ is at most $C\beta e^{-\lambda}/\lambda$.

Proof. Uniformly for $j \ge 0$ and $0 \le \beta \le 1/2$,

$$|\mathsf{H}_\beta(j) - (j+1)^\beta| \le C\beta(j+1)^\beta.$$

Indeed, the logarithm of their ratio is the integral over $0 \le t \le \beta$ of $\psi(j+1-t) - \log(j+1)$, where $\psi = \Gamma'/\Gamma$. This integrand is uniformly bounded: the recurrence $\psi(x+1) = \psi(x) + 1/x$, a bound on the compact interval $[1/2, 3/2]$, and comparison of harmonic sums with the logarithm prove the assertion. Also the mean value theorem gives

$$|(j+1)^\beta - B^\beta| \le \beta\{(j+1)^\beta + B^\beta\}|\log((j+1)/B)|.$$

Consequently

$$|(1+\beta)\mathsf{H}_\beta(j) - B^\beta| \le C\beta\{(j+1)^\beta + B^\beta\}\{1 + |\log((j+1)/B)|\}. \tag{A.12}$$

Apply Lemma A.5 and $\Delta\mathsf{H}_{1+\beta} = (1+\beta)\mathsf{H}_\beta$. For $V = (N+1)/\lambda$, uniformly over $0 \le q \le 1$ and $\lambda \ge 1$,

$$\mathbb{E}V^q + \mathbb{E}[V^q(\log V)^2] \le C.$$

On $V \ge 1$ this follows from $V^q(\log V)^2 \le C(1+V^3)$ and the first three Poisson moments. On $1/2 \le V \le 1$ it is immediate. On $V < 1/2$, use $|\log V| \le \log\lambda$ and the Poisson Chernoff bound $\mathbb{P}(N < \lambda/2) \le e^{-c\lambda}$; the bounded range $1 \le \lambda \le 4$ is immediate. Expand $\log((N+1)/B) = \log(\lambda/B) + \log V$ in (A.12) to obtain (A.11). Finally $\Gamma(2+\beta)|\sin(\pi\beta)|/\pi \le C\beta$; (A.8) proves the bias assertion. □

In probability units, with $a = 1+\beta$, $\lambda = np$ and $B = nb$, Proposition A.6 gives

$$\mathrm{Var}\{n^{-a}\mathsf{H}_a(N) - b^\beta N/n\} \le C\beta^2 n^{-2a}\lambda(\lambda^{2\beta} + B^{2\beta})\{1 + \log^2(\lambda/B)\}.$$

Thus the common family preserves the full β^2 cancellation; an order-independent pilot-tail estimate alone would not do so.

Controlling the negative values. The signed transform will be used on coordinates that the pilot declares frequent. Nevertheless their main count can occasionally be small. We must bound the resulting negative contribution, not assume it is absent. The following lemma supplies an exponentially small bound.

Lemma A.7 (The signed low-count correction)**.** *Let $a = m + \gamma > 1$, $m = \lfloor a \rfloor$, $0 < \gamma < 1$, $h = \gamma(1-\gamma)$, and $\lambda \ge 2(m+1)$. Then*

$$\mathbb{E}\big[|\mathsf{H}_a(N)|\mathbf{1}\{N \le m-1\}\big] \le Che^{-\lambda}\lambda^{m-1}.$$

Every negative value is included in this range; at integer orders there are no negative values.

Proof. For $j \le m-1$, the reciprocal-Gamma recurrence gives

$$\frac{1}{|\Gamma(j+1-a)|} = \frac{\gamma}{\Gamma(1-\gamma)} \prod_{\ell=1}^{m-j-1} (\ell+\gamma) \le Ch(m-j)!.$$

Here $\gamma/\Gamma(1-\gamma) = h/\Gamma(2-\gamma)$ and Γ is bounded away from zero on $[1,2]$. Taking the Poisson expectation cancels the numerator $j!$. Moreover

$$\sum_{j=0}^{m-1} \lambda^j (m-j)! = \lambda^{m-1} \sum_{\ell=1}^{m} \frac{\ell!}{\lambda^{\ell-1}} \le 2\lambda^{m-1},$$

because successive terms have ratio at most $(m+1)/\lambda \le 1/2$. The denominator of $\mathsf{H}_a(j)$ is positive for $j \ge m$, which proves the sign assertion. □

When a floor removes the distinction between the transforms. Some estimators replace all sufficiently small count values by a fixed positive floor. If the floor is at least the order, the signed and nonnegative transforms give exactly the same result, as the next lemma shows. This is a deterministic identity, with no probabilistic assumptions.

Lemma A.8 (Exact agreement after a count floor)**.** *Let $a \geq 1$ and $b \geq a$. For every integer $j \geq 0$,*

$$\max\{\mathsf{H}_a(j), b^a\} = \max\{\mathsf{G}_a(j), b^a\}.$$

Proof. At integer orders the two count families agree. Otherwise write $a = m + \gamma$, $0 < \gamma < 1$. They agree when $j \geq m + 1$. For $0 \leq j \leq m$, the Gamma recurrence gives

$$|\mathsf{H}_a(j)| = \frac{j!}{\Gamma(1-\gamma)} \prod_{\ell=j+1}^{m} (a - \ell) \leq j!(m-j)! \leq m! \leq a^a \leq b^a.$$

The bound $\Gamma(1-\gamma) \geq 1$ follows, for example, by Jensen's inequality applied to $T^{-\gamma}$ with $T \sim \mathrm{Exp}(1)$. The truncated count is zero in this range, proving the claim. □

Above the original cutoff, H_a equals G_a and retains its pointwise lower bound $(j-a)_+^a$. Its additional value at $j = m$ is nonnegative; Lemma A.7 controls the remaining signed correction. These identities therefore extend the heavy-count framework. The subsequent sections construct and calibrate the corresponding polynomial estimator for rare symbols.

A.6 The likelihood-overlap identity used in the lower bounds

For probability laws P, Q, with likelihood ratio $L = dP/dQ$, define $\chi^2(P, Q) = \mathbb{E}_Q(L-1)^2$. Total variation obeys $\mathrm{TV}(P, Q) \leq \frac{1}{2}\sqrt{\chi^2(P, Q)}$. Thus a small second moment of a mixture likelihood ratio shows that the mixture is difficult to distinguish from the reference law. The next calculation gives that second moment explicitly for Poisson observations. In the iid application, the coordinate index e is a symbol and normally $\nu_e = n$.

Lemma A.9 (Exponential overlap and hidden locations)**.** *Let independent $N_e \sim \mathrm{Poi}(\nu_e \theta_e^*)$ be the reference experiment, with $\theta_e^* > 0$. For alternatives θ, ϕ, their likelihood ratios satisfy*

$$\mathbb{E}_* L_\theta L_\phi = \exp\left\{ \sum_e \nu_e \frac{(\theta_e - \theta_e^*)(\phi_e - \theta_e^*)}{\theta_e^*} \right\}. \tag{A.13}$$

Suppose m alternatives have disjoint perturbation supports and $\sum_e \nu_e (\theta_e^{(s)} - \theta_e^*)^2/\theta_e^* \leq J$. Write P_s for the law of alternative s and $\overline{P} = m^{-1}\sum_{s=1}^m P_s$. Their uniform mixture satisfies

$$\chi^2(\overline{P}, P_*) \leq (e^J - 1)/m. \tag{A.14}$$

Proof. For one coordinate, the likelihood ratio is

$$L_\theta(N) = \exp\{-\nu(\theta - \theta^*)\}(\theta/\theta^*)^N.$$

The formula includes $\theta = 0$ by continuity, with $0^0 = 1$. Multiplying two such ratios and using $\mathbb{E}_* t^N = \exp\{\nu\theta^*(t-1)\}$ gives

$$\mathbb{E}_* L_\theta L_\phi = \exp\left\{ -\nu(\theta + \phi - 2\theta^*) + \nu\theta^*(\theta\phi/(\theta^*)^2 - 1) \right\} = \exp\left\{ \nu \frac{(\theta - \theta^*)(\phi - \theta^*)}{\theta^*} \right\}.$$

Independence multiplies the coordinate overlaps, proving (A.13). Off-diagonal overlaps of alternatives equal one by disjointness. Expanding the squared mixture likelihood and subtracting one proves (A.14). □

The same kernel gives the iid moment cancellation. For a finite signed measure ω supported on a bounded subset of $[0,\infty)$ and $u_* > 0$, termwise expansion of (A.13) gives exactly

$$\sum_{j\geq 0} \frac{\left[\int e^{-\lambda u}(\lambda u)^j/j!\, d\omega(u)\right]^2}{e^{-\lambda u_*}(\lambda u_*)^j/j!} = \sum_{\ell\geq 0} \frac{\lambda^\ell}{\ell! u_*^\ell}\left[\int (u-u_*)^\ell\, d\omega(u)\right]^2. \tag{A.15}$$

Bounded support justifies all exchanges by absolute domination by the exponential series. For instance, if $\omega = \mu_+ - \mu_-$ and the probability measures μ_+, μ_- have equal moments through degree L, then $\int (u-u_*)^\ell\, d\omega(u) = 0$ for $0 \leq \ell \leq L$. The right side starts at degree $L+1$. The left side measures a weighted squared difference between their mixed Poisson count probabilities. This is the exact reason moment matching hides differences from the data. Thus the hidden-location calculation and iid moment matching are evaluations of the same independent-count likelihood kernel.

A.7 The order-two example

Imagine first that the number of observations is Poisson with mean n. Then the counts N_i of the symbols are independent and $N_i \sim \mathrm{Poi}(np_i)$. Section A explains this reduction and the return to a fixed sample size.

A.7.1 An unbiased estimator of the collision probability

At $\alpha = 2$, $P_2(p) = \sum_i p_i^2$ is the probability that two independent observations are equal. Since $\mathbb{E}[N_i(N_i-1)] = (np_i)^2$, the estimator

$$Z = \frac{1}{n^2}\sum_{i=1}^{k} N_i(N_i-1)$$

satisfies $\mathbb{E}Z = P_2(p)$. The product $N_i(N_i-1)$ counts ordered pairs of distinct observations of symbol i.

Why not use $\sum_i (N_i/n)^2$? Because $\mathbb{E}N_i^2 = (np_i)^2 + np_i$, its expectation is $P_2 + 1/n$. Subtracting N_i removes that bias exactly.

A.7.2 Its exact variance and a uniform sample bound

The falling-factorial identity

$$(N)_2^2 = (N)_4 + 4(N)_3 + 2(N)_2$$

gives, for a Poisson count with mean λ, $\mathrm{Var}((N)_2) = 4\lambda^3 + 2\lambda^2$. Independence therefore yields

$$\mathrm{Var}\, Z = \frac{4P_3(p)}{n} + \frac{2P_2(p)}{n^2}. \tag{A.16}$$

Since $\max_i p_i \le \sqrt{P_2}$, $P_3 \le (\max_i p_i)P_2 \le P_2^{3/2}$. Combining this with $P_2 \ge 1/k$ gives

$$\frac{\mathrm{Var}\, Z}{P_2^2} \le \frac{4}{n\sqrt{P_2}} + \frac{2}{n^2 P_2} \le \frac{4\sqrt{k}}{n} + \frac{2k}{n^2}. \tag{A.17}$$

Fix $\eta = 1 - e^{-\varepsilon} \in (0,1)$. If $n = C_\varepsilon\sqrt{k}$, Chebyshev's inequality implies

$$\mathbb{P}\{|Z - P_2| > \eta P_2\} \le \frac{4/C_\varepsilon + 2/C_\varepsilon^2}{\eta^2}.$$

Choose C_ε to make this bound at most $1/4$. On the complementary event,

$$|\log Z - \log P_2| \le \max\{-\log(1-\eta), \log(1+\eta)\} = \varepsilon.$$

Thus $-\log Z$ estimates H_2 to accuracy ε. Poissonization can be implemented from a fixed pool of a constant multiple of n observations, with exponentially small overflow probability. This proves $S_2(k, \varepsilon, 1/3) = O_\varepsilon(\sqrt{k})$ for large k. The hidden-coordinate argument in Theorem C.2 proves the matching lower bound.

What this example teaches. The estimator does not estimate each p_i accurately. It estimates a sum directly, and its variance is compared to the square of that sum. For a general integer order, replace $(N)_2$ by $(N)_r$. For a noninteger order, no finite polynomial equals x^α; the next step is polynomial approximation with simultaneous variance control. That is the central additional difficulty.

B A direct proof for every integer order

Before constructing a polynomial approximation for noninteger orders, it is useful to solve the integer case directly. The estimator is particularly simple: count ordered collisions. The subtlety is in the proof when the integer order grows with the alphabet size.

Throughout this section, $r \geq 2$ is an integer, $P_r(p) = \sum_i p_i^r$, and $H_r(p) = -(r-1)^{-1} \log P_r(p)$, as in (2.2). Under Poissonization, the counts are independent with $N_i \sim \mathrm{Poi}(np_i)$. Define

$$\widehat{P}_r = n^{-r} \sum_{i=1}^{k} (N_i)_r. \tag{B.1}$$

It is nonnegative and unbiased: $\mathbb{E}\widehat{P}_r = P_r(p)$. Here $(N_i)_r$ counts ordered selections of r occurrences of symbol i. The estimator does not need to know which symbols are rare or frequent. The partition used below is only a device for analyzing its lower tail.

B.1 Why a constant relative-variance bound is too restrictive

For fixed r, one usually proves that $\widehat{P}_r/P_r$ is close to one. When r grows, that requirement becomes unnecessarily strong. The exact condition needed for entropy accuracy ε is

$$e^{-\varepsilon(r-1)} P_r \leq \widehat{P}_r \leq e^{\varepsilon(r-1)} P_r. \tag{B.2}$$

Indeed, taking logarithms and dividing by $r-1$ gives an error at most ε. The allowed multiplicative interval widens with r.

Here is a concrete reason not to insist on bounded relative variance. Fix $D > 0$, put $a = k^{-1+1/r}$, and consider

$$p_1 = a, \qquad p_i = \frac{1-a}{k-1} \quad (i > 1), \qquad n = Dr/a = Drk^{1-1/r}.$$

Since $a \geq 1/k$, the remaining probabilities are at most $1/k$. Consequently $a^r \leq P_r(p) \leq 2a^r$. The $j = 1$ term of the exact factorial variance formula gives

$$\frac{\mathrm{Var}(\widehat{P}_r)}{P_r(p)^2} \geq \frac{r^2 a^{2r-1}}{4na^{2r}} = \frac{r}{4D}. \tag{B.3}$$

Thus the relative variance diverges along any sequence $r \to \infty$ with fixed D. This does not rule out the entropy guarantee (B.2). The proof below uses the full interval: a second-moment bound controls the upper tail, while positivity and a weighted Jensen inequality control the lower tail.

B.2 The estimator and the uniform guarantee

Theorem B.1 (Raw collision estimator, uniformly over integer orders)**.** *Fix $0 < \varepsilon \leq 1/2$. For every $k \geq 2$ and integer $r \geq 2$, let*

$$D = D_\varepsilon = \frac{100e^{2\varepsilon}}{\varepsilon} \left(\frac{4096}{\varepsilon^2} + 1 \right), \qquad n = Drk^{1-1/r}.$$

The estimator (B.1), based on independent $N_i \sim \mathrm{Poi}(np_i)$, satisfies (B.2) with probability at least $11/12$, uniformly over $p \in \Delta_k$. Therefore $-(r-1)^{-1} \log \widehat{P}_r$ estimates $H_r(p)$ within ε on this event. Define the estimate arbitrarily when $\widehat{P}_r = 0$. Using $\lceil 2n \rceil$ ordinary iid observations instead gives success probability at least $2/3$. The constant D_ε is independent of both r and k; its displayed value is chosen for convenience, not numerical efficiency.

Proof. **Step 1: obtain a variance bound, allowing it to grow with r.** Write $P = P_r(p)$ and $K_r = k^{1-1/r}$. Jensen's inequality gives $P \geq k^{1-r} = K_r^{-r}$. Applying Lemma A.1 to $Q(x) = x^r$ yields

$$\mathrm{Var}(\widehat{P}_r) = \sum_{j=1}^{r} \binom{r}{j}^2 j!\, n^{-j} P_{2r-j}(p).$$

For $q \geq r$, monotonicity of vector norms gives $P_q \leq P^{q/r}$. Moreover $\binom{r}{j}^2 j! = (r)_j^2/j! \leq r^{2j}/j!$. Hence

$$\frac{\mathrm{Var}(\widehat{P}_r)}{P^2} \leq \sum_{j=1}^{r} \frac{1}{j!} \left(\frac{r^2}{nP^{1/r}}\right)^j \leq e^{r/D} - 1, \tag{B.4}$$

where $nP^{1/r} \geq Dr$. We will compare this exponential growth to the exponential tolerance in (B.2).

Step 2: control the upper tail at the actual tolerance. Set $y = \varepsilon(r-1)$. For $u, y \geq 0$, $e^u - 1 \leq ue^u$ and $e^y - 1 \geq ye^{y/2}$; the latter follows from $2\sinh(y/2) \geq y$. Chebyshev's inequality therefore gives

$$\Pr\{\widehat{P}_r > e^y P\} \leq \frac{e^{r/D} - 1}{(e^y - 1)^2} \leq \frac{r}{Dy^2} e^{r/D - y} \leq \frac{2}{D\varepsilon^2} \leq \frac{1}{100}. \tag{B.5}$$

Indeed, $D \geq 2/\varepsilon$ and $r \leq 2(r-1)$ imply $r/D \leq y$, while $r/(r-1)^2 \leq 2$. This is why a relative variance growing like $e^{r/D}$ is harmless for the upper tail.

Step 3: separate rare and frequent coordinates in the proof. Put $\lambda_i = np_i$ and define

$$T = \frac{4096r}{\varepsilon^2}, \qquad \mathcal{L} = \{i : \lambda_i \leq T\}, \qquad \mathcal{H} = \{i : \lambda_i > T\}.$$

Let $L = \sum_{i \in \mathcal{L}} p_i^r$ and $H = \sum_{i \in \mathcal{H}} p_i^r$, so that $L + H = P$. Correspondingly write $\widehat{P}_r = F + J$, where $F = n^{-r} \sum_{i \in \mathcal{L}} (N_i)_r$ and $J = n^{-r} \sum_{i \in \mathcal{H}} (N_i)_r$. We control the two contributions by different inequalities.

Step 4: the rare contribution has a small additive error. For $N \sim \mathrm{Poi}(\lambda)$, the factorial product identity gives

$$\mathbb{E}(N)_r^2 = \sum_{j=0}^{r} \binom{r}{j}^2 j! \lambda^{2r-j} \leq \sum_{j=0}^{r} \binom{r}{j} r^j \lambda^{2r-j} = \lambda^r (\lambda + r)^r.$$

The inequality uses $\binom{r}{j} j! = (r)_j \leq r^j$. Independence and $\lambda_i \leq T$ on $\mathcal{L}$ now imply

$$\mathbb{E}F = L, \qquad \frac{\mathrm{Var}\, F}{P^2} \leq \frac{(T+r)^r L}{n^r P^2} \leq \left(\frac{T+r}{Dr}\right)^r = A^r, \qquad A = \frac{4096/\varepsilon^2 + 1}{D}. \tag{B.6}$$

The second inequality uses $L \leq P$ and $n^r P \geq (Dr)^r$. Set $\eta = \varepsilon(r-1)/2$ and $d = e^{-\eta}(1 - e^{-\eta})$. Because $1 - e^{-\eta} \geq \eta e^{-\eta}$, another application of Chebyshev gives

$$\Pr\{F < L - dP\} \leq \frac{A^r}{d^2} \leq \frac{4(Ae^{2\varepsilon})^r}{\varepsilon^2 (r-1)^2} \leq \frac{4}{10000}.$$

For the middle inequality, $d^2 \geq \eta^2 e^{-4\eta}$ and $4\eta = 2\varepsilon(r-1) \leq 2\varepsilon r$. For the last one, $Ae^{2\varepsilon} = \varepsilon/100$ and $r \geq 2$.

Step 5: a weighted average controls the frequent contribution. If $H = 0$, then $J = 0$ and the claimed bound below is automatic. Otherwise define weights $w_i = p_i^r/H$ for $i \in \mathcal{H}$, and put

$$Z = \sum_{i \in \mathcal{H}} w_i \frac{N_i}{\lambda_i}.$$

The weights are nonnegative and sum to one. Thus

$$\mathbb{E}Z = 1, \qquad \operatorname{Var} Z = \sum_{i \in \mathcal{H}} \frac{w_i^2}{\lambda_i} \le \frac{1}{T}.$$

Except on an event of probability at most $256/(\varepsilon^2 T)$, we have $Z \ge 1 - \varepsilon/16$. For any integer $x \ge 0$, $(x)_r \ge (x - r)_+^r$. Since $z \mapsto (z_+)^r$ is convex, Jensen's inequality then gives

$$\begin{aligned} \frac{J}{H} &\ge \sum_{i \in \mathcal{H}} w_i \left(\frac{N_i - r}{\lambda_i} \right)_+^r \ge \left(Z - r \sum_{i \in \mathcal{H}} \frac{w_i}{\lambda_i} \right)_+^r \\ &\ge (1 - \varepsilon/8)^r \ge e^{-\varepsilon r/4} \ge e^{-\eta}. \end{aligned}$$

Here $r/T \le \varepsilon/16$, $\log(1 - \varepsilon/8) \ge -\varepsilon/4$ for $0 < \varepsilon \le 1/2$, and $r \ge 2$. The exceptional probability is at most $256/(\varepsilon^2 T) = 1/(16r) \le 1/32$. Notice that we did not require every frequent count to concentrate: one weighted average was sufficient.

Step 6: combine the lower-tail estimates. On the two successful events,

$$\widehat{P}_r = F + J \ge L - dP + e^{-\eta} H \ge (e^{-\eta} - d)P = e^{-2\eta} P.$$

The middle inequality uses $L + H = P$ and $L \ge e^{-\eta} L$. Since $2\eta = \varepsilon(r - 1)$, this is the required lower bound. Combining it with (B.5), the total failure probability is at most $1/100 + 4/10000 + 1/32 < 1/12$.

Step 7: return to a fixed number of iid observations. Take $m = \lceil 2n \rceil$ observations, and independently draw $M \sim \mathrm{Poi}(n)$. If $M \le m$, use the first M observations to form the counts in (B.1); otherwise return an arbitrary answer. Coupling to an unlimited iid sequence adds failure probability at most

$$\Pr\{M > m\} \le \Pr\{M > 2n\} \le e^{-n/3} < 1/12.$$

The Poisson Chernoff bound gives the middle inequality; the displayed choice of D_ε ensures the last one. Total failure is below $1/6$, and hence below $1/3$, proving the fixed-sample assertion. □

What this direct argument establishes. For every integer $r \ge 2$, the same raw collision estimator uses $O_\varepsilon(rk^{1-1/r})$ samples. This is the relevant upper rate below the high-order threshold $r \approx \log k$; the lower bound and the $k \log k$ high-order estimator are proved separately. The argument above does not claim that raw collisions attain the saturation rate when $r \gg \log k$.

C Two simpler bounds: large orders and a hidden large coordinate

Before entering the noninteger approximation argument, we prove the large-order upper bound and a lower bound that applies to every $\alpha \geq 3/2$. These proofs use only the preceding count identities. They explain both the integer-type scale and its saturation.

C.1 Why the upper bound saturates at high orders

For large α, the norm $\|p\|_\alpha$ is driven by the largest probabilities. At sample size $Ck \log k$, all empirical coordinates can be controlled to error proportional to $p_i + 1/k$. The additive $1/k$ term is harmless because $\|p\|_\alpha \geq k^{1/\alpha - 1}$. The following proof makes this observation precise and explains why the Gamma-transform order is capped even when the target order is very large.

Proposition C.1 (The high-order and min-entropy upper bound)**.** *For iid $p \in \Delta_k$, real $\alpha \geq \max\{2, \log k\}$, $L = \log(4k/\delta)$, $s = \min\{\alpha, L\}$, and $n = CkL$, use*

$$\widehat{P} = \sum_i [\mathsf{g}_s(N_i)/n]^\alpha, \qquad N_i \sim \mathrm{Poi}(np_i).$$

Its logarithm estimates iid Rényi entropy with fixed error and confidence using $O(k \log k)$ samples. The same construction with $s = L$ and a maximum in place of the sum estimates min-entropy.

Proof. **Step 1: control all counts.** For $N_i \sim \mathrm{Poi}(np_i)$, a Poisson Bernstein bound gives

$$\mathbb{P}\{|N_i - np_i| > \sqrt{2np_iL} + 2L/3\} \leq 2e^{-L}.$$

Since $n = CkL$, division by n and the inequality $\sqrt{p_i/k} \leq (p_i + 1/k)/2$ show that, for sufficiently large $C = C(\eta)$, the right-hand deviation is at most $\eta(p_i + 1/k)$. A union bound over k symbols gives, with failure probability at most $2ke^{-L} \leq \delta/2$, $|N_i/n - p_i| \leq \eta(p_i + 1/k)$ for all i, after fixing C large. **Step 2: account for the count transform.** By Lemma A.3, $s/n \leq 1/(Ck) \leq \eta/k$ and thus

$$|\mathsf{g}_s(N_i)/n - p_i| \leq \eta p_i + 2\eta/k.$$

Step 3: estimate the norm, then take its logarithm. Write $\widehat{v}_i = \mathsf{g}_s(N_i)/n$. By the triangle inequality,

$$|\|\widehat{v}\|_\alpha - \|p\|_\alpha| \leq \|\widehat{v} - p\|_\alpha \leq \eta\|p\|_\alpha + 2\eta k^{1/\alpha - 1}.$$

The ℓ_α triangle inequality and $\|p\|_\alpha \geq k^{1/\alpha - 1}$ imply relative norm error at most 3η. For $\eta \leq 1/12$, logarithmic norm error is at most 6η and entropy error is at most 12η. The ℓ_∞ argument is identical, using $\|p\|_\infty \geq 1/k$. Choose $\eta \leq \min\{1/12, \varepsilon/12\}$. For min-entropy use $-\log \max_i \widehat{v}_i$; for finite order use $-\log \widehat{P}/(\alpha - 1)$. The entropy factor $\alpha/(\alpha - 1)$ is at most two. Taking a Poisson sample from a fixed pool of a constant multiple of n observations adds only an exponentially small overflow probability, absorbed in the unused failure budget. Thus a fixed sample size $O_{\varepsilon,\delta}(k \log k)$ suffices at any fixed confidence. □

C.2 One hidden-coordinate calculation for the iid collision scale

This lower bound hides one enlarged coordinate among several candidates. The entropy changes substantially, but the data reveal too little about which candidate was changed. The reservoir consists of the remaining coordinates; changing its total mass keeps every vector normalized. The proof is uniform in the order. For noninteger orders it gives the collision obstruction only; the additional approximation obstruction requires the random-prior construction later.

Theorem C.2 (Order-uniform collision and saturation lower bound)**.** *Fix $\varepsilon > 0$. For all sufficiently large k, uniformly over every finite real $\alpha \geq 3/2$, iid Rényi entropy estimation with accuracy ε and failure probability at most $1/3$ requires*

$$\Omega_\varepsilon\left(\ell k^{1-1/\ell}\right), \qquad \ell = \min\{\alpha, \log k\}.$$

The same $\Omega_\varepsilon(k \log k)$ conclusion holds for min-entropy.

Proof. **Step 1: construct the null and alternatives.** Set $m = \lfloor e^{\ell/64} \rfloor$, $q = k^{-1+1/\ell}$, and $B = e^{2\varepsilon+2}$. The null distribution gives probability q to each of m distinguished coordinates and distributes its remaining mass uniformly among the other $k - m$ coordinates. Alternative $s \leq m$ changes only the sth distinguished probability to Bq and reduces the reservoir total by $(B-1)q$. All distributions are well defined for large k, uniformly in α: $mq \leq k^{1/64-1/3} \to 0$ and $q \leq k^{-1/3}$.

Step 2: verify entropy separation. The reservoir probabilities under the null are at most $1/k$ because $q \geq 1/k$. Also $q^\alpha \geq k^{1-\alpha}$ because $\alpha \geq \ell$. Thus

$$P_\alpha(p^{(0)}) \leq (m+1)q^\alpha, \qquad P_\alpha(p^{(s)}) \geq B^\alpha q^\alpha.$$

Since $m + 1 \leq 2e^{\alpha/64}$, the entropy separation is at least

$$\frac{\alpha \log B - \log(m+1)}{\alpha - 1} \geq \frac{\alpha(2\varepsilon + 2 - 1/64) - \log 2}{\alpha - 1} > 2\varepsilon.$$

For min-entropy the null and alternative maxima are q and Bq, so their separation is $\log B > 2\varepsilon$; here use $\ell = \log k$.

Step 3: show that the observation laws are close. Consider independent Poisson symbol counts with total mean n. Relative to the null, the diagonal and off-diagonal likelihood overlaps in Lemma A.9 are

$$\mathbb{E}_0 L_s^2 = e^{u+v}, \quad \mathbb{E}_0 L_s L_t = e^v \quad (s \neq t), \qquad u = n(B-1)^2 q, \quad v = \frac{n(B-1)^2 q^2}{1 - mq}.$$

The off-diagonal term accounts exactly for the shared reservoir; it is not set to one. Therefore the uniform mixture obeys

$$\chi^2(\overline{P}, P_0) = e^v \frac{e^u - 1}{m} + e^v - 1.$$

Put $y = \log m$. If $n \leq c(1+y)/q$, then $u \leq \kappa(1+y)$, where $\kappa = c(B-1)^2$. For $0 < \kappa \leq 1/2$,

$$\frac{e^{\kappa(1+y)} - 1}{e^y} \leq \kappa e^\kappa (1+y) e^{-(1-\kappa)y} \leq 4\kappa.$$

Moreover $v \leq \kappa(1+y)q/(1-mq) \to 0$ uniformly over the orders, since $y \leq \log k/64$ and $q \leq k^{-1/3}$. Choose c sufficiently small, then k sufficiently large, to make the total variation at most $\frac{1}{2}\sqrt{\chi^2} < 1/8$. For any test between a null law and an alternative mixture, the sum of its two error probabilities is at least $1 - \mathrm{TV}$. An entropy estimator accurate to ε would provide such a test because the target intervals are disjoint. Thus its worst-case failure probability is at least $(1 - 1/8)/2 = 7/16$, contradicting a uniform bound of $1/3$.

Step 4: read off the sample scale. Since $1 + \log\lfloor e^{\ell/64} \rfloor \asymp \ell$ for $\ell \geq 3/2$, the Poisson lower scale is $\ell k^{1-1/\ell}$. A fixed-sample estimator using s observations can be run on a $\mathrm{Poi}(2s)$ sample except with probability at most $e^{-s/4}$; this transfers the lower scale to fixed sample sizes. More explicitly, apply the comparison at a fixed small constant multiple of the diverging lower scale and use monotonicity to cover all smaller fixed sample sizes. □

D Choosing the light-symbol polynomial and its accuracy margins

The elementary count identities explain how to estimate a polynomial. We must now choose the polynomial. This is the most technical deterministic part of the i.i.d. proof. Its output is a polynomial whose expectation lies strictly inside the required multiplicative accuracy interval. The probabilistic section will show that the observed statistic stays close to this expectation.

The four steps in this section. First, solve a two-variable optimization problem exactly. Second, use that solution to choose the sample size and two probability cutoffs. Third, construct polynomial approximations that behave correctly near both cutoffs. Fourth, add the pointwise bounds over the alphabet. Keeping these steps separate distinguishes an algebraic identity from a statistical sample-complexity guarantee.

Notation and the target. Throughout, $\alpha > 1$ is noninteger, $\beta = \alpha - 1$, $m = \lfloor \alpha \rfloor$, $\gamma = \alpha - m$, $\delta = 1 - \gamma$, and $h = \gamma\delta$. Thus γ and δ are the distances to the adjacent integers. They are not error probabilities. Write $P = \sum_i p_i^\alpha$ and $P_* = k^{1-\alpha} \le P$. The full logarithmic tolerance is $E = 2\varepsilon\beta$ and $\kappa = e^{-E}$. We seek a corrected mean μ satisfying $\kappa P < \mu < P$, with explicit positive gaps from both endpoints. These gaps will later absorb sampling noise and pilot-selection errors.

Quantity	Role in the construction
a	Lower probability cutoff; the aggregate contribution below a is paid by a deterministic correction.
b	Upper end of the polynomial-approximation interval for light symbols.
L	Degree scale of the polynomial for light symbols.
n	Mean number of observations used for the main Poisson counts.
$E_0 < E$	Tolerance used by the deterministic approximation; $E - E_0$ is reserved for subsequent errors.
$X \ge 1$	Dimensionless factor multiplying the integer-type sample scale $\alpha k^{1-1/\alpha}$.
u	A small positive regularization parameter that provides additional room for fluctuations.

The symbol Q^2 below is a scalar appearing in the root equation. Later $Q(p)$ denotes a polynomial; the argument p distinguishes these uses. Likewise, the profile $u_s(x)$ is a function and is distinct from the regularization parameter u.

D.1 How the polynomial becomes an observable estimator

Before constructing the profiles, it is useful to see what they are supposed to accomplish. The unknown quantity is $P_\alpha(p) = \sum_i p_i^\alpha$. A polynomial $Q(x) = \sum_j q_j x^j$ is useful because its expectation can be reproduced from counts: replacing p_i^j by $(N_i)_j/n^j$ gives an unbiased estimate of $Q(p_i)$. The difficulty is therefore to approximate the fractional power using integer powers, while keeping their statistical fluctuations under control.

1. Why use the two adjacent integers? Write $\alpha = m + \gamma = m + 1 - \delta$, where $0 < \gamma, \delta < 1$. Consider, first, two illustrative profiles

$$F_-(x) = x^\gamma R_-(x), \qquad F_+(x) = x^\delta R_+(x),$$

where R_- and R_+ are ordinary polynomials. If $R_-(x) = \sum_{j=0}^{m-1} a_j x^j$ and $R_+(x) = \sum_{j=0}^{d_+} b_j x^j$, then direct expansion gives

$$p^\alpha F_-(a/p) = \sum_{j=0}^{m-1} a_j a^{\gamma+j} p^{m-j}, \tag{D.1}$$

$$p^\alpha F_+(p/b) = \sum_{j=0}^{d_+} b_j b^{-\delta-j} p^{m+1+j}. \tag{D.2}$$

For the first identity, the exponent of p is $\alpha-\gamma-j = m-j$; for the second it is $\alpha+\delta+j = m+1+j$. Thus the first block supplies degrees $1,\dots,m$ and the second supplies degrees $m+1, m+2, \dots$. The fractional exponents cancel exactly. The estimator will use ordinary integer-order falling factorials, not a hypothetical unbiased statistic for a fractional collision.

These two identities explain the construction, but the actual polynomial needs an additional transition between the blocks. Moreover, the reverse profile in the full construction may have a polynomial part of degree m, which also produces a constant term. This is intentional: for $m = 1$, the near-one profile gives an affine polynomial with a nonzero constant. The statistic therefore retains $q_0(N_i)_0 = q_0$, including for unseen symbols. The known alphabet bound makes these zero-count contributions observable.

2. The actual coupled polynomial. The expression used in the proof is the polynomial from (D.50), here called Q_0 to distinguish it from its final rescaling:

$$\begin{aligned} Q_0(p) = p^\alpha \big[& A' u_\gamma(a/p)\{1 - J_1(p/b)\} \\ & + B' v_{\delta,\omega_0}(p/b) + \omega_0(1-B') J_\delta(p/b)\big]. \end{aligned} \tag{D.3}$$

Here $u_\gamma(x) = x^\gamma R_-(x)$ is the reverse profile; $J_\delta(x) = x^\delta R_\delta(x)$ and $v_{\delta,\omega_0}(x) = x^\delta V_{\delta,\omega_0}(x)$ have ordinary polynomial factors. Also J_1 is an ordinary polynomial. Hence every term in (D.3) is an ordinary polynomial, with its value at $p = 0$ defined by that polynomial extension. Multiplication by $1 - J_1$ can create degrees above m; the useful separation in (D.1)–(D.2) describes the starting blocks, not a claim that the full coupled polynomial has disjoint degree supports.

The profiles and their constants will be constructed below. Their roles can already be stated. The reverse profile controls the lower end of $[a, b]$. The forward profile controls its upper end. The factor $1 - J_1$ turns off the reverse contribution as p increases, while the last term connects the polynomial to the interior value $\omega_0 p^\alpha$. The coefficients $A' = Ae^{-M\gamma}$ and $B' = Be^{-M\delta}$ balance the profiles; A, B are the envelope weights in (D.7), and M is a fixed damping constant. Their detailed normalization determines the sample scale, so merely knowing that a polynomial approximation exists would not suffice.

3. The algorithm uses counts, not the unknown probabilities. The numbers a and b are deterministic design parameters. They are not thresholds at which the algorithm can inspect p_i. Instead, it uses an independent pilot count to select a branch. The following steps specify the construction for noninteger $1 < \alpha < \log k$; the integer and high-order cases were treated above.

Step 1. Choose deterministic parameters. The inputs are k, α, ε and an i.i.d. sample from p. Use Lemma D.4 to calculate the root X and E_0, u, L, n, a, b. In particular,

$$n = D\alpha k^{1-1/\alpha} X, \qquad L = \lceil \alpha(C_L + c_L \log X) \rceil, \qquad b = c_b L/n, \qquad a = \lambda \frac{bhz^{1/h}}{m^2 L^2},$$

where $G_\gamma(z) = e^{-E_0}$ and $h = \gamma\delta$. All constants in this rule are fixed in terms of ε alone. Construct Q_0 above with $\omega_0 = e^{-E_0/8}$, and set

$$s = (1 + c_u u)^{-1}, \qquad Q = sQ_0 = \sum_{j=0}^{d} q_j p^j, \qquad \omega = s\omega_0.$$

These operations depend on the known inputs, not on p.

Step 2. Form independent main and pilot counts. Use Poisson splitting to obtain independent $N_i \sim \mathrm{Poi}(np_i)$ and $Y_i \sim \mathrm{Poi}(c_p np_i)$. Select the light branch using the observable indicator

$$I_i = \mathbf{1}\{Y_i \le c_p B\alpha\}.$$

The constants c_p and B are part of the construction. Their threshold is fixed by α; it is not a test of whether the unknown probability lies in $[a, b]$.

Step 3. Compute the two candidate contributions. For each of the k symbols, calculate

$$\widehat{Q}_i = \sum_{j=0}^{d} q_j \frac{(N_i)_j}{n^j}, \qquad \widehat{G}_i = \omega n^{-\alpha} \mathsf{H}_\alpha(N_i), \qquad \mathsf{H}_\alpha(j) = \frac{\Gamma(j+1)}{\Gamma(j+1-\alpha)}.$$

Use the entire reciprocal-Gamma convention at its zeros. The statistic H_α is signed at some low counts; it is not the nonnegative truncated Gamma statistic. Its low-count values are included in the proof.

Step 4. Combine the contributions and the known affine total. Put $\chi = \min\{\alpha - 1, 1\}$, $P_* = k^{1-\alpha}$, and $E_a = \chi d_\varepsilon C_0^m (ka)^{\alpha-1}$, and return the power-sum statistic

$$\widehat{P} = 2E_a P_* + q_1 + \sum_{i=1}^{k} \left[I_i \left(\widehat{Q}_i - q_1 N_i/n \right) + (1 - I_i) \left(\widehat{G}_i - q_1 N_i/n \right) \right]. \tag{D.4}$$

For $\widehat{P} > 0$, the entropy output is $-\log \widehat{P}/(\alpha - 1) - \varepsilon$; define it arbitrarily otherwise. The proof establishes positivity on the success event. A fixed sample pool of size $\lceil 4(1 + c_p)n \rceil$ implements the Poisson splitting; the exponentially small overflow probability is absorbed into the stated failure probability.

4. Why subtract the linear term? Let $T_i = I_i \widehat{Q}_i + (1 - I_i)\widehat{G}_i$. After omitting the deterministic repair $2E_a P_*$, (D.4) equals

$$q_1 + \sum_i (T_i - q_1 N_i/n) = \sum_i T_i + q_1 \left(1 - \sum_i N_i/n \right).$$

Because $\sum_i p_i = 1$, the added term has mean zero. Its variance is q_1^2/n, since $\sum_i N_i \sim \mathrm{Poi}(n)$. This is a control variate: we replace the noisy estimate of a known linear functional by its exact value. The proof bounds the variance of the resulting residual; it does not assume that adding any centered term must reduce variance.

Near order one, write $\alpha = 1 + \beta$. For any known $t > 0$,

$$P_{1+\beta}(p) = t^\beta + \beta \sum_i \frac{p_i^{1+\beta} - t^\beta p_i}{\beta}. \tag{D.5}$$

For $p_i > 0$, the quotient tends to $p_i \log(p_i/t)$ as $\beta \downarrow 0$, and its value at $p_i = 0$ is zero. For the specific scale $t = b/N_{\mathrm{ker}}^2$, where $N_{\mathrm{ker}} \asymp L$ is the kernel degree parameter, the near-one construction has $q_1 = t^\beta + O_\varepsilon(\beta t^\beta)$. Subtracting the affine part therefore exposes a residual of order β. The detailed proof retains this factor in bias and its square in variance, matching the interval width $(1 - e^{-2\varepsilon\beta})P = O_\varepsilon(\beta P)$.

5. What do the margins pay for? Write $P = P_\alpha(p)$ and $\kappa = e^{-2\varepsilon(\alpha-1)}$. The analysis uses

$$\mathcal{M} = E_a P_* + \kappa u P + \chi\kappa\mathcal{B}, \qquad \mathcal{B} = \sum_i p_i^\alpha \min(Lnp_i, 1). \tag{D.6}$$

This quantity depends on p and is used only in the proof. The baseline $E_a P_*$ absorbs the aggregate error from $p_i < a$; the deterministic correction uses part of this allowance, while the upper approximation leaves room for that positive correction. The term $\kappa u P$ is an additional interior reserve supplied by the root calibration and final rescaling. The term $\chi\kappa\mathcal{B}$ provides count-dependent room that controls higher-degree fluctuations and pilot errors. The variance calculations use their sum, rather than assign every error to exactly one term.

The proof first constructs an ideal mixed quantity in $[\kappa P + c_\varepsilon \mathcal{M}, P - c_\varepsilon \mathcal{M}]$. It then shows that the observed statistic is close to this quantity: conditional polynomial fluctuations are small relative to $\mathcal{M}$, genuine heavy counts concentrate, and the total contribution of pilot mistakes is small. These bounds imply $\kappa P \le \widehat{P} \le P$ with high probability. Thus the deterministic construction, the pilot, and the noise bounds have separate roles, and each is needed before taking the logarithm.

D.2 Exact overlap normalization at the original accuracy

The purpose of the next lemma is to combine two endpoint approximations without wasting any of the requested entropy accuracy. Think of u and v temporarily as two numbers in $[0, 1]$, representing the normalized values of those approximations. Their interaction is described by the weighted product $u^\delta v^\gamma$. We choose weights A, B so that $Au + Bv$ has maximum exactly one when that product is at most z. The same weights give an explicit minimum when the product is at least z. The ratio between these two extrema determines how large an approximation interval can be tolerated. This scalar calculation alone is not an estimator or a variance bound.

Lemma D.1 (Exact two-arm envelope)**.** *Let $0 < \gamma < 1$, $\delta = 1 - \gamma$, $h = \gamma\delta$, and $0 < z < 1$. Set*

$$A = \frac{1 - z^{1/\gamma}}{1 - z^{1/h}}, \qquad B = \frac{1 - z^{1/\delta}}{1 - z^{1/h}}, \qquad G_\gamma(z) = \frac{z A^\delta B^\gamma}{\delta^\delta \gamma^\gamma}. \tag{D.7}$$

Then

$$\max_{\substack{0 \le u,v \le 1 \\ u^\delta v^\gamma \le z}} (Au + Bv) = 1, \tag{D.8}$$

$$\min_{\substack{0 \le u,v \le 1 \\ u^\delta v^\gamma \ge z}} (Au + Bv) = G_\gamma(z). \tag{D.9}$$

Moreover, G_γ is a continuous strictly increasing bijection from $(0, 1)$ onto $(0, 1)$.

Proof. **Step 1: compute the maximum.** For fixed u, the largest admissible v in (D.8) is $\min\{1, z^{1/\gamma} u^{-\delta/\gamma}\}$. On $0 \le u \le z^{1/\delta}$ the maximum is at the right endpoint; on $z^{1/\delta} \le u \le 1$ the resulting objective is convex in u. Both boundary values equal one, since $Az^{1/\delta} + B = A + Bz^{1/\gamma} = 1$.

Step 2: show that the desired overlap level can be chosen uniquely. To verify monotonicity, write $z = e^{-ht}$, $t > 0$, and $F_\gamma(t) = -\log G_\gamma(e^{-ht})$. An exact cancellation gives

$$G_\gamma(e^{-ht}) = \frac{[\sinh(\delta t/2)/\delta]^\delta [\sinh(\gamma t/2)/\gamma]^\gamma}{\sinh(t/2)}. \tag{D.10}$$

Let $J(x) = x \coth x$ for $x > 0$ and $J(0) = 1$. Its derivatives satisfy

$$0 < J'(x) = \coth x - \frac{x}{\sinh^2 x} < 1, \qquad J''(x) = \frac{2(x \coth x - 1)}{\sinh^2 x} > 0. \tag{D.11}$$

The first strict inequalities follow respectively from $\sinh x \cosh x > x$ and $1 - e^{-2x} < 2x$. For $q = t/2$, differentiation of (D.10) yields

$$F'_\gamma(t) = \frac{J(q) - \delta J(\delta q) - \gamma J(\gamma q)}{t} > 0. \tag{D.12}$$

Also $F_\gamma(0+) = 0$ and $F_\gamma(\infty) = \infty$. This proves the asserted bijection and, in particular, $G_\gamma(z) < 1$.

Step 3: compute the minimum. Weighted AM–GM, applied to Au/δ and Bv/γ, gives

$$Au + Bv \geq \left(\frac{Au}{\delta}\right)^\delta \left(\frac{Bv}{\gamma}\right)^\gamma \geq \frac{z A^\delta B^\gamma}{\delta^\delta \gamma^\gamma}.$$

This is the lower bound in (D.9). Equality occurs at $u_* = \delta G_\gamma(z)/A$ and $v_* = \gamma G_\gamma(z)/B$. Concavity of $1 - e^{-x}$ gives $A \geq \delta$ and $B \geq \gamma$, so these two coordinates lie in $(0, 1)$; their weighted product is z. □

Why the next estimate is needed. If γ is very close to zero or one, then h is small. A bounded error in a logarithm can become a very large multiplicative error after division by h. The next lemma therefore tracks the correction on the scale h, uniformly as the order approaches an integer.

Lemma D.2 (Uniform overlap correction)**.** *Write*

$$H(\gamma) = -\gamma \log \gamma - \delta \log \delta, \qquad \ell(t) = -\log(1 - e^{-t}), \qquad D_\gamma(t) = \delta \ell(\delta t) + \gamma \ell(\gamma t) - \ell(t).$$

For every $t > 0$,

$$F_\gamma(t) = ht - H(\gamma) + D_\gamma(t), \qquad 2[J(t/4) - 1] \leq \frac{F_\gamma(t)}{h} \leq t. \tag{D.13}$$

For every fixed $\eta > 0$, uniformly over $t \geq \eta$ and $0 < \gamma < 1$,

$$\frac{D_\gamma(t)}{h} = \log\left(1 + \frac{1}{ht}\right) + O_\eta(1). \tag{D.14}$$

Proof. The identity follows directly from (D.7). The numerator in (D.12) equals

$$\delta \int_{\delta q}^{q} J'(s)\, ds + \gamma \int_{\gamma q}^{q} J'(s)\, ds.$$

It is at most $2hq$ by (D.11). Choosing the larger of δ, γ, its corresponding integral is at least $hqJ'(q/2)$. Thus $J'(t/4)/2 \leq F'_\gamma(t)/h \leq 1$; integration proves (D.13).

Let $a = \min\{\gamma, \delta\}$ and $b = 1 - a \geq 1/2$. The exact identity

$$\frac{D_\gamma(t)}{h} = \frac{\ell(bt) - \ell(t)}{a} + \frac{\ell(at) - \ell(t)}{b} \tag{D.15}$$

has first term between zero and $t/(e^{bt} - 1) \leq 1/b \leq 2$. The second term differs from $\ell(at)$ by $(a/b)\ell(at) - (1/b)\ell(t)$. For $t \geq \eta$, the latter difference is bounded in absolute value by $2\ell(\eta) + \log(1 + 2/\eta)$: use $\ell(x) \leq \log(1 + 1/x)$ and the increasing function $a \mapsto a \log(1 + c/a)$. Finally,

$$|\ell(x) - \log(1 + 1/x)| \leq \log 2, \qquad |\log(1 + 1/(at)) - \log(1 + 1/(ht))| \leq \log 2.$$

The first bound follows from $x/(1 + x) \leq 1 - e^{-x} \leq \min\{x, 1\}$, and the second from $h = ab$. These estimates prove (D.14). □

From overlap to an allowable interval size. We choose z_* so that the minimum of the linear combination equals e^{-E}. The relevant interval length will involve $z_*^{-1/h}$. The following calculation retains both its exponential factor and its prefactor; dropping the prefactor would lose uniformity near order one.

Theorem D.3 (Capacity with the exact accuracy)**.** *Fix $\varepsilon > 0$. Let $\alpha > 1$ be noninteger, and define*

$$\beta = \alpha - 1, \quad \gamma = \alpha - \lfloor \alpha \rfloor, \quad \delta = 1 - \gamma, \quad h = \gamma\delta, \quad E = 2\varepsilon\beta, \quad \Psi = \gamma^{1+1/\delta}\delta^{1+1/\gamma}.$$

The unique solution $z_* \in (0, 1)$ of $G_\gamma(z_*) = e^{-E}$ satisfies

$$z_*^{-1/h} = \Theta_\varepsilon\left(\frac{\beta}{\alpha}(\gamma^\gamma\delta^\delta)^{-1/h}e^{E/h}\right) = \Theta_\varepsilon\left(\frac{\beta h}{\alpha\Psi}e^{E/h}\right). \tag{D.16}$$

All constants are uniform over noninteger $\alpha > 1$.

Proof. Write $z_* = e^{-ht_*}$, so $F_\gamma(t_*) = E$. Since $\beta \geq \gamma$, $E/h = 2\varepsilon\beta/(\gamma\delta) \geq 2\varepsilon$. Equation (D.13) therefore gives $t_* \geq 2\varepsilon$. Convexity of J implies that $[J(t/4) - 1]/t$ is increasing. Consequently, with $c_\varepsilon = [J(\varepsilon/2) - 1]/\varepsilon > 0$,

$$c_\varepsilon h t_* \leq E \leq h t_*.$$

Equation (D.14), with $\eta = 2\varepsilon$, now gives

$$D_\gamma(t_*)/h = \log(1 + 1/E) + O_\varepsilon(1), \qquad e^{-D_\gamma(t_*)/h} = \Theta_\varepsilon\left(\frac{E}{1 + E}\right) = \Theta_\varepsilon\left(\frac{\beta}{\alpha}\right).$$

Exponentiating $t_* = E/h + H(\gamma)/h - D_\gamma(t_*)/h$ proves the first equality in (D.16). The second uses $(\gamma^\gamma\delta^\delta)^{1/h} = \Psi/h$. □

How the scalar calculation predicts the sample scale. At this point we have solved only a two-number optimization problem. The following calculation explains which cutoffs a polynomial should achieve. It is conditional: the existence and the shape of that polynomial are proved later in the section.

Conditional polynomial-capacity calibration. The algebraic consequence for a common polynomial construction is as follows. Suppose its lower cutoff and endpoint capacity obey

$$a \asymp_\varepsilon \frac{\beta}{\alpha k}, \qquad \frac{b}{a} \asymp_\varepsilon \frac{m^2 L^2}{h z_*^{1/h}}, \qquad m \asymp \alpha. \tag{D.17}$$

Here m and L denote the two endpoint degree scales; these relations are assumptions about the construction, not conclusions of the scalar lemmas. Then (D.16) gives, with both factors β/α retained,

$$\begin{aligned} b &\asymp_\varepsilon \underbrace{\frac{\beta}{\alpha k}}_{\text{lower cutoff}} \frac{m^2 L^2}{h} \underbrace{\frac{\beta h}{\alpha \Psi} e^{E/h}}_{\text{overlap normalization}} \\ &= \frac{m^2}{\alpha^2} \frac{\beta^2 L^2}{k\Psi} e^{E/h} \asymp_\varepsilon \frac{\beta^2 L^2}{k\Psi} e^{E/h}. \end{aligned} \tag{D.18}$$

If also $b \asymp L/n$, the resulting deterministic balance is

$$n \asymp_\varepsilon \frac{k\Psi}{\beta^2 L} e^{-E/h}. \tag{D.19}$$

Putting $K = \log k$, $n = \alpha k^{1-1/\alpha} x$, and $L \asymp \alpha(1 + \log x)$, with $x \geq 1$, rewrites this balance as

$$x(1 + \log x) \asymp_\varepsilon Q^2, \qquad Q^2 = \frac{\Psi}{\alpha^2 \beta^2} \exp\left(\frac{K}{\alpha} - \frac{E}{h}\right). \tag{D.20}$$

Thus the scalar calculation produces the original-accuracy exponential factor and the two near-one normalization factors in the root. It does not prove (D.17) for actual polynomial profiles, nor supply the stochastic regularizer by itself. Those tasks are verified in Sections D.3–E.3.

D.3 One parameter rule for the regularized upper bound

The deterministic balance above does not yet account for random noise. The root definition below includes a further term that supplies room for that noise. It specifies one collection of parameters for every noninteger order. The proof now checks that these numerical choices have the properties required by the polynomial construction and its later variance analysis. In particular, increasing the final sample multiplier D makes the lower cutoff a sufficiently small without changing the requested accuracy ε.

Lemma D.4 (Regularized-root parameter rule)**.** *Fix $0 < \varepsilon < 1/2$. For a noninteger $1 < \alpha < \log k$, set*

$$\begin{gathered} \beta = \alpha - 1, \quad m = \lfloor \alpha \rfloor, \quad \gamma = \alpha - m, \quad \delta = 1 - \gamma, \quad h = \gamma\delta, \quad E = 2\varepsilon\beta, \\ \Psi = \gamma^{1+1/\delta} \delta^{1+1/\gamma}, \qquad Q^2 = \frac{\Psi}{\alpha^2 \beta^2} \exp\{(\log k)/\alpha - E/h\}, \\ X = \inf\left\{ x \geq 1 : \frac{\alpha h}{\beta} x^\alpha \left[1 + \log \frac{x(1 + \log x)}{Q^2}\right]_+ \geq 1 \right\}, \qquad v = \frac{\beta}{\alpha} X^{-\alpha}. \end{gathered}$$

The constants in the following definitions are design parameters. They will be fixed in an order stated below, using ε only; they do not depend on k or α. Fix positive constants $C_h, c_b, \lambda, c_L, u_0$,

$C_L \geq 1$, and $0 < c_\omega < 1$. For $D \geq 1$, define

$$\Delta = \min\{v + C_h h, E/2\}, \qquad E_0 = E - \Delta, \qquad u = \min\{v + h, u_0\}, \qquad \omega = e^{-c_\omega E_0}, \tag{D.21}$$

$$n = D\alpha k^{1-1/\alpha} X, \qquad L = \lceil \alpha(C_L + c_L \log X) \rceil, \qquad b = \frac{c_b L}{n}, \qquad a = \lambda \frac{b h z^{1/h}}{m^2 L^2}, \tag{D.22}$$

where $z \in (0,1)$ is the unique solution of $G_\gamma(z) = e^{-E_0}$. Put

$$\sigma = 1 + \log \frac{X(1 + \log X)}{Q^2} - \frac{v}{h}, \qquad w = \sigma + \frac{(v - E/2)_+}{h}. \tag{D.23}$$

Then $\sigma \geq 0$, and $\sigma = 0$ if $X > 1$. There are constants $r_-, r_+ > 0$, independent of k, α, D and depending only on the other fixed constants and ε, such that

$$\frac{r_-}{D} e^{-w} \leq \frac{a}{\beta/(\alpha k)} \leq \frac{r_+}{D} e^{-w}. \tag{D.24}$$

In addition,

$$\frac{a m^2 L^2}{b} = \lambda h z^{1/h} \leq \lambda/4. \tag{D.25}$$

Thus any fixed requirement $a \leq c_a \beta/(\alpha k)$ holds by increasing D. Fixed choices of C_L and then D also ensure $a \leq b \leq 1$ and $nb \geq 2B\alpha$ for any prescribed fixed pilot constant B.

Proof. **Step 1: extract the identity supplied by the root.** The root-defining expression is continuous and strictly increasing once its positive part is nonzero. Hence the inequality at its infimum is an equality unless $X = 1$. This proves the assertions about σ and the exact identity

$$\frac{Q^2}{X} = (1 + \log X) e^{1 - v/h - \sigma}. \tag{D.26}$$

Step 2: evaluate the actual lower cutoff. Since $E/2 \leq E_0 \leq E$, Theorem D.3 applies with accuracy in $[\varepsilon/2, \varepsilon]$, uniformly on that compact interval. It gives

$$z^{1/h} \asymp_\varepsilon \frac{\alpha \Psi}{\beta h} e^{-E_0/h}.$$

Substitution into (D.22), followed by (D.26), yields

$$\frac{a}{\beta/(\alpha k)} \asymp_\varepsilon \frac{\lambda c_b}{D} \frac{\alpha^3 (1 + \log X)}{m^2 L} e^{1 - \sigma - (v - \Delta)/h}. \tag{D.27}$$

The degree factor is between $1/\max(C_L + 1, c_L)$ and $4/\min(C_L, c_L)$. Also $v - \Delta = \max\{-C_h h, v - E/2\}$, so

$$e^{-w} \leq e^{-\sigma - (v-\Delta)/h} \leq e^{C_h} e^{-w}.$$

These bounds prove (D.24). **Step 3: verify the remaining scale restrictions.** Equation (D.25) is immediate from the definition. Choose C_L so that $\lambda/(4m^2 L^2) \leq 1$ and $c_b C_L \geq 2B$. Finally,

$$b \leq \frac{c_b}{D} \frac{C_L + 1 + c_L \log X}{X} \leq \frac{c_b (C_L + 1 + c_L)}{D},$$

which proves the remaining restrictions after increasing D. □

A uniform lower bound for the error budget. The actual value of a can be smaller than its nominal scale. A later variance calculation is easier to express using a baseline depending only on D and the order. The next lemma shows that any discrepancy between that nominal baseline and the actual one is paid by the regularization margin $\kappa u P$. It also checks that the reserved logarithmic width produces a multiplicative gap of that size.

Lemma D.5 (Actual baseline and the reserved width)**.** *Use Lemma D.4 with $\alpha \geq 3/2$. Let $C_0 \geq 1$, $0 < d_\varepsilon \leq 1$, and define*

$$\kappa = e^{-E}, \qquad u_* = \min\{\varepsilon/2, 1/3, u_0\}, \qquad \rho_- = r_-/3, \quad \rho_+ = r_+,$$
$$E_{\mathrm{base}} = \min\{\beta, 1\}\, d_\varepsilon C_0^m (\rho_-/D)^\beta, \qquad E_{\mathrm{actual}} = \min\{\beta, 1\}\, d_\varepsilon C_0^m (ka)^\beta.$$

If $D \geq \rho_- C_0^2 e^{2\varepsilon}$, then

$$E_{\mathrm{base}} \leq E_{\mathrm{actual}} + \kappa u/u_*. \tag{D.28}$$

Consequently, for $P_* = k^{1-\alpha}$, $P = \sum_i p_i^\alpha$, and any $\mathcal{B} \geq 0$,

$$E_{\mathrm{base}} P_* + \kappa u P + \kappa \mathcal{B} \leq (1 + 1/u_*)\{E_{\mathrm{actual}} P_* + \kappa u P + \kappa \mathcal{B}\}. \tag{D.29}$$

Furthermore, for a fixed $c > 0$ depending only on the stated constants,

$$\Delta \geq cu, \qquad e^{-E_0} - e^{-E} \geq c\kappa u. \tag{D.30}$$

Proof. Here $\beta \geq 1/2$, $\beta/\alpha \geq 1/3$, and $m \leq 2\beta$. Thus $ka \leq \rho_+/D$, and $w = 0$ implies $ka \geq \rho_-/D$. The condition on D gives $E_{\mathrm{base}} \leq \kappa$. If $w > 0$, then either $\sigma > 0$, which forces $X = 1$ and $v \geq 1/3$, or $v > E/2 = \varepsilon\beta \geq \varepsilon/2$. Equivalently, the single bound

$$\mathbf{1}\{w > 0\} \leq u/u_*$$

holds. Combining it with the cutoff lower bound when $w = 0$ proves (D.28); using $P_* \leq P$ proves (D.29). Finally,

$$\Delta \geq \min\{\min(1, C_h)(v + h), \varepsilon/2\} \geq cu.$$

Since $e^{-E_0} - e^{-E} = \kappa(e^\Delta - 1)$, this proves (D.30). □

Choice of the baseline constant and count multiplier. The factor $\min\{\beta, 1\}$ is between $1/2$ and one in Lemma D.5; it also makes the actual baseline vanish at the order-one endpoint. The prefactor d_ε can be fixed after C_0, for example at a constant multiple of c_ε/C_0^2, to fit the deterministic baseline bound. This changes only fixed factors in the noise normalization.

There is an important arithmetic point if D is chosen last. For a low degree $1 \leq j \leq m$, write $\theta = j/\alpha$ and $\eta = \alpha - j$. A coefficient bound with $ka \leq \rho_+/D$, combined with the factorial second-moment estimate and normalization by $(E_{\mathrm{base}} + \kappa u P/P_*)^2$, gives the exact power

$$D^{-2\eta - j + \beta(2-\theta)} = D^{-2+j/\alpha} \leq D^{-1}. \tag{D.31}$$

Here $u \geq \min\{1/3, u_0\} X^{-\alpha}$ absorbs the corresponding X powers. The remaining baseline factors are bounded by $4 d_\varepsilon^{-2} C_0^{-m} e^{C\alpha}$ after the factorial coefficient sums are evaluated. For the high-degree block the analogous factor is

$$\frac{D^{-\alpha}}{E_{\mathrm{base}}} \leq 2 d_\varepsilon^{-1} D^{-1} C_0^{-m} \rho_-^{-\beta}.$$

Thus one first fixes profile and pilot constants, chooses C_L and c_L for the degree bounds, determines ρ_-, ρ_+, and chooses C_0 to absorb the exponential coefficient costs. One then fixes d_ε and chooses D last. Substituting $c_a \asymp D^{-1}$ into a cruder bound and subsequently choosing D in terms of c_a would lose this arithmetic. The displayed calculations are normalization identities. The coefficient and conditional-variance estimates that use them are proved in Sections D.4 and E.1.

Why order one requires a separate calculation. As $\alpha \downarrow 1$, the target power sum approaches the known value $\sum_i p_i = 1$, while the admissible multiplicative error is proportional to $\beta = \alpha - 1$. Bounds that merely stay finite are insufficient: the relevant approximation errors must vanish at least linearly in β. The next lemma records precisely that cancellation for the same parameter rule.

Lemma D.6 (Order-one calibration of the same parameters)**.** *Use Lemma D.4 with $1 < \alpha \le 3/2$. Let $N \asymp L$, with fixed comparison constants, and set $t = b/N^2$. For the degree-one reverse profile, let $\tau = c(1+\beta)a/\beta$ with fixed $c > 0$. Then*

$$\frac{k}{nL} \le C_\varepsilon/D, \qquad kt \le C_\varepsilon/D, \qquad c_\varepsilon \le \tau/t \le C_\varepsilon. \tag{D.32}$$

The exact amplitudes A, B of (D.7) satisfy $B \le C_\varepsilon\beta$ and $1 - A \le C_\varepsilon\beta$. In particular, $A - \omega = O_\varepsilon(\beta)$, and

$$\min\{1-\omega, \omega-\kappa\} \ge c_{\varepsilon,c_\omega}\beta. \tag{D.33}$$

All constants may depend on the fixed profile constants in Lemma D.4.

Proof. Here $m = 1$, $\gamma = \beta$, and $\delta = 1 - \beta \ge 1/2$. Cancellation of the near-one factors gives

$$\frac{Q^2}{k^{1/\alpha}} = \frac{\beta^{\beta/\delta}\delta^{1+1/\beta}}{\alpha^2} e^{-2\varepsilon/\delta} \asymp_\varepsilon 1. \tag{D.34}$$

The logarithm is uniformly bounded, using $\beta|\log\beta| \le 1/e$ and $\log(1-\beta) \ge -\beta/(1-\beta)$. Equation (D.26) gives $Q^2 \le eX(1+\log X)$, including $X = 1$. Therefore

$$k^{1/\alpha} \le C_\varepsilon X(1+\log X), \qquad \frac{k}{nL} \le \frac{C_\varepsilon(1+\log X)}{D\alpha L} \le C_\varepsilon/D.$$

This also proves the bound for kt.

Write $z = e^{-ht_*}$, so $F_\gamma(t_*) = E_0$. Since $E_0/h \in [\varepsilon, 4\varepsilon]$, (D.13) yields

$$\varepsilon \le t_* \le T_\varepsilon, \qquad J(T_\varepsilon/4) = 1 + 2\varepsilon.$$

In particular $z^{1/h}$ lies in a fixed positive compact interval. Equation (D.22) gives the exact relation

$$\frac{\tau}{t} = c\lambda\frac{N^2}{L^2}\alpha\delta z^{1/h}.$$

Since $3/4 \le \alpha\delta = 1 - \beta^2 \le 1$, this proves the remaining assertion in (D.32). Finally,

$$B = \frac{1-e^{-\beta t_*}}{1-e^{-t_*}} \le C_\varepsilon\beta, \qquad 1 - A = e^{-\delta t_*}B \le C_\varepsilon\beta.$$

The estimates for ω follow from $\varepsilon\beta \le E_0 \le 2\varepsilon\beta$ and $0 < c_\omega < 1$. □

Correctly centered localized bias. Suppose the common coupled polynomial satisfies $\sup_{0\le p\le b}|Q(p) - \omega p^\alpha| \le C_\varepsilon\beta t^\alpha$ in the interval of Lemma D.6. Then its aggregate light-domain bias, divided by $P_* = k^{1-\alpha}$, is at most

$$C_\varepsilon\beta(kt)^\alpha \le C_\varepsilon\beta/D.$$

Increasing D places this inside the original-accuracy margins in (D.33), without changing ε in the root. This assertion uses the target ωP_α.

The remaining compact parameter range. The ratio E/h distinguishes a long overlap interval from one with bounded width. When it is bounded and $\alpha \geq 3/2$, all order parameters remain in a compact set. Ordinary fixed-constant bounds are then sufficient. This observation simplifies later proofs without introducing a different estimator.

Lemma D.7 (Bounded interpolation width)**.** *Fix $0 < M < \infty$. Within the same parameter rule, suppose $\alpha \geq 3/2$ and $E/h \leq M$. Then, with constants depending on ε, M and the fixed profile parameters,*

$$Q^2 \asymp_{\varepsilon,M} k^{1/\alpha}, \qquad \frac{k}{nL} \leq C_{\varepsilon,M}/D, \qquad kt \leq C_{\varepsilon,M}/D, \qquad u \geq c_{\varepsilon,M} > 0, \tag{D.35}$$

where $t = b/N^2$ and $N \asymp L$.

Proof. The hypothesis and $h \leq 1/4$ imply

$$1/2 \leq \beta \leq M/(8\varepsilon), \qquad h \geq E/M \geq \varepsilon/M.$$

Thus α is bounded above and both γ, δ are bounded away from zero. The positive factor $\Psi e^{-E/h}/(\alpha^2\beta^2)$ lies in a fixed positive compact interval, proving the first assertion. Equation (D.26), including its clipped-root inequality, gives $Q^2 \leq eX(1 + \log X)$. The proof of (D.32) then applies verbatim to $k/(nL)$ and kt. Finally, $u = \min\{v + h, u_0\} \geq \min\{\varepsilon/M, u_0\} > 0$. This verifies additional margins of the same construction; it does not select a different estimator. □

D.4 Constructing the polynomial explicitly

What the construction must accomplish. We need a polynomial in p, even though p^α itself is not a polynomial. The identity $\alpha = m + \gamma = m + 1 - \delta$ suggests two representations: multiply a function of a/p behaving like $(a/p)^\gamma$ by p^α, or multiply a function of p/b behaving like $(p/b)^\delta$ by p^α. The resulting powers of p are integers. The first term controls the lower endpoint and the second controls the upper endpoint. Smooth polynomial cutoffs join them.

We give one polynomial construction. All constants below may depend on the fixed accuracy $0 < \varepsilon < 1/2$, but not on the alphabet size or the order. Write

$$\beta = \alpha - 1, \quad m = \lfloor \alpha \rfloor, \quad \gamma = \alpha - m, \quad \delta = 1 - \gamma, \quad h = \gamma\delta, \qquad E = 2\varepsilon\beta.$$

For the deterministic bound first suppose $\alpha \geq 3/2$ and $E/2 \leq E_0 \leq E$. Put $\kappa_0 = e^{-E_0}$ and $\omega = e^{-E_0/8}$. For $0 < z < 1$ define

$$A(z) = \frac{1 - z^{1/\gamma}}{1 - z^{1/h}}, \qquad B(z) = \frac{1 - z^{1/\delta}}{1 - z^{1/h}}, \qquad G_\gamma(z) = \frac{zA(z)^\delta B(z)^\gamma}{\delta^\delta \gamma^\gamma}.$$

Choose z from $G_\gamma(z) = \kappa_0$. In the following proof we use $z \leq \kappa_0$, $A, B \leq 1$, and $A, B \geq 1 - z$; these also follow directly from these formulas and the monotonicity properties of G_γ. Fix

$$q = e^{-\varepsilon/16}, \qquad q_L \in (q, 1).$$

In particular $q > \kappa_0$ and $q\omega > \kappa_0$ with a fixed relative margin, because $E_0 \geq \varepsilon/2$.

The elementary building block. Let U_j be the Chebyshev polynomial of the second kind, defined by $U_0(t) = 1$, $U_1(t) = 2t$, and $U_{j+1}(t) = 2tU_j(t) - U_{j-1}(t)$. The kernel below is concentrated near $x = 0$, on the scale N^{-2}. The normalized integral J_s is a polynomial times x^s that rises from zero to one. The variable $y = N^2x$ measures position relative to the concentration scale. These facts allow us to build endpoint approximations while controlling both their size and their coefficients.

Lemma D.8 (Elementary endpoint-kernel estimates)**.** *For $N \geq 2$ and $d \in \{2, 6\}$, set*

$$K_{N,d}(x) = \Big[U_{N-1}(\sqrt{1-x})/N\Big]^{2d}, \quad y = N^2x, \quad I_s = \int_0^1 x^{s-1}K_{N,d}(x)\,dx.$$

For $0 < s \leq 1$, the following estimates are uniform in N, s:

$$0 \leq K_{N,d}(x) \leq \min(1, y^{-d}), \qquad e^{-Cy} \leq K_{N,d}(y/N^2) \leq e^{-cy} \quad (y \leq y_0), \tag{D.36}$$

$$1 - K_{N,d}(y/N^2) \leq Cy \quad (y \leq 1), \qquad \sup_{y \geq y_0} K_{N,d}(y/N^2) \leq \rho < 1, \tag{D.37}$$

$$sN^{2s}I_s = \exp(O(s)), \qquad J_s(x) := I_s^{-1}\int_0^x t^{s-1}K_{N,d}(t)\,dt = y^s\exp(O(s)) \quad (y \leq 1), \tag{D.38}$$

$$1 - J_s(x) \leq Csy^{s-d} \quad (y \geq 1), \qquad J_s(x) \geq e^{-Cs} \quad (y \geq 1), \tag{D.39}$$

$$\max_{0 \leq x \leq 1} x^s K_{N,d}(x) = N^{-2s}s^s\exp(O(s)). \tag{D.40}$$

The derivative of the normalized kernel density with respect to $s \in [1/2, 1]$ is bounded in L^1; in particular $0 \leq J_\delta - J_1 \leq C(1-\delta)$ for $1/2 \leq \delta \leq 1$. The kernel and the polynomial $J_s(x)/x^s$ have coefficient sums at most C^{N+1}.

Proof. Put $\theta = \arcsin\sqrt{x}$. The elementary identity

$$U_{N-1}(\cos\theta) = \frac{\sin(N\theta)}{\sin\theta}$$

gives the tail bound because $N\sin\theta = \sqrt{y}$. The modulus of the quotient divided by N is at most one: it is the modulus of an average of N unit complex numbers from the geometric sum for $\sin(N\theta)/\sin\theta$. This inequality is strict for $0 < \theta \leq \pi/2$ when $N \geq 2$. For y sufficiently small, Taylor's formula with its uniform remainder gives

$$\frac{\sin(N\theta)}{N\sin\theta} = 1 - \frac{(N^2-1)\theta^2}{6} + O(N^4\theta^4) = 1 - \frac{1-N^{-2}}{6}y + O(y^2).$$

Since $1 - N^{-2} \in [3/4, 1)$, this proves the two local exponential bounds and the linear deficit. The lower exponential bound also holds up to $y = 1$ after changing its constant: on that interval $N\theta \leq N\arcsin(1/N) \leq \pi/3$, so the sine quotient is bounded away from zero.

For the gap, first use $K \leq y^{-d}$ beyond a fixed large R. On $[y_0, R]$, the quotient converges uniformly as $N \to \infty$ to $\sin\sqrt{y}/\sqrt{y}$, whose modulus is strictly less than one. For each of the finitely many remaining N, continuity and the strict geometric-sum inequality give the same conclusion on its nonempty scaled interval. Taking their maximum proves a uniform $\rho < 1$.

After rescaling, let $M_s = N^{2s}I_s = \int_0^{N^2} y^{s-1}K_{N,d}(y/N^2)\,dy$. The upper bounds give

$$sM_s \leq 1 + s\int_1^\infty y^{s-1-d}\,dy \leq 1 + Cs.$$

For the lower bound, integrate e^{-Cy} on $[0, 1]$ and use Jensen's inequality under the probability density sy^{s-1}:

$$sM_s \geq s\int_0^1 y^{s-1}e^{-Cy}\,dy \geq \exp[-Cs/(s+1)] \geq e^{-Cs}.$$

This is why the normalization error is $O(s)$ rather than a fixed constant. The same argument after rescaling $[0, y]$ proves the truncated-integral estimate. Integrating the inverse-power bound from y to infinity proves (D.39).

For the sharp maximum, evaluate at $y = c_0 s$ for a fixed small c_0 to obtain the lower bound. On the local interval, the upper bound is at most $\max_{y\ge 0} y^s e^{-cy} = (s/(ce))^s$. Outside that interval the uniform gap and inverse-power tail imply $\sup y^s K \le \rho' < 1$ for all sufficiently small s. This follows by first fixing a large tail cutoff and then decreasing s on the remaining compact interval. Since $s^s \to 1$, this region cannot violate the asserted upper bound. For s bounded below, the trivial uniform bound on $y^s K$ is absorbed into e^{Cs}. The same argument applies to any kernel with a local exponential bound, a uniform gap away from zero, and a tail Cy^{-a} for fixed $a > 1$; its constants may depend on the constants in those bounds.

For the exponent derivative, the rescaled normalized density is $y^{s-1}K(y/N^2)/M_s$. Its mean absolute logarithm is uniformly bounded for $s \in [1/2, 1]$: integrate $y^{-1/2}|\log y|$ below one and use the inverse-power tail above one. Differentiating a normalized integral is a covariance with $\log y$, giving the asserted uniform derivative bound. The sign follows because the likelihood ratio for the smaller exponent is decreasing in y.

Finally, $U_{n+1}(t) = 2tU_n(t) - U_{n-1}(t)$ bounds coefficient sums by an exponential in N. Taking a fixed even power and replacing its even monomials in $\sqrt{1-x}$ by powers of $1-x$ preserves such a bound. For integration, the jth coefficient is divided by $j+s$ and then multiplied by $I_s^{-1} \le CsN^{2s}$; the factor s cancels the only possible singularity at $j = 0$. Polynomial factors in N are absorbed into C^{N+1}. □

The two endpoint profiles. A *profile* here is simply a function taking values in $[0, 1]$. The low profile must rise with the correct fractional power near zero, remain above a fixed positive level after that transition, and retain a definite deficit below one away from its maximum. The high profile has a similar local rise but approaches a prescribed level ω in its tail. Both are a fractional power multiplied by an ordinary polynomial. These properties will make their combination a polynomial in the original probability p.

The low profile u_s is a normalized raised-floor Chebyshev profile of ordinary polynomial degree at most m after division by x^s. Let $\ell \asymp m$ denote its endpoint resolution. A fixed floor parameter strictly between q_L and one gives the following bounds, with constants depending only on ε:

$$0 \le u_s \le 1, \qquad u_s(x) \le e^{Cs}(\ell^2 x/s)^s, \tag{D.41}$$

$$u_s(x) \ge \min\{q, e^{-Cs}(\ell^2 x/s)^s\}, \tag{D.42}$$

$$1 - u_s(x) \ge c\min(\ell^2 x, 1) \quad (s \le s_0,\ \ell^2 x \ge Cs). \tag{D.43}$$

Its coefficient sum is at most C^{m+1}. These are fixed-floor bounds: the floor is not allowed to tend to one with α.

For completeness the degree-one completion has no boundary ambiguity. Choose the unique $c_s \in (s/(1+s), 1)$ satisfying

$$(1 - c_s)c_s^s(1+s)^{1+s}/s^s = q_L,$$

and set

$$u_s(x) = \frac{x^s(1 - c_s x)}{a_s}, \qquad a_s = \frac{s^s}{c_s^s(1+s)^{1+s}}.$$

Its maximum is attained at $s/[c_s(1+s)] < 1$, so a_s is the exact normalizer. In particular $c_s \to 1 - q_L > 0$ as $s \downarrow 0$. Continuity of the unique solution makes c_s uniformly bounded above and away from zero for $0 < s \le 1/2$, with constants depending only on ε. Thus the common near-one

tangent-scale comparison is valid for this variable c_s as well. The finitely many degrees below the Chebyshev construction are covered by this same completion; their fixed resolution ratios are absorbed in $e^{\pm Cs}$. For larger degrees one may use $K = [U_{\ell-1}(\sqrt{1-x})/\ell]^4$, its normalized integral J_s, and

$$u_s = x^s K/a_s + f_L J_s(1-K), \qquad a_s = \max_x \frac{x^s K}{1 - f_L J_s(1-K)}, \quad q_L < f_L < 1.$$

Its degree is at most $4(\ell-1)$, so $\ell = \lfloor m/4 \rfloor + 1$ is admissible. We verify all three low-profile properties explicitly. Write $d_L = 1 - f_L > 0$ and $D_s = 1 - f_L J_s(1-K)$. Then

$$\frac{K}{D_s} \le \frac{K}{d_L + f_L K}.$$

The right side has local exponential decay with a positive d_L-dependent constant, a uniform gap away from zero, and a y^{-2} tail. Lemma D.8 therefore gives $a_s \le \ell^{-2s} s^s e^{C_\varepsilon s}$. Since $D_s \le 1$, the ordinary sharp maximum gives the reverse bound. The defining maximum proves $u_s \le 1$ exactly. For $y = \ell^2 x \le s$, its first term is $(y/s)^s e^{O_\varepsilon(s)}$ and its added term is at most Cy^{s+1}; relative to $(y/s)^s$ the latter is $O(s)$. This proves the sharp bounds there; the global upper follows from $u_s \le 1$.

If $y \ge C_f s$, sharp normalization gives $x^s/a_s \ge 1$ after increasing C_f. Hence

$$u_s \ge K + f_L J_s(1-K) \ge f_L - (1-J_s)(1-K).$$

For $y \le 1$, the product on the right is at most $Cs\, y(1 + |\log y|) \le Cs$; for $y \ge 1$ it is again at most Cs. Thus $u_s \ge q$ for all sufficiently small s on this range. For the other s, take C_q sufficiently large and use $J_s \to 1$, $K \to 0$ uniformly in the scaled tail to get $u_s \ge q$ when $y \ge C_q s$. On $s \le y \le C_q s$, if $s \le y_0/C_q$, the local kernel lower bound makes the first term at least $(y/s)^s e^{-C_\varepsilon s}$. Otherwise s is bounded below, and $u_s \ge c_\varepsilon > 0$ on $y \ge s$: both J_s and x^s/a_s are bounded below there, so their weighted contributions through K and $1-K$ give this assertion. Increasing C_ε absorbs $(y/s)^s \le C_q$ on this compact range. This proves (D.42).

For the deficit, the ordinary sharp normalizer $a_s^{(0)}$ is at most a_s, so

$$u_s \le x^s K/a_s^{(0)} + f_L(1-K).$$

On $C_f s \le y \le y_0$, the excess of the first term over K is bounded by $K\{\exp(Cs + s\log(y/s)) - 1\}$. Choose C_f so that $s[C + \log(y/s)] \le \eta y$ with η small relative to d_L. The excess is then at most $d_L(1-K)/2$, giving a deficit at least $c_\varepsilon y$. On $y \ge y_0$, that excess tends uniformly to zero as $s \downarrow 0$: use the uniform gap on compact intervals and the inverse-power tail, or the bound $Cs(1+|\log s|)\sqrt{K}$. Since $1-K$ is bounded below there, sufficiently small s gives a fixed deficit. This proves (D.43). The coefficient statement follows from the final part of Lemma D.8 and $a_s^{-1} \le C\ell^2$.

For the degree-one completion, c_s stays in a fixed positive compact interval as $s \downarrow 0$, and $a_s = s^s \exp(O_\varepsilon(s))$. The same local estimates follow directly from $1 - c_s x$. Past its interior maximum the profile is decreasing to $q_L > q$, so the floor argument is immediate; away from zero its limit is $1 - (1-q_L)x$, giving the asserted small-s deficit by the same $s\log(x/s)$ comparison. The finitely many resolution factors below $\ell = 2$ only alter C_ε.

For the high profile choose $N \asymp L$, where L is the light-polynomial budget, and let

$$R(x) = \left[U_{N-1}(\sqrt{1-x})/N\right]^6, \quad K = R^2, \quad J_s(x) = \frac{\int_0^x t^{s-1}K(t)\,dt}{\int_0^1 t^{s-1}K(t)\,dt}.$$

With a fixed $f \in (0,1)$ sufficiently close to one, put

$$\begin{aligned} W(x) &= f + (1-f)(1-R(x))^2, \\ a_{s,\omega} &= \sup_x \frac{x^s K(x)}{1 - \omega J_s(x)(1-K(x))W(x)}, \\ v_{s,\omega}(x) &= x^s K(x)/a_{s,\omega} + \omega J_s(x)(1-K(x))W(x). \end{aligned} \tag{D.44}$$

The value of a ratio at a zero of K is its continuous extension. One has, uniformly for $0 < \omega \le 1$,

$$a_{s,\omega} = N^{-2s} s^s \exp(O_\varepsilon(s)), \qquad 0 \le v_{s,\omega} \le 1, \tag{D.45}$$

$$e^{-Cs}(y/s)^s \le v_{s,\omega}(x) \le e^{Cs}(y/s)^s \quad (y = N^2 x \le s), \tag{D.46}$$

$$v_{s,\omega}(x) \ge c\omega \quad (y \ge s), \qquad v_{s,\omega}(x) \ge q\omega \quad (y \ge C_q s), \tag{D.47}$$

$$|v_{s,\omega}(x) - \omega J_s(x)| \le C \min\{y^s, y^{-3}\}. \tag{D.48}$$

The upper sharp bound extends to all y, using $v \le 1$. The coefficient sum of $v_{s,\omega}/x^s$ is at most C^{L+1}. We choose the fixed degree constant large enough that $N^2 \ge 2C_q$.

Here is a normalization argument retaining the factor s in the exponent. For $\omega = 1$ the denominator is at least

$$K + (1-f)(1-K)(2R - R^2),$$

so its kernel quotient is bounded by

$$\frac{R}{R + (1-f)(1-R^2)(2-R)}.$$

This has local exponential decay, a uniform gap away from zero, and a y^{-3} tail. The sharp maximum is therefore $N^{-2s} s^s \exp(O_\varepsilon(s))$. For general ω, the normalizer lies between the ordinary sharp normalizer and this one. Exact boundedness follows from the definition. The polynomial degree is at most $18(N-1)$ after division by x^s. The difference in (D.48) equals

$$x^s K/a_{s,\omega} - \omega J_s K - \omega(1-f) J_s (1-K)(2R - R^2),$$

which proves its two tail bounds using $K \le y^{-6}$ and $R \le y^{-3}$. The floor $q\omega$ follows, for small s and $y \ge C_f s$, from $x^s/a_{s,\omega} \ge 1$ and

$$v_{s,\omega} \ge \omega[K + J_s(1-K)W] \ge \omega(1 - (1-f) - Cs).$$

For the other s, increase C_q and use the integral and kernel tails. In addition $J_s \ge q$ for $y \ge C_q s$, after enlarging C_q. Indeed $J_s \ge y^s e^{-Cs}$ for $y \le 1$, while $1 - J_s \le Csy^{s-6}$ for $y \ge 1$.

The following distinction is necessary. A global bound $v_{s,\omega} \ge \min\{q\omega, e^{-Cs}(y/s)^s\}$ is asserted only when ω is bounded below by a fixed positive constant. In that case it follows by the same compact-interval argument as for the low profile. For arbitrary $\omega \downarrow 0$ we use only (D.46) and (D.47). A claim of the stronger global envelope uniformly down to $\omega = 0$ would be false at interior kernel zeros.

Combining the profiles. The scalar envelope determines the weights A, B and the product scale z. Small exponential adjustments A', B' leave a usable strict gap below one. The factor $1 - J_1$ turns off the reverse profile as p increases; the remaining terms approach ωp^α. The proposition gives two-sided bounds for every $p \in [a,b]$ and a separate bound below a. These are deterministic statements about functions, before any data are sampled.

Proposition D.9 (One coupled deterministic bound)**.** *There are fixed $M, \lambda, C_\varepsilon, c_\varepsilon > 0$ with the following property. Set $A' = Ae^{-M\gamma}$, $B' = Be^{-M\delta}$, and choose $0 < a < b \le 1$ by the exact capacity equation*

$$a = \lambda \frac{bhz^{1/h}}{m^2 L^2}. \tag{D.49}$$

Define one polynomial by

$$Q(p) = p^\alpha \left[A' u_\gamma(a/p)(1 - J_1(p/b)) + B' v_{\delta,\omega}(p/b) + \omega(1 - B') J_\delta(p/b) \right]. \tag{D.50}$$

The expression extends polynomially to $p = 0$: the reverse degree is at most m, and $\alpha + \delta = m + 1$. For $a \le p \le b$, write $Y = N^2 p/b$ and $Z = \ell^2 a/p$. Then

$$\frac{Q(p)}{p^\alpha} \ge \kappa_0 e^{-C_\varepsilon h} + c_\varepsilon \kappa_0 \min(Y, 1), \tag{D.51}$$

$$\frac{Q(p)}{p^\alpha} \le 1 - c_\varepsilon \min\{h + Y + \min(Z, 1), 1\}. \tag{D.52}$$

Below a, its absolute contribution and the target are bounded by $C_\varepsilon^{m+1} a^\alpha$ per coordinate. The constants do not change with the fractional part or with m.

Proof. **Step 1: reduce the interaction to the scalar envelope.** Write $u = u_\gamma(a/p)$, $v = v_{\delta,\omega}(p/b)$, $H = J_1(p/b)$, $J = J_\delta(p/b)$, and $T = Q(p)/p^\alpha$. The degree ratios ℓ/m and N/L are bounded above and below by fixed positive constants. Thus $ZY = \lambda(\ell/m)^2 (N/L)^2 h z^{1/h}$. A sufficiently small fixed $\lambda = \lambda_\varepsilon$ gives

$$u^\delta v^\gamma \le z, \qquad u \le e^{C\gamma} z^{1/\delta} (\delta/Y)^\gamma. \tag{D.53}$$

In the central interval $Z \le \gamma$, $Y \le \delta$, the reverse inequality holds up to e^{-Ch} in the product. Maximizing a linear functional under $u^\delta v^\gamma \le z$ gives $Au + Bv \le 1$: the two endpoint values are exactly one, and the functional restricted to the product curve is convex.

Step 2: establish a strict upper gap. The proof separates fractional parts close to zero, close to one, and away from both endpoints. This is only an analysis of the same polynomial; its definition does not change. Fix a sufficiently small fractional threshold $s_0 = s_0(\varepsilon)$ and then choose M sufficiently large. If $\gamma \le s_0$ and $Y \ge \gamma$, (D.53) gives $u \le e^{-\varepsilon/4}$ after decreasing s_0, since $E_0 \ge \varepsilon/2$. Also $J - H \le C\gamma$. Hence

$$T \le \max(e^{-\varepsilon/4}, e^{-\varepsilon/16}) + C\gamma + e^{-M(1-s_0)},$$

which has a fixed deficit. If $Y < \gamma$, choose a fixed $r \in (\kappa_{\max}, 1)$, where $\kappa_{\max} = e^{-\varepsilon/2}$. When $u \le r$, the bounds $v, J \le C\gamma$ give a fixed deficit. When $u > r$, the product constraint gives $v \le e^{-c/\gamma}$; on this range $J \le Cv$. Therefore

$$T \le e^{-M\gamma} u + Ce^{-c/\gamma} \le 1 - c[(1 - u) + \gamma].$$

The low-profile deficit then supplies $\min(Z, 1)$; when $Z < C\gamma$ it is already controlled by γ.

For small δ, the high profile has

$$1 - v \ge c_\varepsilon \min(Y, 1) \quad (Y \ge C_\varepsilon \delta). \tag{D.54}$$

To verify this, bound v by the ordinary sharp bump plus $\omega(1 - K)$. On $C\delta \le Y \le y_0$, its excess over K is at most an arbitrarily small multiple of Y, while $(1 - \omega)(1 - K) \ge c_\varepsilon Y$. Outside this

neighborhood the sharp bump converges uniformly to K as $\delta \downarrow 0$ and $1-K$ is bounded below. This proves (D.54). On $Y \geq C_\varepsilon \delta$, the low term is at most $Ce^{-\varepsilon/(2\delta)}$, while

$$B'v + \omega(1-B')J \leq 1 - B'c_\varepsilon \min(Y,1) - (1-B')(1-\omega).$$

The exponential term is absorbed. Since $u \geq c\min(Z,1)$ for $\gamma \geq 1/2$, the same margin controls $\min(Z,1)$. On $Y < C_\varepsilon\delta$, if $u \leq c_1\delta$ then $1-B' \geq 1-e^{-M\delta} \geq \delta/2$ supplies a linear deficit. If $u > c_1\delta$, the product constraint gives $v \leq e^{-\varepsilon/4}$, and the low amplitude $A' \leq e^{-M\gamma}$ gives a fixed deficit. Finally, when both fractional parts are bounded below, $T \leq e^{-Ms_0} + e^{-Ms_0} + \omega$ gives a fixed deficit. This proves (D.52) for $\alpha \geq 3/2$.

Step 3: control the loss introduced by the cutoff. We next record the cancellation needed for the lower bound. Since $1-B = Az^{1/\delta} \leq e^{-E_0/\delta}$, concavity of the logarithm gives

$$\log \frac{1-B'}{1-B} \geq (1-e^{-M\delta})E_0/\delta \geq (1-e^{-M})E_0.$$

Take $M \geq 1$, so that $\omega(1-B') \geq 1-B$. Consequently

$$T \geq A'u + B'v - (A'u - 1 + B)_+ H, \qquad (A'u - 1 + B)_+ H \leq C_\varepsilon \kappa_0 h. \tag{D.55}$$

For the second inequality, use (D.53) and $1-B = Az^{1/\delta}$. If $Y \leq \delta$, put $r = Y/\delta$ and use $H \leq CY$ together with $\max_{0\leq r\leq 1}(r^\delta - r) = \gamma\delta^{\delta/\gamma} \leq \gamma$. For $Y \geq \delta$ the loss is at most $C\gamma(1-B)$. When $\delta \geq 1/2$ this is $O(\kappa_0 h)$; otherwise use $(1-B)/\kappa_0 \leq \kappa_0^{\gamma/\delta} \leq C_\varepsilon\delta$.

Step 4: reserve a term that supplies the local lower gap. Reserve a fixed small fraction of the high arm by putting $B'' = B'e^{-\eta\delta}$, with fixed $\eta > 0$. On $Y \leq C_q\delta$ and bounded E_0, the sharp and compact-interval profile bounds give

$$(B' - B'')v \geq c_\varepsilon \kappa_0 Y. \tag{D.56}$$

For $Y \leq \delta$ this follows from $\delta v \geq cY$; for $\delta \leq Y \leq C_q\delta$ use the fixed positive floor, with ω bounded below. On $Y \geq C_q\delta$, $J \geq q$ and $v \geq q\omega$, so

$$T \geq q\omega \geq (1+c_\varepsilon)\kappa_0. \tag{D.57}$$

Step 5: prove the main lower bound uniformly in the order. It remains to justify the base term without assuming false global sharp bounds as $\omega \downarrow 0$. Choose a fixed sufficiently large $E_* = E_*(\varepsilon)$. If $E_0 \geq E_*$, a saturated low profile ($Z \geq \gamma$) supplies $A'u(1-H) \geq c_\varepsilon$, because then $Y \leq \delta$ and $1-H$ is bounded below. A saturated high profile ($Y \geq \delta$) supplies $B'v \geq c_\varepsilon\omega$. Choose E_* so both quantities are at least $2\kappa_0$. These cases give the base and local gain. In the remaining central interval both sharp lower bounds apply; weighted AM–GM, after the reserve, gives $A'u + B''v \geq \kappa_0 e^{-Ch}$. The reserve is at least $c_\varepsilon\kappa_0 Y$ there.

If $E_0 \leq E_*$, then m is bounded and ω is bounded below. Set

$$U = e^{-C\gamma}(Z/\gamma)^\gamma, \qquad V = e^{-C\delta}(Y/\delta)^\delta.$$

On $Y \leq C_q\delta$, the valid global comparisons are $u \geq \min(q, U)$ and $v \geq \min(q\omega, V)$. If a saturated low term satisfies $A'q \geq \kappa_0$, the base sum is already at least κ_0. Otherwise $A'q < \kappa_0$ forces $\gamma \geq c_\varepsilon$: as $\gamma \downarrow 0$, $A' \to 1$ uniformly for $z \leq \kappa_{\max}$, whereas $q > \kappa_{\max}$. Since $p \geq a$, $Z \leq \ell^2$ and m is bounded. Thus $U \leq C_\varepsilon$, and replacement of U by its floor costs only $\delta \log C_\varepsilon = O_\varepsilon(h)$ in weighted AM–GM. The same argument for $B''q\omega < \kappa_0$ forces $\delta \geq c_\varepsilon$, because $B'' \to 1$ uniformly as $\delta \downarrow 0$

and $q\omega/\kappa_0 > 1 + c_\varepsilon$. On the bounded Y interval, $V \le C_\varepsilon$, so its floor costs $\gamma \log C_\varepsilon = O_\varepsilon(h)$. If its saturated term already exceeds κ_0 no comparison is needed. In every case

$$A'u + B''v \ge \kappa_0 e^{-C_\varepsilon h}.$$

For sufficiently small h, subtracting the loss in (D.55) merely changes the constant in the exponential; retain the reserve separately. If h is bounded below, $1 - H$ is bounded below on $Y \le C_q$ by our choice $N^2 \ge 2C_q$. For an explicit uniform justification, integrate the scaled kernel over $[C_q, 2C_q]$. As $N \to \infty$ it converges uniformly there to $(\sin\sqrt{y}/\sqrt{y})^{12}$, whose integral is positive. The finitely many remaining admissible N also have positive integrals, and the normalized total integral is uniformly bounded. This proves a fixed positive tail bound. Thus the base part of T is at least a fixed positive multiple of $A'u + B''v$, again absorbed by $e^{-C_\varepsilon h}$; the reserve is unchanged. Together with (D.57) this proves (D.51). The proof used the same polynomial and the same parameters throughout.

Step 6: bound the portion below the lower cutoff. Finally, below $p = a$ the reverse polynomial has coefficients at most $C^{m+1}a^{\alpha-j}$ for $0 \le j \le m$. Since $p/b \le a/b$, $0 \le 1 - H \le 1$, and the high profiles are bounded, its absolute value and p^α are at most $C^{m+1}a^\alpha$. This proves the final assertion. □

D.5 From pointwise approximation to a power-sum estimate

Why a deterministic correction is possible. There can be as many as k probabilities below a, so their errors must be added. The upper bound on the approximation has deliberately left a positive gap on the other coordinates. The following calculation shows that this gap is large enough to pay for a known correction that covers all the below-cutoff errors. This is why the estimator includes a deterministic additive term instead of requiring an accurate approximation at every arbitrarily small probability.

Summed affine baseline. Let $P_* = k^{1-\alpha}$, $a_0 = ka \le c_\varepsilon$, and $P = \sum_i p_i^\alpha$. For $\alpha \ge 2$, at least half the mass is on $p_i \ge 1/(2k) \ge a$, and hence

$$\sum_{p_i \ge a} p_i^\alpha \min(\ell^2 a/p_i, 1) \ge 2^{1-\alpha} \min(\ell^2 ka, 1/2) P_*.$$

For $3/2 \le \alpha < 2$, put $s = kp$ and $\theta = N^2/(bk)$. The weighted sum of $Z + Y$ has pointwise factor

$$\ell^2 a_0 s^{\beta-1} + \theta s^{\beta+1} \ge (\ell^2 a_0)^{(\beta+1)/2} \theta^{(1-\beta)/2} \ge c_\varepsilon a_0^\beta,$$

where the last step uses capacity, $\beta = \gamma \ge 1/2$, and $z \ge \kappa_0/2 \ge e^{-2\varepsilon\beta}/2$. If $Z + Y \ge 1$, truncation still gives the same conclusion on $s \ge 1/2$. Consequently the upper margin pays

$$E_a P_*, \qquad E_a = d_\varepsilon C_0^m (ka)^\beta,$$

with sufficiently small fixed $d_\varepsilon > 0$ and sufficiently small ka, and it dominates $C^{m+1} ka^\alpha$. This justifies a deterministic correction of size $2E_a P_*$ while retaining fixed portions of the upper margins. For a common formula including $\alpha \downarrow 1$, replace E_a by $\min(\beta, 1) d_\varepsilon C_0^m (ka)^\beta$.

The order-one divided difference. Near $\alpha = 1$, the reverse profile has a particularly transparent interpretation: it is a tangent line to $p \mapsto p^{1+\beta}$. The exact tangent identity below exposes a factor β in every nonlinear error. The known linear aggregate $\sum_i p_i = 1$ can therefore be removed before estimating the remainder.

For $\alpha = 1 + \beta \le 3/2$, use the same degree-one profile, whose exact reverse identity is

$$p^\alpha u_\beta(a/p) = (1+\beta)\tau^\beta p - \beta\tau^{1+\beta} =: T_\tau(p), \qquad \tau = c_\beta(1+\beta)a/\beta.$$

Suppose the common capacity rule gives τ/t in a fixed positive compact interval, where $t = b/N^2$, and gives $A' = 1 + O_\varepsilon(\beta)$ and $B' = O_\varepsilon(\beta)$. Then the very same polynomial satisfies

$$\sup_{0 \le p \le b} |Q(p) - \omega p^\alpha| \le C_\varepsilon \beta t^\alpha. \tag{D.58}$$

Subtract $\omega[T_\tau(1-H) + p^\alpha J_\delta]$. Its tangent-cutoff error is $O(\beta t^\alpha)$. The remaining terms are $(A' - \omega)T_\tau(1-H)$ and $B' p^\alpha (v_{\delta,\omega} - \omega J_\delta)$; localization bounds each by $O_\varepsilon(\beta t^\alpha)$. This includes the interval $p < a$. If $Q = \sum_j q_j p^j$, then

$$q_0 = -A'\beta\tau^{1+\beta}, \qquad q_1 = A'[(1+\beta)\tau^\beta + \beta\tau^{1+\beta} J_1'(0)/b], \qquad |q_1 - t^\beta| \le C_\varepsilon \beta t^\beta.$$

For $j \ge 2$, coefficientwise cancellation at $\beta = 0$ gives $|q_j| \le \beta C_\varepsilon^{L+1} b^{\alpha - j}$. The known constant and linear aggregates are therefore removed before the common factorial-variance calculation.

What the deterministic proposition supplies to the statistical theorem. For any light/heavy assignment with light coordinates at most b, replace heavy coordinates by a mean centered at ωp_i^α. The proposition gives the baseline and local margins after paying the below-cutoff error. Rescaling both parts by $(1 + c_u u)^{-1}$ supplies an additional upper margin of order uP. If $E - E_0 \ge C_{\mathrm{bias}} h + c_{\mathrm{reg}} u$, its lower counterpart follows directly from (D.51). When this inequality is not available, one must use the same polynomial's localization toward ωp^α and the exact root calibration to bound the aggregate error. The present proposition does not by itself establish that final calibration, the pilot errors, or the sampling variance; these are separate hypotheses to be verified for the single common estimator.

The deterministic output used in the sampling proof. The next statement collects all preceding estimates into the precise form needed later. The sets $\mathcal{L}$ and $\mathcal{H}$ form an arbitrary partition of the coordinates into light and heavy groups. The only restriction is $p_i \le b$ for light coordinates. Thus the statement can later be applied after conditioning on an independent pilot sample, provided that pilot classification has succeeded. The quantity $\mathcal{M}$ is an explicit error budget: it is a sum of a below-cutoff baseline, a regularization gap, and a local gap.

Corollary D.10 (Bias transfer for the single corrected expression)**.** *Let $\kappa = e^{-E}$, $\chi = \min(\beta, 1)$,*

$$\mathfrak{u} = \min\{h + (\beta/\alpha)X^{-\alpha}, u_0\}, \quad r = (1 + c_u \mathfrak{u})^{-1}, \quad E_a = \chi d_\varepsilon C_0^m (ka)^\beta.$$

For any assignment to light and heavy coordinates, with every light coordinate at most b, define the oracle mean

$$\mu = r\left[\sum_{i \in \mathcal{L}} Q(p_i) + \omega \sum_{i \in \mathcal{H}} p_i^\alpha\right] + 2E_a P_*.$$

Use the same Q, radius rule, and coefficients for all orders. Put

$$B_p = \sum_i p_i^\alpha \min(N^2 p_i/b, 1), \qquad \mathcal{M} = E_a P_* + \kappa \mathfrak{u} P + \chi\kappa B_p.$$

Suppose the root calibration has established the following quantitative facts, with constants depending only on ε:

1. *$ka \le c_\varepsilon/D$, and $E/2 \le E_0 \le E$.*
2. *For $\alpha \ge 3/2$ with E/h above a fixed sufficiently large threshold, $E - E_0 \ge C_{\mathrm{bias}}h + c_{\mathrm{reg}}\mathfrak{u}$, where C_{bias} is chosen after the deterministic proposition and $c_{\mathrm{reg}} > 0$ is a fixed, possibly small, accuracy-dependent constant.*
3. *If $\alpha \ge 3/2$ and E/h is bounded by that threshold, then α is bounded, γ, δ are bounded below, and $kt \le C_\varepsilon/D$ for $t = b/N^2$.*
4. *For $1 < \alpha \le 3/2$, one has $kt \le C_\varepsilon/D$, τ/t in a fixed positive compact interval, $A' = 1 + O_\varepsilon(\beta)$, $B' = O_\varepsilon(\beta)$, and $\mathfrak{u} = \Theta_\varepsilon(\beta)$.*

Then, choosing $c_u, d_\varepsilon > 0$ sufficiently small and subsequently D sufficiently large, one has

$$\boxed{\kappa P + c_\varepsilon \mathcal{M} \le \mu \le P - c_\varepsilon \mathcal{M}.}$$

The enumerated statements are calibration hypotheses, not consequences of polynomial approximation alone.

Proof. The upper baseline bounds and ka small imply that $2E_aP_*$ pays the below-cutoff error and consumes only a fixed part of the upper baseline. For $\alpha \ge 3/2$ under the second calibration inequality, apply (D.51)–(D.52). Heavy coordinates have mean ωp_i^α with a fixed relative margin from both endpoints; their gaps pay any omitted B_p or baseline terms. Rescaling supplies an upper gap proportional to $\mathfrak{u}P$. The calibration inequality makes $r\kappa_0 e^{-C_\varepsilon h} \ge \kappa(1 + c\mathfrak{u})$, so the corresponding lower gap is retained. Since $\chi \ge 1/2$, the result is the stated margin.

To verify the other calibration range with the same polynomial, observe on $a \le p \le b$ that

$$|Q(p) - \omega p^\alpha| \le Cp^\alpha \min\{1, (t/p)^3\}.$$

This follows directly from $T - \omega = A'u(1-H) + \omega(J-1) + B'(v - \omega J)$. For $\nu = \min(\alpha, 2)$, Hölder's inequality bounds its aggregate, divided by P, by $C(kt)^\nu (P/P_*)^{-\nu/\alpha}$. The below-$a$ contribution is at most $C^{m+1}(ka)^\alpha P_*$. Under the third calibration statement these errors tend uniformly to zero as D increases. In that range α lies in a fixed compact interval and $\beta \ge 1/2$, so the center $r\omega$ remains a fixed positive distance from both κ and one after c_u is chosen small. Choose d_ε small and then D large. Since $\mathcal{M} \le C_\varepsilon P$, these fixed gaps imply the claimed inequality.

Finally, under the fourth calibration statement, (D.58) bounds the aggregate error by $C_\varepsilon\beta(kt)^\alpha P_*$. The center $r\omega$ is separated from both κ and one by fixed positive multiples of β, after choosing c_u small. The correction is at most $Cd_\varepsilon\beta P$ because $m = 1$. Taking d_ε small and D large leaves gaps of order βP. Here $\mathcal{M} \le C_\varepsilon\beta P$; the factor χ in its last term is essential. This gives the same asserted inequality without changing the estimator or its parameter rule. □

What has now been proved, and what remains. Once the listed calibration conditions are checked, the corrected oracle mean is at distance at least $c_\varepsilon\mathcal{M}$ from both endpoints of $[\kappa P, P]$. The next section replaces that oracle mean by observable count statistics. It must show that polynomial-count fluctuations, heavy-count errors, and pilot mistakes jointly use less than this available distance. No sample-complexity conclusion follows from the deterministic bounds alone.

E From the polynomial approximation to a sampling guarantee

The preceding section constructed a polynomial whose values are close enough to p^α on the range where small probabilities matter. We now replace its monomials by factorial statistics and prove that the resulting random quantity stays inside the required power-sum interval. There are two different statements to establish:

1. the deterministic approximation has room on both sides of the interval $[e^{-2\varepsilon(\alpha-1)}P_\alpha(p), P_\alpha(p)]$;
2. count fluctuations and mistakes made by the pilot consume less than that room with sufficiently high probability.

A small approximation error alone does not establish the second statement. Large polynomial coefficients can amplify count noise. This is why we first bound the coefficients of the actual polynomial, rather than replace it by an unspecified polynomial of the same degree.

The objects used throughout this section. Write $\alpha = m + \gamma$, where $m = \lfloor \alpha \rfloor$ and $0 < \gamma < 1$, and put $\beta = \alpha - 1$ and $\delta = 1 - \gamma$. The previous section has specified a sample scale n, a polynomial parameter L, and probability cutoffs $0 < a < b$. The actual polynomial has degree $d \le C_d L$ and is written $Q(p) = \sum_{j=0}^{d} q_j p^j$. For every symbol we generate two independent counts,

$$N_i \sim \mathrm{Poi}(np_i), \qquad Y_i \sim \mathrm{Poi}(c_p np_i),$$

where N_i is used to estimate the contribution and Y_i is used only to select the branch. The indicator $I_i = \mathbf{1}\{Y_i \le c_p B\alpha\}$ selects the polynomial branch. The Gamma branch is used when $I_i = 0$. All counts are independent across symbols as well. Here B, c_p are fixed constants selected in the proof, and not unknown features of the distribution.

Why the proof is split into several parts. For $\alpha \ge 3/2$, the approximation has a margin that controls the conditional variance of the polynomial, pilot errors, and the Gamma branch separately. For α close to one, the desired interval shrinks at rate $\beta = \alpha - 1$; the proof must therefore retain a factor β in bias and a factor β^2 in variance. The estimator is unchanged, but its analysis uses the cancellation $\sum_i p_i = 1$ explicitly. After these estimates are established, Theorem E.13 assembles them into one uniform upper bound.

A useful identity to keep in mind. If $\widehat{Q}_i = \sum_j q_j (N_i)_j / n^j$, then $\mathbb{E}\widehat{Q}_i = Q(p_i)$. Moreover, the sum of the linear terms can be replaced by the known number q_1:

$$q_1 + \sum_i \left(\widehat{Q}_i - q_1 N_i/n\right) = \sum_i \widehat{Q}_i + q_1 \left(1 - \sum_i N_i/n\right).$$

The added term has mean zero. Its role is particularly important near order one, where the target is almost the known linear functional $\sum_i p_i$. We call this replacement the *affine anchor*.

E.1 Conditional count noise for the common polynomial

This section evaluates the count variance for one polynomial and one pilot rule. The deterministic approximation margins are hypotheses of Theorem E.4; they are not consequences of a variance bound. Throughout, logarithms are natural and $(w)_j = w(w-1)\cdots(w-j+1)$.

Step 1: control the coefficients before controlling variance. The kernels used in the approximation have many coefficients, but their coefficients decay factorially after the relevant endpoint scale is removed. This permits summation of the variances of all monomials. The first lemma records that decay; the next transports it to the coupled polynomial actually used by the estimator. As usual, $[z^j]P$ denotes the coefficient of z^j in a polynomial P.

Lemma E.1 (Endpoint coefficients)**.** *For $N \geq 2$, put*

$$B_N(z) = \frac{1 - T_N(1-2z)}{2N^2 z}, \qquad K_N(z) = B_N(z)^2.$$

The singularity at zero is removable, $K_N(0) = 1$, and

$$|[z^\ell]K_N(z)| \leq \frac{(16N^2)^\ell}{(2\ell)!}. \tag{CC1}$$

The same type of estimate, with a different absolute constant, holds for any fixed power of K_N and for a fixed finite number of products of its normalized integral profiles.

In particular, the low endpoint profiles used in the coupled construction have polynomial part P of degree at most m satisfying

$$|[z^\ell]P| \leq Cm^{2\gamma}\frac{(Cm^2)^\ell}{(\ell!)^2}, \qquad 0 \leq \ell \leq m, \quad 0 < \gamma < 1. \tag{CC2}$$

For $Q_-(p) = p^{m+\gamma}F(a/p)$, this implies

$$|[p^{m-\ell}]Q_-| \leq Cm^{2\gamma}a^{\gamma+\ell}\frac{(Cm^2)^\ell}{(\ell!)^2}. \tag{CC3}$$

Constants may depend on the fixed profile parameters, but not on m, the fractional part, a, or the alphabet size.

Proof. The Chebyshev recurrence, or induction in that recurrence, gives

$$[z^\ell]B_N(z) = (-1)^\ell \frac{2^{2\ell+1}}{(2\ell+2)!}\prod_{r=1}^{\ell}(N^2 - r^2), \qquad 0 \leq \ell < N.$$

Thus $|[z^\ell]B_N| \leq (4N^2)^\ell/(2\ell+1)!$. Convolution and the identity

$$\sum_{i=0}^{\ell}\frac{1}{(2i+1)!(2\ell-2i+1)!} = \frac{2^{2\ell+1}}{(2\ell+2)!} \leq \frac{4^\ell}{(2\ell)!}$$

prove (CC1). Each additional fixed product changes only the constant, by the even binomial convolution.

For a normalized integral profile

$$J_s(z) = I_s^{-1}\int_0^z u^{s-1}K(u)\,du, \qquad I_s = \int_0^1 u^{s-1}K(u)\,du,$$

its polynomial part has coefficients $k_\ell/[I_s(\ell+s)]$. The endpoint normalization $(sI_s)^{-1} = N^{2s}e^{O(s)}$ therefore preserves (CC1), up to the prefactor CN^{2s}. For the actual raised-floor low profile, let $f \in (0,1)$ be its fixed floor parameter and define

$$A_{s,f} = \max_{0\leq z\leq 1}\frac{z^s K(z)}{1 - fJ_s(z)(1-K(z))}, \qquad \frac{F_s(z)}{z^s} = \frac{K(z)}{A_{s,f}} + f\frac{J_s(z)}{z^s}(1-K(z)).$$

Its proved normalization is $A_{s,f} = N^{-2s}[s/(1-f)]^s e^{O_f(s)}$. Because $s/(1-f) \le 1/(1-f)$ and $s^{-s} \le e^{1/e}$, one has $A_{s,f}^{-1} \le C_f N^{2s}$ and $N^{2s} \le C_f A_{s,f}^{-1}$. The kernel term therefore has coefficient bound $C_f N^{2s}(CN^2)^\ell/(2\ell)!$. For the product with $1-K$, convolve the integral-polynomial estimate with (CC1); the even binomial convolution only changes the base constant. The added product has zero constant coefficient, so the leading coefficient remains exactly $A_{s,f}^{-1}$. This proves (CC2) when N is a fixed fraction of m. For the finitely many smaller available degrees, use the actual fixed-floor degree-one profile. For the prescribed $q_L \in (0,1)$, let $c_s \in (s/(1+s), 1)$ be the unique solution of

$$(1-c_s)c_s^s(1+s)^{1+s}/s^s = q_L.$$

The left side decreases continuously from one to zero on this interval. Its stationary point $s/[c_s(1+s)]$ lies in $(0,1)$, so its exact normalization is

$$F_s(z) = z^s(1-c_s z)/A_s, \qquad A_s = \frac{s^s}{c_s^s(1+s)^{1+s}}.$$

Both polynomial coefficients are bounded by $4e^{1/e}$, since $c_s \le 1$, $s^{-s} \le e^{1/e}$, and $(1+s)^{1+s} \le 4$. This is the same prescribed-floor degree-one profile as in the approximation construction; no different profile is substituted. Substitution of a/p proves (CC3). □

Lemma E.2 (Coefficients of the coupled polynomial)**.** *Write $\alpha = m+\gamma$, $\delta = 1-\gamma$, and consider the actual coupled polynomial*

$$Q(p) = A'Q_-(p)\{1 - J_1(p/b)\} + p^\alpha\{B' v_\delta(p/b) + \omega(1-B')J_\delta(p/b)\}. \tag{CC4}$$

Assume that its amplitudes are bounded, that $a \le b$, $nb = c_b L$, that all high profiles have degree $O(L)$ and endpoint scale L^2, and that

$$Z = am^2L^2/b \le Z_0. \tag{CC5}$$

Here c_b, Z_0 and the profile constants are fixed. Then, writing $Q = \sum_{j=0}^d q_j p^j$, $d \le C_d L$, one has

$$|q_j| \le Cm^{2\gamma}a^{\alpha-j}\frac{(Cm^2)^{m-j}}{((m-j)!)^2}, \qquad 1 \le j \le m, \tag{CC6}$$

$$|n^{-\delta}q_{m+1}| \le C(L/\delta)^\delta, \tag{CC7}$$

$$|q_j| \le e^{CL}b^{\alpha-j}, \qquad m+1 \le j \le d. \tag{CC8}$$

These bounds hold for the omega-tail profile with $\omega = e^{-c_0E_0}$ and attenuated amplitudes $A' = Ae^{-M\gamma}$, $B' = Be^{-M\delta}$.

Proof. The coefficient bound for the normalized cutoff is

$$J_1(z) = \sum_{t\ge1} v_t z^t, \qquad |v_t| \le (CL^2)^t/(t!)^2.$$

For a coefficient of degree $m-\ell$ in the cross term Q_-J_1, use $(\ell+t)! \ge \ell!t!$ to bound its absolute value by

$$Cm^{2\gamma}a^{\gamma+\ell}\frac{(Cm^2)^\ell}{(\ell!)^2}\sum_{t\ge1}\frac{(CZ)^t}{(t!)^4}.$$

The last sum is bounded under (CC5); the pure high arm starts at degree $m+1$. This proves (CC6).

The cross coefficient at degree $m+1$, expressed in count units, is at most

$$C\left(\frac{am^2n}{\gamma}\right)^{\gamma}\frac{L^2}{nb}\sum_{\ell\ge 0}\frac{(CZ)^{\ell}}{(\ell!)^2((\ell+1)!)^2}\le CL^{\delta}(Z/\gamma)^{\gamma}\le C(L/\delta)^{\delta}.$$

We used $am^2n=c_bZ/L$ and $\gamma^{-\gamma}\le e^{1/e}$. The pure high arm has the same leading bound by its sharp normalization. This proves (CC7). Rescaling $p=bz$ gives (CC8): the reverse coefficients contribute factors $(a/b)^{\gamma+\ell}\le 1$, and all polynomial coefficient sums are at most C^L.

For the omega-tail normalization, the denominator in the defining maximum lies between the ordinary bump denominator and the localized unit-tail denominator. Its normalizer therefore lies between the corresponding two normalizers, both equal to $N^{-2s}s^se^{O(s)}$. The added term $\omega J_s(1-K)W$ is a fixed product of the same kernel and integral polynomials. It preserves all the bounds just proved. Bounded scalar amplitudes and their attenuation also preserve these bounds. □

Step 2: use the pilot to suppress excessive polynomial noise. The polynomial is designed for small probabilities. Without a pilot, its factorial moments could be large at a frequent symbol. Independence of Y and N lets us multiply the factorial second moment by the small probability that a frequent symbol is incorrectly assigned to this branch. In the following lemma j is an integer, as required by the falling factorial.

Lemma E.3 (A pilot bound for factorial moments)**.** *Fix $B,c_p>0$. Let $N\sim\mathrm{Poi}(\lambda)$ and $Y\sim\mathrm{Poi}(c_p\lambda)$ be independent, and put $I=\mathbf{1}\{Y\le c_pB\alpha\}$. For $1\le j\le\alpha$,*

$$\mathbb{E}[I(N)_j^2]\le(C_{B,c_p}\alpha)^j\lambda^j. \tag{CC9}$$

Proof. The factorial identity gives

$$\mathbb{E}(N)_j^2=\sum_{r=0}^{j}\binom{j}{r}^2 r!\lambda^{2j-r}\le\lambda^j(\lambda+j)^j.$$

For $\lambda\le 2B\alpha$ this proves the assertion. For larger means, $\mathbb{P}(I=1)\le e^{-c_p\lambda/8}$, and $\sup_{\lambda\ge0}e^{-c_p\lambda/8}(\lambda+j)^j\le(C_{c_p}j)^j$. Independence finishes the proof. □

Step 3: measure noise against the available margin. Let $P=\sum_i p_i^{\alpha}$ and $P_*=k^{1-\alpha}$. The inequality $P\ge P_*$ follows from convexity, so $A=P/P_*\ge 1$ measures how far the power sum is above its smallest possible value. The margin below has three terms. The first is the deterministic correction for probabilities below a. The second is the room reserved by the scalar-root calibration. The third rewards coordinates whose expected counts are large enough to provide additional room. We need a variance bound proportional to the square of this entire margin, uniformly over the unknown p.

Conditioning on the pilot makes all selectors I_i fixed; it does not remove count noise in N_i. The theorem bounds that conditional noise. Pilot randomness itself is handled later, so this theorem should not be read as a bound on the full variance of the hybrid estimator.

Theorem E.4 (Conditional count bound with the actual cutoff)**.** *Fix $\varepsilon>0$ and $\alpha\ge 3/2$ noninteger. Set*

$$P_*=k^{1-\alpha},\quad P=\sum_i p_i^{\alpha},\quad \kappa=e^{-2\varepsilon(\alpha-1)},\quad n=D\alpha k^{1-1/\alpha}X,$$

$$L = \lceil \alpha(C_L + c_L \log X) \rceil, \qquad nb = c_b L, \qquad u \geq c_u X^{-\alpha}, \quad X \geq 1.$$

Assume the coefficients satisfy (CC6)–(CC8), $nb \geq 2B\alpha$, and that fixed positive ρ_-, ρ_+ obey $ka \leq \rho_+/D$. Let

$$E_b = \chi_\alpha d_\varepsilon C_0^m (\rho_-/D)^{\alpha-1}, \qquad E_a = \chi_\alpha d_\varepsilon C_0^m (ka)^{\alpha-1}, \quad \chi_\alpha = \min\{\alpha - 1, 1\},$$

where $1/2 \leq \chi_\alpha \leq 1$, and assume the calibrated baseline comparison

$$E_b \leq E_a + C_\varepsilon \kappa u. \tag{CC10}$$

Define the actual margin

$$\mathcal{M} = E_a P_* + \kappa u P + \kappa \mathcal{B}, \qquad \mathcal{B} = \sum_i p_i^\alpha \min(Lnp_i, 1). \tag{CC11}$$

For independent counts $N_i \sim \mathrm{Poi}(np_i)$ and $Y_i \sim \mathrm{Poi}(c_p np_i)$, let

$$I_i = \mathbf{1}\{Y_i \leq c_p B\alpha\}, \qquad \widehat{Q} = \sum_i I_i \sum_{j=0}^{d} q_j (N_i)_j / n^j.$$

For any fixed $\eta > 0$, the constants can be chosen so that

$$\mathbb{E}\,\mathrm{Var}(\widehat{Q} \mid Y) \leq \eta \mathcal{M}^2. \tag{CC12}$$

A valid order of choices is: fix the profiles, pilot constants, c_b and C_L; choose c_L sufficiently small; determine the calibration constants ρ_-, ρ_+; choose C_0 sufficiently large; choose the fixed positive baseline prefactor d_ε; and choose D last. In particular, the proof permits the actual cutoff to be proportional to D^{-1} or arbitrarily smaller.

Proof. **3a. Introduce a comparison margin.** The actual cutoff a can be arbitrarily small near an integer, which makes direct normalization inconvenient. The baseline comparison replaces it by a controlled reference quantity without changing the margin by more than a constant factor. Write $A = P/P_* \geq 1$ and $\mathcal{M}_b = P_*(E_b + \kappa u A) + \kappa \mathcal{B}$. Equation (CC10) gives $\mathcal{M}_b \leq (1 + C_\varepsilon)\mathcal{M}$.

3b. Bound degrees below the real order. These degrees satisfy $j < \alpha$. Their sums $P_j = \sum_i p_i^j$ can be bounded by P through the power-mean inequality, and their coefficients contain positive powers of the small cutoff a. For $j \leq m$, put $\ell = m - j$, $\xi = \alpha - j$, and $\theta = j/\alpha \in (0, 1)$. Power means give $P_j \leq k^{1-j} A^\theta$. Lemma E.3 and (CC6) bound the degree-j conditional variance, divided by P_*^2, by

$$C m^{4\gamma} \frac{(Cm^2)^{2\ell}}{(\ell!)^4} (\rho_+/D)^{2\xi} \left(\frac{C}{DX}\right)^j k^{j/\alpha - 1} A^\theta. \tag{CC13}$$

For $0 < \theta < 1$,

$$\sup_{A>0} \frac{A^\theta}{(E_b + \kappa u A)^2} \leq E_b^{\theta-2} (\kappa u)^{-\theta}.$$

Also $X^{-j} u^{-\theta} \leq c_u^{-1}$ and $k^{j/\alpha - 1} \leq 1$. Retaining the sample multiplier exactly yields

$$D^{-2\xi - j + (\alpha-1)(2-\theta)} = D^{-2+j/\alpha} \leq D^{-1}. \tag{CC14}$$

All powers of the fixed ρ_-, ρ_+ and of κ^{-1} are $e^{O_\varepsilon(\alpha)}$; the baseline contributes $\chi_\alpha^{\theta-2} d_\varepsilon^{\theta-2} \leq 4 d_\varepsilon^{-2}$ and $C_0^{m(\theta-2)} \leq C_0^{-m}$. Finally,

$$\frac{m^{2\ell}}{(\ell!)^2} \leq e^{2\ell} \binom{m}{\ell}^2.$$

Indeed $(m)_\ell \ge (m/e)^\ell$, by the geometric mean of the largest ℓ numbers in $\{1,\dots,m\}$. Taking square roots in (CC13), summing by the L^2 triangle inequality, and using the binomial theorem costs only $e^{C\alpha}$. Since $m \ge \alpha/2$, choosing C_0 sufficiently large gives

$$\mathbb{E}\,\mathrm{Var}(\widehat{Q}_{\le m} \mid Y)/\mathcal{M}_b^2 \le C_{d_\varepsilon} D^{-1} e^{-c\alpha}. \tag{CC15}$$

The degree zero is conditionally deterministic.

3c. Bound degrees above the real order. For $j > \alpha$, the previous power-mean comparison is no longer the useful direction. We instead split expected counts into very small and larger means. The leading degree receives a sharp bound; the remaining degrees are controlled using the pilot and a geometric series. This split explains why the coefficient of degree $m+1$ was bounded separately in the preceding lemma. For the high block put $r = m+1$, $H = nb$, and $\tau = e^{-C_\tau L}$, with C_τ fixed sufficiently large. Define $A_> = P_*^{-1}\sum_{np_i>\tau} p_i^\alpha$. We claim

$$\frac{\mathbb{E}\,\mathrm{Var}(\widehat{Q}_{\ge r} \mid Y)}{P_*^2} \le \frac{e^{C\alpha}}{(DX)^\alpha} A + e^{CL}\left(\frac{c_b L}{D\alpha X}\right)^\alpha A_>. \tag{CC16}$$

For $\lambda = np \le \tau$, use $\mathbb{E}(N)_j^2 \le j!\lambda^j e^{2\sqrt{j\lambda}}$. The leading coefficient (CC7) is harmless because $(L/\delta)^{2\delta}\tau^\delta \le C$. For $j \ge r+1$, use $j! \le (r+1)! d^{j-r-1}$ and (CC8). The L^2 sum is a geometric series with ratio at most $\sqrt{d\lambda}/H$; increasing C_L makes it bounded. Its square is at most $e^{CL} r!\lambda^{r+1} \le r!\lambda^\alpha$ by the choice of C_τ. Thus the small-mean part is at most $e^{C\alpha} r!\lambda^\alpha$ in count units. Summing and using $r! \le e^{C\alpha}\alpha^\alpha$ proves the first term of (CC16).

For $\tau < \lambda \le H$ and $j \ge r > \alpha$,

$$H^{2\alpha-2j}\mathbb{E}(N)_j^2 \le H^\alpha \lambda^\alpha (1+d/H)^j.$$

All coefficient and summation costs are e^{CL}. For $\lambda > H \ge 2B\alpha$, the pilot probability is at most $e^{-c_p\lambda/8}$ and $\mathbb{E}(N)_j^2 \le \lambda^{2j} e^{j^2/\lambda}$. After division by $H^\alpha\lambda^\alpha$, the extra factor is

$$e^{-c_p\lambda/8 + j^2/\lambda}(\lambda/H)^{2j-\alpha} \le e^{CL},$$

because $j \le C_d L$ and the supremum of $C\log z - c'z$ on $z \ge 1$ is finite. This proves (CC16).

3d. Compare both bounds with the margin. A variance estimate is useful only after its ratio to $\mathcal{M}^2$ is small. The following calculation makes the dependence on the final sample multiplier D explicit. The constants in the construction are chosen before D; a negative power of D therefore genuinely permits us to make the noise small. To normalize its first term, use $\sup_{A>0} A/(E_b + \kappa u A)^2 \le (4E_b\kappa u)^{-1}$. Since $X^\alpha u \ge c_u$, its bound is a fixed multiple of

$$d_\varepsilon^{-1} D^{-1} C_0^{-m} \rho_-^{-(\alpha-1)} e^{C\alpha}.$$

For the second term, $\mathcal{B}/P_* \ge \tau A_>$; hence its bound is

$$d_\varepsilon^{-1} D^{-1} C_0^{-m} \rho_-^{-(\alpha-1)} e^{C\alpha + C'L}\left(\frac{c_b L}{\alpha X}\right)^\alpha.$$

Writing $s = \log X$, one has

$$\log(L/\alpha) \le \log(C_L+1) + \frac{c_L}{C_L+1} s.$$

Choose $c_L[C' + 1/(C_L+1)] \le 1/2$. The last bound becomes $d_\varepsilon^{-1} D^{-1} C_0^{-m} \rho_-^{-(\alpha-1)} e^{C''\alpha} X^{-\alpha/2}$. The stated choice of C_0 absorbs every fixed exponential in α, and D then makes the remaining fixed prefactors as small as required. Combining the two blocks and the comparison of $\mathcal{M}_b$ with $\mathcal{M}$ proves (CC12). □

Step 4: justify replacing the random linear aggregate. The total main-sample count has mean n and variance n. Thus the anchor introduced at the start of this section is centered and its variance can be computed exactly. This calculation prevents the anchor from becoming an unaccounted source of noise.

Lemma E.5 (The common affine anchor)**.** *Under the low coefficient bound,*

$$|q_1| \le C^m (ka)^{\alpha-1} P_*.$$

Consequently, replacing the random global affine aggregate by its known value changes the unanchored estimator by $q_1(1-\sum_i N_i/n)$, whose expectation is zero and whose variance is q_1^2/n. Relative to the actual margin (CC11), this is at most $C_{d_\varepsilon}\mathcal{M}^2/n$ after fixing C_0 sufficiently large. This estimate is independent of how small a becomes.

Proof. Set $j = 1$ in (CC6); $m^{2\gamma}(Cm^2)^{m-1}/((m-1)!)^2 \le C'^m$. Since $a^{\alpha-1} = (ka)^{\alpha-1}P_*$, the first assertion follows. The total count has law $\mathrm{Poi}(n)$, proving the variance identity. Finally $\mathcal{M} \ge \frac{1}{2} d_\varepsilon C_0^m (ka)^{\alpha-1} P_*$. □

What changes when α approaches one? The logarithmic target divides by $\beta = \alpha - 1$. A variance estimate that is merely $O(P^2)$ would therefore be insufficient. Subtracting the nearly linear part exposes coefficients of order β in all remaining terms. The next lemma states the resulting variance bound; Section F verifies its coefficient hypotheses for the actual polynomial and controls the entire hybrid estimator.

Lemma E.6 (The same polynomial near order one)**.** *Suppose $\alpha = 1+\beta$, $0 < \beta \le 1/2$, and set $t = b/N_0^2$, where N_0 is comparable to L. Assume the actual coupled polynomial has the coefficient cancellation*

$$kt \le C_\varepsilon, \qquad |q_1 - t^\beta| \le C_\varepsilon \beta t^\beta, \qquad |q_j| \le \beta e^{CL} b^{\alpha-j} \quad (j \ge 2). \tag{CC17}$$

With the same pilot and L, n, b as above, the conditional variance of the light residual after subtracting $t^\beta N_i/n$ is at most $\eta\beta^2 P^2$, for fixed sufficiently small c_L, sufficiently large D, and sufficiently large k. The same conclusion holds when the single anchor used at every order is q_1.

Proof. The linear residual contributes at most $C_\varepsilon \beta^2 t^{2\beta}/n \le C_\varepsilon \beta^2 P^2/n$. Every remaining degree satisfies $j \ge 2 > \alpha$, so the medium- and large-mean argument in (CC16) applies at every mean, without a small-mean split. It bounds the variance, divided by $\beta^2 P^2$, by

$$e^{CL}\left(\frac{c_b L}{D\alpha X}\right)^\alpha.$$

The same logarithmic estimate used in the proof of Theorem E.4 makes this uniformly small. The degree zero is conditionally deterministic. Changing the global anchor from t^β to q_1 adds the centered quantity $(q_1 - t^\beta)(1-\sum_i N_i/n)$, whose variance is at most $C_\varepsilon\beta^2P^2/n$ by (CC17). □

Scope of the bound. The coefficient assumptions above are preserved by the omega-tail modification and the common global anchor. The calibrated identity $am^2L^2/b = \lambda h z^{1/h} \le \lambda/4$ verifies (CC5), and the actual-cutoff comparison verifies (CC10). The section therefore supplies the conditional count-variance component of the common proof. The remaining pilot and heavy-count estimates are proved next; the mean margin, including both saturation intervals, is combined with these estimates in Theorem E.13.

E.2 Pilot errors and the analytic heavy count

There are two possible pilot errors. A *false-heavy* assignment uses the Gamma statistic at a symbol with a small true mean. A *false-light* assignment uses the polynomial outside its intended interval. Neither error can simply be ruled out symbol by symbol: a union bound would introduce an unnecessary dependence on k. Instead we bound their summed contribution in expectation, then use one probability bound for that aggregate.

False-heavy assignments. The analytic statistic can be signed at low counts. Consequently we bound its absolute first moment and its second moment, rather than assume positivity. The pilot upper tail provides enough powers of λ to offset even the nonzero value at count zero.

Lemma E.7 (False-heavy moments for the analytic count)**.** *For real $\alpha \geq 3/2$, define*

$$H_\alpha(j) = \frac{\Gamma(j+1)}{\Gamma(j+1-\alpha)}, \qquad j = 0, 1, \ldots,$$

using the entire reciprocal-Gamma value at integer orders. Given any fixed $c_* > 0$, fixed constants B, c_p can be chosen so that, for independent $N \sim \mathrm{Poi}(\lambda)$ and $Y \sim \mathrm{Poi}(c_p\lambda)$, with $J = \mathbf{1}\{Y > c_pB\alpha\}$, and every $0 \leq \lambda \leq B\alpha/2$,

$$\mathbb{E}[J|H_\alpha(N)|] \leq e^{-c_*\alpha}\lambda^\alpha \min(\lambda, 1), \tag{CC18}$$

$$\mathbb{E}[JH_\alpha(N)^2] \leq e^{-c_*\alpha}\alpha^\alpha\lambda^\alpha \min(\lambda, 1). \tag{CC19}$$

Proof. Write $\alpha = m + \gamma$. For $j \leq m$, reciprocal-Gamma recurrence and $\Gamma(1-\gamma) \geq 1$ give

$$|H_\alpha(j)| \leq j!(m-j)!/\Gamma(1-\gamma) \leq m! \leq \alpha^\alpha.$$

At integer orders the same bound follows from the factorial statistic. For $j \geq m+1$, $0 \leq H_\alpha(j) = G_\alpha(j) \leq j^\alpha$. For any $q \geq 1$, the bound $x^q \leq (q/(et))^q e^{tx}$, with $t = q/(\lambda+q) \leq 1$, and $e^t - 1 \leq t + t^2$, gives $\mathbb{E}N^q \leq e^{Cq}(\lambda+q)^q$. Therefore

$$\mathbb{E}|H_\alpha(N)| \leq e^{C\alpha}(\lambda+\alpha)^\alpha, \qquad \mathbb{E}H_\alpha(N)^2 \leq e^{C\alpha}(\lambda+\alpha)^{2\alpha}. \tag{CC20}$$

For $\alpha \leq \lambda \leq B\alpha/2$, the pilot upper tail is at most $\exp\{-c_pB\alpha(\log 2 - 1/2)\}$. After division by the power scales in (CC18)–(CC19), the other factors in (CC20) are bounded by $e^{C\alpha}(CB)^\alpha$. Increasing B, c_p proves the assertions on this range. For $\lambda < \alpha$, set $r = \lambda/\alpha < 1$. The pilot probability is at most $(er/B)^{c_pB\alpha}$, and both normalized moments are bounded by

$$e^{C\alpha}(e/B)^{c_pB\alpha} r^{(c_pB-1)\alpha}.$$

Choose $c_pB \geq 3$. The last power is at most $\min(\lambda, 1)$, and the preceding factor is at most $e^{-c_*\alpha}$ after the fixed constants are increased. At $\lambda = 0$, the pilot never selects the heavy branch. □

Corollary E.8 (Aggregated false-heavy error)**.** *For independent coordinates, restrict the wrongly selected heavy statistic to $np_i \leq B\alpha/2$. Its absolute expected value is at most $e^{-c_*\alpha}\mathcal{B}$, and its unconditional variance, including pilot randomness, is at most*

$$e^{-c_*\alpha}(\alpha/n)^\alpha P.$$

Under the calibration in Theorem E.4, this variance is an arbitrarily small fixed multiple of $\mathcal{M}^2$ after the stated choices of constants.

Proof. Sum (CC18) after multiplying by $n^{-\alpha}$; $\min(np_i, 1) \leq \min(Lnp_i, 1)$. Independence and second moments give the variance bound from (CC19). Its normalization is exactly the first high-degree term in (CC16), and uses the same baseline comparison and exact D^{-1} calculation. □

False-light assignments. A degree-d polynomial may grow rapidly for $p > b$. However, the probability that the pilot calls such a symbol light decays exponentially in np. Exponential decay dominates polynomial growth once the fixed pilot multiplier is large enough.

Lemma E.9 (False-light mean beyond the interval)**.** *Suppose the complete polynomial has degree $d \le C_d L$ and rescaled coefficient sum bounded by $e^{CL} b^\alpha$. Let $nb = c_b L \ge 2B\alpha$. For any prescribed fixed $c_* > 0$, choosing the fixed pilot multiplier c_p sufficiently large gives, for all $p \ge b$,*

$$\mathbb{P}\{\mathrm{Poi}(c_p np) \le c_p B\alpha\} |Q(p)| \le e^{-c_* L} p^\alpha. \tag{CC21}$$

Proof. Put $\lambda = np$ and $z = \lambda/(nb) \ge 1$. The polynomial bound and pilot lower tail give a ratio to p^α at most

$$e^{CL - c_p \lambda/8} \{z^{-\alpha} + z^{d-\alpha}\}.$$

Its logarithm is bounded by $C'L + C_d L \log z - c_p c_b L z/8$. The supremum over $z \ge 1$ tends to minus infinity, after division by L, as c_p increases. This proves the claim. □

The genuine heavy contribution. For a frequent symbol, the Gamma statistic behaves approximately like the αth power of its count. It is more efficient to control the ℓ_α norm of the entire count vector than to require an accurate estimate of every coordinate. The next proof uses the triangle inequality for that norm. This is also why it has no alphabet-sized union bound.

Lemma E.10 (One high-mean concentration bound)**.** *For every fixed $0 < t < 1/8$ and $0 < \rho < 1$, there is $B = B(t, \rho)$ such that, for every real $\alpha \ge 3/2$ and independent Poisson counts of means $\lambda_i \ge B\alpha$,*

$$\mathbb{P}\left\{ (1-2t)^\alpha \sum_i \lambda_i^\alpha \le \sum_i H_\alpha(N_i) \le (1+2t)^\alpha \sum_i \lambda_i^\alpha \right\} \ge 1 - \rho. \tag{CC22}$$

The bound also holds conditionally for any subset selected by independent pilot counts, provided all means in that subset satisfy the displayed lower bound.

Proof. Poisson Chernoff bounds imply

$$\mathbb{P}(|N - \lambda| \ge z) \le 2e^{-z^2/[2(\lambda+z)]} \le 2e^{-z^2/(4\lambda)} + 2e^{-z/4}.$$

Integration of $q \int_0^\infty z^{q-1} \mathbb{P}(|N - \lambda| \ge z)\, dz$ gives

$$(\mathbb{E}|N - \lambda|^q)^{1/q} \le C(\sqrt{q\lambda} + q), \qquad q \ge 2.$$

For $3/2 \le q < 2$, Jensen's inequality and the Poisson variance give the same estimate. Let $S = \sum_i \lambda_i^\alpha$ and $R = S^{1/\alpha}$. Since $\lambda_i \ge B\alpha$,

$$\mathbb{E} \sum_i |N_i - \lambda_i|^\alpha \le (C/\sqrt{B})^\alpha S.$$

Markov's inequality makes the ℓ_α norm of the count error at most $tR/2$, except with probability at most $\rho/3$, uniformly for $\alpha \ge 3/2$, once B is large.

Write $g_\alpha(j) = G_\alpha(j)^{1/\alpha}$. The pointwise inequalities $(j - \alpha)_+ \le g_\alpha(j) \le j$ imply

$$\|g_\alpha(N) - \lambda\|_\alpha \le \|N - \lambda\|_\alpha + \alpha s^{1/\alpha} \le (t/2 + 1/B)R \le tR,$$

where s is the number of coordinates. Thus the truncated Gamma sum lies in $[(1-t)^\alpha S, (1+t)^\alpha S]$ on this event. The analytic and truncated statistics differ only when $N_i \le m$; their absolute difference is at most α^α there. For $B \ge 4$, Poisson lower tails give

$$\mathbb{E}\sum_i |H_\alpha(N_i) - G_\alpha(N_i)| \le B^{-\alpha} e^{-cB\alpha} S.$$

The gap between $(1-t)^\alpha S$ and $(1-2t)^\alpha S$ is at least $t(1-2t)^\alpha S$, and the upper gap is larger. Increase B so that Markov's inequality puts this signed correction below the gap with probability at least $1-\rho/3$. This proves (CC22). Conditioning on an independent pilot only fixes the subset, so the same proof applies. □

The multiplicative factor ω deliberately places the heavy estimate inside the permissible interval. This bias is useful: it reserves room on both sides for random fluctuations. The following corollary checks that the room is uniform for $\alpha \ge 3/2$.

Corollary E.11 (The actual omega heavy arm)**.** *Let $E = 2\varepsilon(\alpha-1)$, $E/2 \le E_0 \le E$, $\omega = e^{-c_0 E_0}$ for a fixed $0 < c_0 < 1$, and $\kappa = e^{-E}$. The constants in Lemma E.10 can be chosen so that its omega-scaled statistic lies, with probability at least $1-\rho$, in*

$$[(\kappa + c_\varepsilon \kappa)S, (1 - c_\varepsilon \kappa)S].$$

Proof. Put $d_0 = \min(c_0/2, 1-c_0) > 0$. The logarithm $-c_0 E_0$ has distance at least $d_0 E$ from both endpoints $-E$ and 0. Since $\alpha/(\alpha-1) \le 3$, choose $t = t(\varepsilon, c_0) > 0$ so that the logarithmic error in (CC22) is at most $d_0 E/2$. The resulting bounds are $\kappa e^{d_0 E/2}$ and $e^{-d_0 E/2}$. Because $\alpha - 1 \ge 1/2$, both have the asserted margins after choosing a fixed positive c_ε. □

Connection to the deterministic bound. These lemmas control signed low-count leakage, count variance, and true-heavy concentration for the specified statistic. The complementary polynomial and aggregate bounds are established in Section D.4. Theorem E.13 combines those deterministic bounds with the stochastic estimates here.

The remaining deterministic boundary: $p < a$. The approximation guarantee on $[a, b]$ says nothing about symbols below a. There may be k such symbols. We therefore bound their total contribution by k times a uniform bound, then compare that quantity with the deterministic baseline already included in the estimator.

Lemma E.12 (The complete polynomial below its low cutoff)**.** *For the coupled polynomial (CC4), suppose $a \le b$, $0 \le J_1, J_\delta, v_\delta \le 1$ on $[0,1]$, and its amplitudes are bounded. Then*

$$|Q(p)| \le C^m a^\alpha \qquad (0 \le p \le a).$$

Consequently, the total absolute contribution of these coordinates is at most $C^m k a^\alpha$. For $\alpha \ge 3/2$, relative to the actual baseline $E_a P_*$ in Theorem E.4, its ratio is at most

$$\frac{2}{d_\varepsilon}(C/C_0)^m \, ka.$$

This tends uniformly to zero as the final multiplier D is increased, since $ka \le \rho_+/D$.

Proof. Substitute $p = ay$, $0 \le y \le 1$, in the reverse polynomial. Its rescaled coefficient sum is bounded by C^m, so $|Q_-(p)| \le C^m a^\alpha$. The cutoff multiplier $1 - J_1$ lies in $[0,1]$. The complete high term is bounded in absolute value by $Cp^\alpha \le Ca^\alpha$, since the high profiles and amplitudes are bounded. Summation and $ka^\alpha = (ka)^\alpha P_*$ prove the assertions. □

E.3 One calibrated estimator and its matching upper bound

The preceding lemmas now have one concrete use: they prove that the following hybrid power-sum estimate lies inside its prescribed interval. The statement includes the full formula so that the estimator can be read without reconstructing it from the technical lemmas. Its four components are a deterministic correction, the known affine total, the polynomial branch for a light pilot, and the Gamma branch for a heavy pilot. In this subsection we treat noninteger orders at least $3/2$; the near-one range is treated separately below.

Theorem E.13 (The common noninteger upper bound)**.** *Fix $0 < \varepsilon < 1/2$. For $k \ge k_\varepsilon$, uniformly over noninteger $3/2 \le \alpha < \log k$, the root X in Lemma D.4 satisfies*

$$S_\alpha(k, \varepsilon, 1/3) \le C_\varepsilon \alpha k^{1-1/\alpha} X.$$

Use the following construction in this range. Use the parameters of Lemma D.4, with $c_\omega = 1/8$, and the polynomial Q_0 of (D.50). Put

$$\chi = \min\{\alpha - 1, 1\}, \quad s = (1 + c_u u)^{-1}, \quad Q = sQ_0 = \sum_{j=0}^{d} q_j p^j, \quad \omega = se^{-E_0/8},$$

$$P_* = k^{1-\alpha}, \qquad E_a = \chi d_\varepsilon C_0^m (ka)^{\alpha-1}.$$

For independent $N_i \sim \mathrm{Poi}(np_i)$ and $Y_i \sim \mathrm{Poi}(c_p np_i)$ define

$$I_i = \mathbf{1}\{Y_i \le c_p B\alpha\}, \quad \widehat{Q}_i = \sum_{j=0}^{d} q_j (N_i)_j / n^j, \quad \widehat{G}_i = \omega n^{-\alpha} \mathsf{H}_\alpha(N_i), \quad \mathsf{H}_\alpha(j) = \frac{\Gamma(j+1)}{\Gamma(j+1-\alpha)}.$$

Use the entire reciprocal-Gamma value at its zeros, and set

$$\widehat{P} = 2E_a P_* + q_1 + \sum_{i=1}^{k} \left[I_i(\widehat{Q}_i - q_1 N_i/n) + (1 - I_i)(\widehat{G}_i - q_1 N_i/n) \right]. \tag{E.1}$$

All constants can be fixed in terms of ε alone so that

$$\Pr_p\{e^{-2\varepsilon(\alpha-1)} P_\alpha(p) \le \widehat{P} \le P_\alpha(p)\} \ge 3/4. \tag{E.2}$$

On a positive output return $-\log \widehat{P}/(\alpha - 1) - \varepsilon$, with an arbitrary output otherwise. A fixed sample pool of size $C_\varepsilon n$ implements the estimator with failure at most $1/3$.

Proof. Write $P = P_\alpha(p)$, $\kappa = e^{-E}$, and introduce the single margin

$$\mathcal{M} = E_a P_* + \kappa u P + \chi \kappa \mathcal{B}, \qquad \mathcal{B} = \sum_i p_i^\alpha \min(Lnp_i, 1). \tag{E.3}$$

The factor χ is necessary: both the logarithmic interval and the affine-anchored fluctuation vanish at order one.

Final assembly, Step 1: establish a deterministic interior. We first verify the deterministic margin and its parameter budget. Let C be the lower-exponent constant in Proposition D.9 and take $C_h \ge 2C + 2$. Since $u \le v + h \le 2\beta$, choose $c_u > 0$ sufficiently small in terms of ε. If $\Delta = v + C_h h$, then $\Delta - Ch \ge u$. If $\Delta = E/2$ and $E \ge 4Ch$, then $\Delta - Ch \ge E/4 = \varepsilon\beta/2 \ge \varepsilon u/4$. Thus, wherever $E \ge 4Ch$, the same reserve pays the lower uP margin and the common rescaling. The pointwise

bound, the summed baseline, and the bound below a consequently give strict margins $c_\varepsilon \mathcal{M}$ for any admissible light/heavy assignment. The baseline is chosen to leave at least $8E_aP_*$ before adding $2E_aP_*$. Such a choice is possible by the explicit summed bounds: first fix C_0, then d_ε sufficiently small, and finally make $ka \le C_\varepsilon/D$ small by increasing D.

For clarity, the allowed heavy aggregate in this assertion may replace $\omega P_{\mathcal{H}}$ by any value with lower margin $c_\varepsilon \kappa(u+\chi)P_{\mathcal{H}}$ above $\kappa P_{\mathcal{H}}$ and upper margin $c_\varepsilon(u+\chi)P_{\mathcal{H}}$ below $P_{\mathcal{H}}$. Here $u \le u_0$ and $\chi \le 1$, so a fixed heavy interior margin dominates the stated multiple of $u+\chi$. Its fixed upper deficit also pays its share of the baseline, since $\min\{h+Y+\min(Z,1),1\} \le 1$. The light coordinates have $p_i \le b$; coordinates below a contribute only the explicitly bounded baseline error. This is exactly the summation used in Corollary D.10.

The complementary inequality $E < 4Ch$ is handled by the other bound for the same polynomial, namely its localization toward ωp^α. For $\alpha \ge 3/2$, this inequality confines α and h to fixed compact ranges, and Lemma D.7 gives $kt \le C_\varepsilon/D$ and $u \ge c_\varepsilon$, with $t = b/N^2$. The localized error, with $\nu = \min\{\alpha, 2\}$, sums to at most $C_\varepsilon(kt)^\nu P$; below a its bound is $C^{m+1}(ka)^\alpha P_*$. The deterministic correction is at most $C_\varepsilon D^{-\beta} P$. All tend to zero uniformly as D increases. Both gaps $1-\omega$ and $\omega-\kappa$ have fixed positive lower bounds here, while $c_\varepsilon P \le \mathcal{M} \le C_\varepsilon P$. An aggregate heavy value within $\eta_\varepsilon P_{\mathcal{H}}$ of $\omega P_{\mathcal{H}}$ is therefore admissible, for a fixed sufficiently small $\eta_\varepsilon > 0$. These two bounds verify one deterministic bound; the polynomial, pilot, root, rescaling, and correction in (E.1) do not change between them.

Final assembly, Step 2: condition on the pilot and build an ideal comparison. The comparison statistic uses polynomial expectations on its light coordinates and a concentrated heavy aggregate on the others. It is chosen only for analysis; the estimator does not know p_i or any of the sets defined below. Choose the true-heavy threshold $B\alpha/2$ with B large. The count budget has $nb = c_b L \ge 2B\alpha$ and $na < B\alpha/2$; the latter follows directly from the capacity equation after fixing the degree constant. For a realization of the pilot define

$$\mathcal{L} = \{i : np_i < B\alpha/2\} \cup \{i : B\alpha/2 \le np_i \le nb,\ I_i = 1\}, \qquad \mathcal{H} = \mathcal{L}^c,$$

$$\mathcal{G} = \{i : I_i = 0,\ np_i \ge B\alpha/2\}, \qquad \mathcal{F} = \{i : p_i > b,\ I_i = 1\}.$$

Thus every light coordinate is at most b, and $\mathcal{H} = \mathcal{G} \dot\cup \mathcal{F}$. Conditional on the pilot, Lemma E.10 applies to $\mathcal{G}$. By choosing its relative-root tolerance sufficiently small, the aggregate

$$V_{\mathcal{H}} = \sum_{i\in\mathcal{G}} \widehat{G}_i + \omega P_{\mathcal{F}}$$

is admissible for the preceding deterministic bound except with a prescribed small fixed probability. Indeed $-\log\omega = E_0/8 + \log(1+c_u u)$ lies strictly between zero and E, with gaps proportional to β. The logarithmic fluctuation of the heavy norm is at most a fixed small multiple of α; $\alpha/\beta \le 3$ converts it into a small part of those gaps. When $E < 4Ch$, α is bounded, so the same fixed choice of tolerance also supplies the stronger absolute error $\eta_\varepsilon P_{\mathcal{H}}$. Increasing B enforces both requirements simultaneously, including for the empty set.

Final assembly, Step 3: bound the difference from the ideal comparison. On this event, the ideal mixed statistic

$$Z_0 = 2E_aP_* + \sum_{i\in\mathcal{L}} Q(p_i) + V_{\mathcal{H}}$$

lies in $[\kappa P + c\mathcal{M}, P - c\mathcal{M}]$, with a fixed $c = c_\varepsilon > 0$. We compare the actual statistic to Z_0. Its centered conditional polynomial fluctuation has unconditional variance $\mathbb{E}\,\mathrm{Var}(\widehat{Q} \mid Y)$ at most $\eta\mathcal{M}^2$ by Theorem E.4 and Lemma D.5. The factor $\chi \in [1/2, 1]$ only changes fixed constants in that

theorem. The affine replacement contributes $q_1(1-\sum_i N_i/n)$, with variance $q_1^2/n \le C_\varepsilon \mathcal{M}^2/n$ by Lemma E.5. These two terms are smaller than a fixed fraction of $c\mathcal{M}$ with arbitrarily small fixed failure probability, by Chebyshev's inequality and the stated choices of C_0, D, k_ε.

The remaining difference consists precisely of

$$\sum_{i\in\mathcal{F}}\{Q(p_i)-\omega p_i^\alpha\} - \sum_{\substack{np_i<B\alpha/2\\ I_i=0}} Q(p_i) + \sum_{\substack{np_i<B\alpha/2\\ I_i=0}} \widehat{G}_i.$$

Lemma E.9 bounds the expectation of the absolute polynomial part of the first term by $Ce^{-cL}\mathcal{B}$; the same pilot lower tail bounds $\omega P_{\mathcal{F}}$, since $Lnp_i \ge 1$ on $p_i > b$. For the second term with $p_i \ge a$, use $|Q(p_i)| \le Cp_i^\alpha$ and the pilot upper tail; its bound is $Ce^{-c\alpha}\mathcal{B}$. The portion below a is at most $C^{m+1}ka^\alpha$, which is an arbitrarily small part of E_aP_* by Lemma E.12 and the choice of D. The last, signed Gamma term has the same exponentially small expectation by Lemma E.7. Choose the pilot constants so that the exponents dominate $\kappa^{-1} = e^{2\varepsilon\beta}$. Markov's inequality then makes their total smaller than the remaining fixed fraction of $c\mathcal{M}$, with a prescribed small fixed failure probability. No union bound over the alphabet is used.

Final assembly, Step 4: combine the events and return to fixed samples. A union bound over these fixed probability budgets proves (E.2), with slack; for example each of the heavy event, the combined centered fluctuation, and the boundary errors can be allocated failure 1/32.

The hierarchy is consistent: fix the profiles and damping, their bias constants and C_h, then the pilot and degree constants; choose c_L sufficiently small; determine the cutoff comparison constants; choose C_0, then d_ε sufficiently small, and finally D and k_ε. The exact powers of D in (CC14) ensure that no earlier constant is required to depend on D. Every choice depends only on the original ε, and $E = 2\varepsilon\beta$ has never been replaced in the root.

Finally the conversion of (E.2) into entropy accuracy is immediate. Independent Poisson main and pilot samples are generated from a fixed pool of $\lceil 4(1+c_p)n\rceil$ iid observations by Poisson thinning; the pool overflow probability is exponentially small in n. The strict probability slack absorbs it, giving the claimed fixed-sample theorem. □

F The near-one upper bound

This section proves the upper bound for $1 < \alpha \le 3/2$, including the random pilot and signed Gamma arm. We use the regularized root and a degree-one low profile. Auxiliary constants are chosen for this range; they need not coincide with those used for orders at least $3/2$.

Why a separate calculation is necessary. When $\beta \downarrow 0$, the admissible interval has width

$$(1 - e^{-2\varepsilon\beta})P = 2\varepsilon\beta P + O(\beta^2 P).$$

Thus the required absolute error in the power sum vanishes with β. The calculation below shows that the estimator's error vanishes at the same rate. The central cancellation is elementary: at $\beta = 0$, the target $p^{1+\beta}$ equals p, whose total mass is known exactly. The technical work is to keep this cancellation through the polynomial, the Gamma statistic, and the random pilot.

Fix $0 < \varepsilon < 1/2$. For $1 < \alpha \le 3/2$, write $\beta = \alpha - 1$, $\delta = 1 - \beta$, $h = \beta\delta$, and $E = 2\varepsilon\beta$. Define

$$\Psi = \beta^{1+1/\delta}\delta^{1+1/\beta}, \qquad Q_*^2 = \frac{\Psi}{\alpha^2\beta^2} e^{(\log k)/\alpha - E/h},$$

$$X = \inf\left\{ x \ge 1 : \frac{\alpha h}{\beta} x^\alpha \left[1 + \log \frac{x(1+\log x)}{Q_*^2} \right]_+ \ge 1 \right\}.$$

Use the following parameterization, with fixed constants chosen below,

$$v_0 = \frac{\beta}{\alpha} X^{-\alpha}, \qquad E_0 = E - \min\{v_0 + C_h h, E/2\},$$
$$n = D\alpha k^{1-1/\alpha} X, \qquad L = \lceil \alpha(C_L + c_L \log X) \rceil, \qquad b = c_b L/n.$$

Let $z \in (0,1)$ solve $G_\beta(z) = e^{-E_0}$, where G_β is the exact envelope function in Lemma D.1, and set

$$a = \lambda \frac{bhz^{1/h}}{L^2}, \qquad A = \frac{1 - z^{1/\beta}}{1 - z^{1/h}}, \qquad B = \frac{1 - z^{1/\delta}}{1 - z^{1/h}},$$

$$A' = Ae^{-M\beta}, \qquad B' = Be^{-M\delta}, \qquad \omega_0 = e^{-c_0 E_0}, \quad 0 < c_0 < 1.$$

Also use the common final rescaling

$$u_0^* = \min\{v_0 + h, u_0\}, \qquad s_0 = (1 + c_u u_0^*)^{-1}, \qquad \omega = s_0\omega_0.$$

Here $u_0 > 0$ is fixed and $c_u > 0$ is chosen sufficiently small in terms of ε, c_0. The numbers $C_h, c_b, \lambda, M, c_0$ are fixed parameters chosen for this construction. Let the integer kernel scale N satisfy $N \asymp L$, with fixed comparison constants, and put $t = b/N^2$.

The common degree-one profile. Fix the same floor $q_L \in (0,1)$ as in the common low construction. For $0 < \beta \le 1/2$, choose the unique $c_\beta \in (\beta/(1+\beta), 1)$ satisfying

$$(1 - c_\beta)c_\beta^\beta (1+\beta)^{1+\beta}/\beta^\beta = q_L.$$

The function on the left decreases strictly from one to zero on this interval. Define

$$A_\beta^{\mathrm{low}} = \frac{\beta^\beta}{c_\beta^\beta (1+\beta)^{1+\beta}}, \qquad u_\beta(x) = \frac{x^\beta(1 - c_\beta x)}{A_\beta^{\mathrm{low}}}, \qquad \tau = \frac{c_\beta(1+\beta)a}{\beta}.$$

The maximum of $x^\beta(1-c_\beta x)$ occurs at $\beta/[c_\beta(1+\beta)] < 1$, so the normalization is exact and $u_\beta(1) = q_L$. Since $1/4 \le A_\beta^{\mathrm{low}} \le 1$,

$$1 - q_L \le c_\beta \le 1 - q_L/4.$$

Thus, as a polynomial identity, including its extension to $p = 0$,

$$p^\alpha u_\beta(a/p) = T_\tau(p), \qquad T_\tau(p) = (1+\beta)\tau^\beta p - \beta\tau^{1+\beta}. \tag{F.1}$$

No differentiation of c_β is needed below.

For clarity, the high profiles used here are the same localized profiles as in the common construction. Write

$$R_N(x) = \Big\{U_{N-1}(\sqrt{1-x})/N\Big\}^6, \quad K_N = R_N^2, \quad W = f + (1-f)(1-R_N)^2, \qquad 0 < f < 1,$$

$$I_s = \int_0^1 x^{s-1}K_N(x)\,dx, \qquad J_s(x) = I_s^{-1}\int_0^x u^{s-1}K_N(u)\,du,$$

$$A_{s,\omega} = \sup_{0\le x<1} \frac{x^s K_N(x)}{1-\omega J_s(x)(1-K_N(x))W(x)}, \quad V_{s,\omega}(x) = \frac{x^s K_N(x)}{A_{s,\omega}} + \omega J_s(x)(1-K_N(x))W(x).$$

Use the continuous extension of the quotient at a zero of K_N. The coupled polynomial is

$$Q(p) = s_0\Big[A'T_\tau(p)\{1 - J_1(p/b)\} + \omega_0 p^\alpha J_\delta(p/b) + B'p^\alpha\{V_{\delta,\omega_0}(p/b) - \omega_0 J_\delta(p/b)\}\Big]. \tag{F.2}$$

It is a polynomial because $\alpha + \delta = 2$. Write $Q(p) = \sum_{j=0}^d q_j p^j$, with $d \le C_d L$.

The parameter $t = b/N^2$ is the effective endpoint scale of the polynomial kernel. In this range, the root simplifies enough to show that n has order $k/\log k$ and that kt can be made small by the sample multiplier D. These two facts will control aggregate bias.

Lemma F.1 (Near-one parameter estimates)**.** *Uniformly over $1 < \alpha \le 3/2$, for sufficiently large k,*

$$X \asymp_\varepsilon k^{1/\alpha}/\log k, \qquad n \asymp_\varepsilon Dk/\log k, \qquad L \asymp c_L \log k, \qquad kt \le C_\varepsilon/D.$$

The ratio τ/t lies in a fixed positive compact interval independent of D, and

$$A' = 1 + O_\varepsilon(\beta), \quad B' = O_\varepsilon(\beta), \quad \omega = 1 + O_\varepsilon(\beta).$$

Constants may depend on the fixed profile and degree constants.

Proof. Direct cancellation gives

$$Q_*^2/k^{1/\alpha} = \frac{\beta^{\beta/\delta}\delta^{1+1/\beta}}{\alpha^2}e^{-2\varepsilon/\delta} \asymp_\varepsilon 1.$$

The logarithm is uniformly bounded, using $\beta|\log\beta| \le 1/e$ and $\log(1-\beta) \ge -\beta/(1-\beta)$. Let $y \ge 1$ solve $y(1+\log y) = Q_*^2/e$. Uniformly, $y \asymp_\varepsilon k^{1/\alpha}/\log k$. The root expression vanishes at y, and at $2y$ is at least $(3/4)(2y)^\alpha \log 2 > 1$ for sufficiently large k. Thus $y \le X \le 2y$, proving the sample and degree comparisons. The bound on kt follows from $t \asymp 1/(nL)$.

Write $z = e^{-ht_*}$. Since $E_0 \in [\varepsilon\beta, 2\varepsilon\beta]$,

$$\varepsilon \le E_0/h \le 4\varepsilon.$$

The overlap inequalities, with $J(x) = x\coth x$, $2[J(t_*/4) - 1] \le F_\beta(t_*)/h \le t_*$ therefore put t_* in a fixed positive compact interval. Consequently

$$\frac{\tau}{t} = c_\beta\lambda\frac{N^2}{L^2}\alpha\delta z^{1/h}$$

is bounded above and below by positive constants, independently of D. Also $B = (1-e^{-\beta t_*})/(1-e^{-t_*}) = O_\varepsilon(\beta)$ and $1 - A = e^{-\delta t_*}B = O_\varepsilon(\beta)$. The attenuations preserve the asserted bounds. Since $u_0^* \le v_0 + h \le 2\beta$, both s_0 and ω equal $1 + O_\varepsilon(\beta)$. □

The decisive near-one approximation estimates. The first inequality below controls bias after summing over symbols. The next two identify the constant and nearly linear coefficients. The last shows that every coefficient of degree at least two contains a factor β. This factor is what ultimately yields variance of order $\beta^2 P^2$.

Lemma F.2 (Localized bias and affine cancellation)**.** *For the polynomial* (F.2)*, uniformly in the near-one range,*

$$\sup_{0\le p\le b} |Q(p) - \omega p^\alpha| \le C_\varepsilon \beta t^\alpha, \tag{F.3}$$

$$|q_0| \le C_\varepsilon \beta t^\alpha, \qquad |q_1 - t^\beta| \le C_\varepsilon \beta t^\beta, \tag{F.4}$$

$$|q_j| \le C_\varepsilon \beta e^{CL} b^{\alpha-j}, \qquad 2 \le j \le d. \tag{F.5}$$

Here C is independent of D.

Proof. The sine identity for U_{N-1} gives $0 \le K_N(x) \le \min\{1, (N^2x)^{-6}\}$ and a positive local lower bound. In the scaled variable $Y = N^2x$, for $1/2 \le s \le 1$ the normalizing integral and its first absolute logarithmic moment are bounded above and below by constants. Thus, with $F_s(Y) = J_s(Y/N^2)$,

$$1 - F_1(Y) \le C\min(1, Y^{-5}), \qquad |F_{1-\beta}(Y) - F_1(Y)| \le C\beta \begin{cases} 1, & Y \le 1, \\ Y^{-5}(1+\log Y), & Y \ge 1. \end{cases}$$

The second estimate follows by differentiating the normalized density in s; for $Y \ge 1$, differentiate its tail integral. The ordinary tangent inequality and the mean value theorem now give

$$\sup_{p\le b} |T_t(p)(1 - J_1(p/b)) + p^\alpha J_\delta(p/b) - p^\alpha| \le C\beta t^\alpha.$$

Indeed, after scaling by t, write the error as

$$\{(1+\beta)Y - \beta - Y^{1+\beta}\}(1 - F_1(Y)) + Y^{1+\beta}(F_{1-\beta}(Y) - F_1(Y)).$$

Each term is uniformly $O(\beta)$ by the preceding tail bounds. Replacing t by τ in the tangent changes this error by $O_\varepsilon(\beta t^\alpha)$ because τ/t is in a fixed positive compact interval and $p(1 - J_1(p/b)) \le Ct$.

The high normalization lies between the ordinary sharp normalization and its tail-one counterpart. For $1/2 \le \delta \le 1$ it follows that $A^{-1}_{\delta,\omega_0} \le CN^{2\delta}$. Its exact defining expression then gives

$$|V_{\delta,\omega_0}(x) - \omega_0 J_\delta(x)| \le C\min\{Y^\delta, Y^{-3}\}, \qquad Y = N^2x.$$

To check the large-Y bound, expand the difference into $x^\delta K_N/A_{\delta,\omega_0} - \omega_0 J_\delta K_N - (1-f)\omega_0 J_\delta(1 - K_N)(2R_N - R_N^2)$. Consequently its product with p^α is bounded by Ct^α. Now subtract ω times the tangent-cutoff expression from (F.2). The remaining coefficients $s_0A' - \omega$ and s_0B' are $O_\varepsilon(\beta)$. This proves (F.3).

The constant and linear coefficients are exactly

$$q_0 = -s_0A'\beta\tau^\alpha, \qquad q_1 = s_0A'\left\{(1+\beta)\tau^\beta + \beta\tau^\alpha \frac{J_1'(0)}{b}\right\}.$$

Since $J_1'(0)/b = 1/(bI_1) \asymp 1/t$, this proves (F.4). For (F.5), rescale by $p = bx$. The kernel coefficients have sum at most C^N by the Chebyshev recurrence. The integrated coefficients are $[x^j]K_N/((j+s)I_s)$; for $1/2 \le s \le 1$, their derivatives in s have sum at most $C^N(1+\log N)$. At $\beta = 0$, the two undamped expressions cancel coefficientwise as $x(1 - J_1) + xJ_1 = x$. The factors $s_0A' - 1$, $\omega - 1$, and s_0B' are $O_\varepsilon(\beta)$; also $\log(\tau/b) = O_\varepsilon(1 + \log N)$. These observations give the stated factor β for every degree at least two. The high term needs no derivative of its maximum normalization because its prefactor B' already contains that factor. All powers and logarithms of N are absorbed into e^{CL}. □

Assemble the near-one estimator. The statistic is the same polynomial/Gamma hybrid as before, with the same global anchor. Its deterministic correction is denoted r_k here to make clear exactly how much correction the proof can absorb. The estimate need not be positive on every outcome; it is positive on the success event, which is all that is needed for the guarantee.

Theorem F.3 (Near-one estimation theorem)**.** *There are fixed choices of the pilot constants B_0, c_p, the degree constants C_L, c_L, and the multiplier D, and a k_ε, such that the following holds for every $k \geq k_\varepsilon$ and every $1 < \alpha \leq 3/2$. Let independently*

$$N_i \sim \mathrm{Poi}(np_i), \qquad Y_i \sim \mathrm{Poi}(c_p np_i), \qquad I_i = \mathbf{1}\{Y_i \leq c_p B_0 \alpha\}.$$

Put

$$\widehat{Q}_i = \sum_{j=0}^{d} q_j (N_i)_j / n^j, \qquad \widehat{G}_i = \omega n^{-\alpha} \mathsf{H}_\alpha(N_i),$$

and use the single global affine convention

$$\widehat{P} = q_1 + r_k + \sum_i \left\{ I_i(\widehat{Q}_i - q_1 N_i/n) + (1 - I_i)(\widehat{G}_i - q_1 N_i/n) \right\}. \tag{F.6}$$

Here the same deterministic repair used by the common construction is allowed whenever $|r_k| \leq \eta_0 \beta k^{-\beta}$, for a sufficiently small fixed $\eta_0 > 0$. In particular this includes

$$r_k = 2\beta d_\varepsilon C_0 (ka)^\beta k^{-\beta},$$

after choosing d_ε sufficiently small, since $ka \leq 1$ for sufficiently large D. A further term bounded by Cka^α may also be included. Then, uniformly over $p \in \Delta_k$,

$$\mathrm{Prob}\{e^{-2\varepsilon\beta} P_\alpha(p) \leq \widehat{P} \leq P_\alpha(p)\} \geq 3/4.$$

Thus $-\beta^{-1} \log \widehat{P} - \varepsilon$, with any definition when $\widehat{P} \leq 0$, estimates H_α within ε. The sample complexity is $O_\varepsilon(k/\log k)$, equivalently $O_\varepsilon(\alpha k^{1-1/\alpha} X)$ uniformly on this range.

Proof. Write $P = \sum_i p_i^\alpha \geq k^{-\beta}$ and $K = \log k$. All constants below are uniform over $0 < \beta \leq 1/2$. The bounds $E_0 \in [\varepsilon\beta, 2\varepsilon\beta]$ and $u_0^* \leq 2\beta$ imply, for example when $c_u \leq (1 - c_0)\varepsilon/4$,

$$\min\{1 - \omega, \omega - e^{-2\varepsilon\beta}\} \geq c_\varepsilon \beta. \tag{F.7}$$

Near-one proof, Step 1: control the mean. We first center the analysis at ωP. The previous display says this center is separated from both endpoints by a constant multiple of βP. First use the anchor t^β in place of q_1 and omit r_k; call this statistic $\widehat{P}_t$. The replacement will be controlled at the end. Let $s(\lambda)$ denote the probability of the light pilot. The analytic Gamma mean identity gives

$$|\mathbb{E}\mathsf{H}_\alpha(\mathrm{Poi}(\lambda)) - \lambda^\alpha| \leq C\beta e^{-\lambda}$$

including $\lambda = 0$, where the value is $1/\Gamma(-\beta)$. For every prescribed $\eta > 0$, fixed sufficiently large B_0, c_p ensure

$$(1 - s(\lambda))e^{-\lambda} \leq \eta \lambda^\alpha \quad (\lambda \geq 0).$$

For $\lambda \geq B_0/2$ use exponential decay; for $1 \leq \lambda < B_0/2$ use the pilot upper tail; for $0 < \lambda < 1$ use $(1 - s(\lambda)) \leq (e\lambda/(B_0\alpha))^{c_p B_0 \alpha}$ with $c_p B_0 \geq 3$. Thus the aggregate Gamma mean error is at most $C\eta\beta P$.

On $p_i \le b$, Lemma F.2 bounds the polynomial mean error by $C_\varepsilon \beta k t^\alpha$. For $p > b$, the affine coefficient bounds imply

$$|Q(p) - t^\beta p| \le C_\varepsilon \beta e^{CL} b^\alpha \{1 + (p/b)^d\}.$$

Choose C_L so $nb = c_b L \ge 2B_0\alpha$. For $\lambda = np \ge nb$, the pilot lower tail gives $s(\lambda) \le e^{-c_p\lambda/8}$. Writing $w = p/b \ge 1$, the exponent controlling the polynomial tail is $CL + C_d L \log w - c_p c_b L w/8$. Increasing c_p makes its supremum arbitrarily negative, uniformly in $L \ge 1$. The remaining affine difference satisfies

$$|t^\beta p - \omega p^\alpha|/p^\alpha \le C_\varepsilon \beta + \beta \log(p/t),$$

and the same pilot exponential absorbs $\log(p/t) = \log w + 2\log N$. Therefore the false-light mean beyond b is at most $C\eta\beta P$. We have proved

$$|\mathbb{E}\widehat{P}_t - \omega P| \le \beta P\{C_\varepsilon (kt)^\alpha + C\eta\}. \tag{F.8}$$

Near-one proof, Step 2: control all randomness. We next control the entire variance, including the random pilot. The discrete Poisson Poincaré inequality used below is $\operatorname{Var} f(Z) \le \lambda \mathbb{E}(f(Z+1) - f(Z))^2$ for $Z \sim \mathrm{Poi}(\lambda)$. Thus a factor β in each discrete increment becomes a factor β^2 in variance. Put $\xi = nt$. The analytic recurrence and the order-one increment bound give, for all integers $j \ge 0$,

$$|\Delta\{\omega \mathsf{H}_\alpha(j) - \xi^\beta j\}| \le C_\varepsilon \beta\{(j+1)^\beta + \xi^\beta\}\{1 + |\log((j+1)/\xi)|\}.$$

For $Z \sim \mathrm{Poi}(\lambda)$ and $V = (Z+1)/(\lambda+1)$, $\mathbb{E}V^q + \mathbb{E}[V^q(\log V)^2] \le C$ uniformly in $\lambda \ge 0$ and $0 \le q \le 1$. For $\lambda \ge 1$ use the usual Poisson lower tail and fixed moments; for $\lambda \le 1$, $V \ge 1/2$ and its fixed moments are bounded. Poisson Poincaré consequently gives

$$\operatorname{Var}\{\omega \mathsf{H}_\alpha(Z) - \xi^\beta Z\} \le C_\varepsilon \beta^2 \lambda\{(\lambda+1)^{2\beta} + \xi^{2\beta}\}\{1 + \log^2((\lambda+1)/\xi)\}.$$

Since $\xi \asymp 1/L$, $\lambda \le n$, and $2\beta \le 1$, summing in probability units yields

$$\sum_i \operatorname{Var}(\widehat{G}_i - t^\beta N_i/n) \le \frac{C_\varepsilon \beta^2 K^2}{n}\{P_{2\alpha-1} + n^{-2\beta} + t^{2\beta}\}. \tag{F.9}$$

The inequality $P_{2\alpha-1} \le P^{2-1/\alpha}$ implies that the right side divided by $\beta^2 P^2$ is at most

$$\frac{C_\varepsilon K^2}{n}\{k^{\beta/\alpha} + (k/n)^{2\beta} + (kt)^{2\beta}\} = o(1),$$

uniformly, because $\beta/\alpha \le 1/3$ and $n \asymp k/K$.

Near-one proof, Step 3: account for switching branches. The preceding estimate concerns the Gamma residual. The actual statistic replaces it by the polynomial residual whenever the pilot selects light. The following two second-moment bounds control that replacement directly, including the selector's randomness. For the pilot correction, the same affine coefficient bounds and factorial second-moment identity prove

$$\mathbb{E}[I_i(\widehat{Q}_i - t^\beta N_i/n)^2] \le C_\varepsilon \beta^2 e^{CL} b^{2\alpha}. \tag{F.10}$$

Indeed, if $\lambda \le H = nb$, then $\mathbb{E}(Z)_j^2 \le \lambda^j(\lambda+j)^j \le H^{2j}(1+d/H)^j$. For $\lambda > H$, multiply $\mathbb{E}(Z)_j^2 \le \lambda^{2j} e^{j^2/\lambda}$ by $s(\lambda) \le e^{-c_p\lambda/8}$. With $\lambda = Hw$ and $j \le C_d L$, the remaining factor has

logarithm at most $L\{C + C\log w - cw\}$ and hence is $e^{O(L)}$. The L^2 triangle inequality across the degrees proves (F.10).

A similar, simpler bound holds for the signed Gamma residual under the light pilot. Reciprocal-Gamma differentiation on $j \geq 1$ and the value at $j = 0$ give

$$|\omega \mathsf{H}_\alpha(j) - \xi^\beta j| \leq C_\varepsilon \beta (j+1)^\alpha \{1 + |\log \xi| + \log(j+1)\}.$$

Fixed Poisson moments and the exponential tail of $s(\lambda)$ then imply

$$\mathbb{E}[I_i(\widehat{G}_i - t^\beta N_i/n)^2] \leq C_\varepsilon \beta^2 n^{-2\alpha}(1 + \log L)^2. \tag{F.11}$$

This estimate includes all signed low-count values.

Now write the statistic exactly as

$$\widehat{P}_t = t^\beta + \sum_i (\widehat{G}_i - t^\beta N_i/n) + \sum_i I_i(\widehat{Q}_i - \widehat{G}_i).$$

Independence across coordinates, the variance triangle inequality, and (F.9)–(F.11) control its full variance, without treating the pilot as deterministic. The additional terms divided by $\beta^2 P^2$ are bounded by

$$C_\varepsilon e^{CL} K^6/k + C_\varepsilon K^3 (1 + \log K)^2/k = o(1)$$

after choosing c_L sufficiently small. Thus $\operatorname{Var} \widehat{P}_t = o(\beta^2 P^2)$ uniformly.

Near-one proof, Step 4: restore the prescribed estimator. Finally, the global q_1 convention changes $\widehat{P}_t$ by

$$(q_1 - t^\beta)\left(1 - \sum_i N_i/n\right).$$

Its mean is zero and its variance is at most $C_\varepsilon \beta^2 t^{2\beta}/n = o(\beta^2 P^2)$. The deterministic repair contributes at most $\eta_0 \beta P$. Any additional $O(ka^\alpha)$ term has ratio to P at most $C_\varepsilon \beta (kt)^\alpha$, because $a/t = O_\varepsilon(\beta)$. Choose η, η_0 small, then D sufficiently large in (F.8), and finally k_ε sufficiently large for the uniform variance bound. Chebyshev's inequality and (F.7) prove the asserted probability interval with strict slack in the failure budget. For a fixed-sample implementation, draw independently $M_0 \sim \mathrm{Poi}((1 + c_p)n)$ and allocate each of the first M_0 observations to the main or pilot sample with probabilities $1/(1 + c_p)$ and $c_p/(1 + c_p)$. Use a pool of $\lceil 4(1 + c_p)n \rceil$ iid observations; if M_0 exceeds the pool, return an arbitrary answer. The ideal counts have exactly the independent laws used above, and the overflow probability is exponentially small in n. The initial strict slack absorbs this additional failure. □

G Lower bounds: two random distributions that the data cannot distinguish

An upper bound supplies an estimator. A lower bound must rule out *every* estimator. We do this by constructing two random choices of an unknown distribution. Their entropies are separated by more than 2ε, but their sample laws are close in total variation. An estimator accurate to ε would then distinguish two statistically indistinguishable hypotheses, a contradiction.

The proof has four stages. First, construct two nonnegative random weights U_0, U_1 with mean one. Their low integer moments agree, whereas their αth moments differ. Second, draw many such weights and normalize them to obtain actual probability vectors. Third, show that normalization preserves the entropy separation. Finally, bound the statistical distance between the resulting count data. The parameter calculation at the end chooses the weights so that this argument works at the same sample scale as the upper bound.

Notation for this section. The subscript $i \in \{0, 1\}$ denotes a hypothesis, not a symbol in the alphabet. The random variable U_i is one unnormalized weight; $A_i = \mathbb{E}U_i^\alpha$ is its power moment. The letter B bounds these weights. Later, $b = k/2$ is the expected number of weights and θ is the sampling intensity per unit weight. Thus θb, not θ, is the expected total count before normalization. A signed measure $\lambda = \lambda^+ - \lambda^-$ is written as the difference of its positive and negative parts; their total masses need not be equal. The notation δ_a means unit point mass at a.

G.1 Why normalization must be analyzed explicitly

The weights do not initially sum to one. A tempting argument would replace their sum by its expectation and ignore the error. That loses uniformity when α approaches one. The following identity keeps exactly the cancellation that is needed in that limit.

Lemma G.1 (Normalization with the affine term removed)**.** *Let $\alpha > 1$, $\beta = \alpha - 1$, and let $U \geq 0$ satisfy $\mathbb{E}U = 1$ and $A = \mathbb{E}U^\alpha < \infty$. Consider either of the following experiments:*

1. *$b = N$ is a positive integer and $U_1, \dots, U_N$ are independent copies of U;*
2. *$b > 0$, $Q \sim \mathrm{Poi}(b)$, and $U_1, \dots, U_Q$ are independent copies of U, independent of Q.*

In the first experiment sums below have N terms, and in the second they have Q terms. Define

$$S = \frac{1}{b}\sum_j U_j, \qquad V = \frac{1}{b}\sum_j U_j^\alpha, \qquad W = V - AS.$$

On $S > 0$, put $p_j = U_j/(bS)$. Then

$$\beta\left\{H_\alpha(p) - \left(\log b - \frac{\log A}{\beta}\right)\right\} = \beta \log S - \log\left(1 + \frac{W}{AS}\right). \tag{G.1}$$

For $0 < m \leq 1$, set

$$t = \min\left\{\frac{1}{4}, \frac{m}{16\beta}\right\}, \qquad u = \frac{m}{16}.$$

On $|S - 1| \leq t$ and $|W| \leq Au$, the absolute entropy error in (G.1) is less than $m/(4\beta)$. If the second moments below are finite, the failure probability is at most

$$\frac{v_1}{bt^2} + \frac{v_\alpha}{bA^2u^2}, \tag{G.2}$$

where, in the independent experiment,

$$v_1 = \mathrm{Var}(U), \qquad v_\alpha = \mathbb{E}(U^\alpha - AU)^2,$$

and, in the compound-Poisson experiment,

$$v_1 = \mathbb{E}U^2, \qquad v_\alpha = \mathbb{E}(U^\alpha - AU)^2.$$

Proof. **Step 1: isolate the normalization error.** The actual probability is $p_j = U_j/(bS)$, so

$$\sum_j p_j^\alpha = \frac{\sum_j U_j^\alpha}{b^\alpha S^\alpha} = b^{1-\alpha}\frac{V}{S^\alpha}.$$

If $S = 1$ and $V = A$, its entropy would be $\log b - (\log A)/\beta$. The difference from this deterministic center is the quantity we must control. Substitute $V = AS + W$ and take logarithms to obtain (G.1). On the stated event $S \geq 3/4$. Hence

$$|\beta \log S| \leq \frac{4}{3}\beta t \leq \frac{m}{12}, \qquad \left|\frac{W}{AS}\right| \leq \frac{m}{12}.$$

The inequality $|\log(1+x)| \leq |x|/(1-|x|)$ for $|x| < 1$ gives

$$\left|\log\left(1 + \frac{W}{AS}\right)\right| \leq \frac{m/12}{1 - m/12} \leq \frac{m}{11}.$$

Since $1/12 + 1/11 < 1/4$, the asserted entropy error follows.

Step 2: control the two random errors separately. In both experiments $\mathbb{E}S = 1$ and $\mathbb{E}W = 0$, because $\mathbb{E}(U^\alpha - AU) = 0$. Independence gives $\mathrm{Var}(S) = \mathrm{Var}(U)/b$ in the first experiment. Conditioning on Q gives $\mathrm{Var}(S) = \mathbb{E}U^2/b$ in the second. Similarly, in either experiment, $\mathrm{Var}(W) = \mathbb{E}(U^\alpha - AU)^2/b$. Chebyshev's inequality and a union bound prove (G.2). □

Remark G.2 (The order-one limit is already included). Define $g_\beta(u) = (u^{1+\beta} - u)/\beta$, with $g_\beta(0) = 0$. There is the exact cancellation

$$U^\alpha - AU = \beta\{g_\beta(U) - (\mathbb{E}g_\beta(U))U\}. \tag{G.3}$$

When the separation margin m is proportional to β, the factor β^2 in v_α cancels the factor m^2 in (G.2). Thus the lemma does not lose a factor β^{-2} as $\alpha \downarrow 1$. For example, if $U \in \{0\} \cup [1, B]$, $0 < \beta \leq 1/2$, and A is bounded, then $|g_\beta(U)| \leq U^{1+\beta}\log B$ and $\mathbb{E}g_\beta(U) \leq A\log B$. Consequently

$$\mathbb{E}\{g_\beta(U) - (\mathbb{E}g_\beta(U))U\}^2 \leq C_A B^3(1 + \log B)^2.$$

This is a direct substitution into one variance formula, not a second normalization argument.

G.2 Constructing moment-matched weights by interpolation

A signed measure that integrates every low-degree monomial to zero encodes moment matching: its positive and negative parts have identical moments. Lagrange interpolation constructs such a measure explicitly. The small perturbation ρ below is deliberate. It permits one integer moment to differ by a controlled amount, which is needed close to an integer order.

For distinct nodes $u_0, \ldots, u_n$, the Lagrange basis is

$$\ell_j(u) = \prod_{i \neq j} \frac{u - u_i}{u_j - u_i}.$$

Every polynomial P of degree at most n satisfies $P(a) = \sum_j \ell_j(a)P(u_j)$. We evaluate this identity at a point a below every node, rather than inside the node interval.

Lemma G.3 (Perturbed interpolation factor)**.** *Let $u_0, \dots, u_n > 0$ be distinct, let $0 < a < \min_j u_j$, and let ℓ_j be their Lagrange basis polynomials. For $\rho \geq 0$, define*

$$\nu_\rho = (1+\rho)\delta_a - \sum_{j=0}^{n} \ell_j(a)(1+\rho a/u_j)\delta_{u_j}, \qquad C_a = \prod_{j=0}^{n}(1-a/u_j).$$

Then

$$\int u^j \, d\nu_\rho = 0 \qquad (1 \leq j \leq n), \tag{G.4}$$

$$\int d\nu_\rho = \rho C_a, \qquad \int u^{-1} \, d\nu_0 = a^{-1} C_a. \tag{G.5}$$

In particular, ν_0 annihilates every polynomial of degree at most n. If $e \in \mathbb{R}$ and P is the positive Jordan mass of $u^{-e}\nu_\rho$, the measure $\lambda = u^{-e}\nu_\rho/P$ has positive mass one and satisfies

$$\int u^{e+j} \, d\lambda = 0 \quad (1 \leq j \leq n), \qquad \int u^e \, d\lambda = \rho C_a / P.$$

Proof. For $1 \leq j \leq n$, interpolation of u^j and u^{j-1} gives

$$(1+\rho)a^j - \sum_i \ell_i(a) u_i^j - \rho a \sum_i \ell_i(a) u_i^{j-1} = 0.$$

If R interpolates $u \mapsto 1/u$ at the nodes, the polynomial $1 - uR(u)$ has the same zeros and constant term as $\prod_i (1 - u/u_i)$ and has degree at most $n+1$. They therefore agree. Evaluating at a proves both remaining identities. Weighting by the strictly positive function u^{-e} and normalizing proves the last assertions. The argument includes $n = 0$, with the first cancellation list empty. □

Why use two interpolation factors? Write a noninteger order as an integer plus a fractional part. One factor will cancel integer moments below that order; the other will cancel integer moments above it. A tensor product combines the two cancellation lists. We separate its positive and negative terms into two *positive* measures, so the final construction gives probability laws rather than formal signed objects.

Lemma G.4 (Tensor interpolation and mean-one priors)**.** *Fix $\alpha > 1$ and $\beta = \alpha - 1$. Let λ_- and λ_+ be finite signed measures on $[a_-, 1]$ and $[a_+, 1]$, respectively, where $0 < a_-, a_+ < 1$. Their positive Jordan masses are one and their negative Jordan masses are $R_-, R_+ \in [0, 1)$, with $R_- + R_+ > 0$. Write $B = (a_- a_+)^{-1}$ and assume the single mean-matching identity*

$$\left(\int u^\beta \, d\lambda_-(u)\right)\left(\int w^{-\beta} \, d\lambda_+(w)\right) = 0. \tag{G.6}$$

Push the following two positive tensor measures forward by $z = a_- w/u$:

$$\mu_1 = \lambda_-^+ \otimes \lambda_+^+ + \lambda_-^- \otimes \lambda_+^-, \qquad \mu_0 = \lambda_-^+ \otimes \lambda_+^- + \lambda_-^- \otimes \lambda_+^+.$$

Then $Z = \int z^{-\beta} d\mu_i$ has the same positive value for $i = 0, 1$. Place mass $(BZ)^{-1} z^{-\alpha} d\mu_i$ at $U_i = Bz$ and complete its total mass to one by an atom at zero. These are probability laws on $\{0\} \cup [1, B]$, with $\mathbb{E}U_i = 1$. For every real $j > 0$,

$$\mathbb{E}U_1^j - \mathbb{E}U_0^j = \frac{B^{j-1}}{Z}\left(\int (u/a_-)^{\alpha-j} \, d\lambda_-(u)\right)\left(\int w^{j-\alpha} \, d\lambda_+(w)\right). \tag{G.7}$$

Writing $A_i = \mathbb{E}U_i^\alpha$, one has exactly

$$A_1 = \frac{B^\beta}{Z}(1 + R_- R_+), \qquad A_0 = \frac{B^\beta}{Z}(R_- + R_+), \qquad \log \frac{A_1}{A_0} = \log \frac{1 + R_- R_+}{R_- + R_+}. \tag{G.8}$$

In particular, an exact accuracy ε and a positive logarithmic margin m are preserved whenever $1 + R_- R_+ \geq e^{2\varepsilon\beta + m}(R_- + R_+)$.

Proof. **Step 1: keep the two hypotheses positive.** The notation "pushforward" means that an atom at (u, w) is moved to the scalar value $z = a_- w/u$, without changing its mass. Both pushforwards are supported on $[a_- a_+, 1] = [1/B, 1]$. Their difference is the pushforward of $\lambda_- \otimes \lambda_+$. Its $z^{-\beta}$ integral vanishes by (G.6). Their common integral Z is positive because both positive measures have positive total mass. **Step 2: reweight to enforce mean one.** At the point $U = Bz$, the assigned mass is proportional to $z^{-\alpha}$. Multiplication by U therefore leaves the power $z^{1-\alpha} = z^{-\beta}$, precisely the quantity whose integrals agree. The constructed nonzero measure has first moment $\int z^{-\beta} d\mu_i / Z = 1$. Its support lies in $[1, B]$, so its total mass is at most one. Completion at zero is therefore legitimate. **Step 3: read every moment from the tensor product.** Direct integration gives

$$\mathbb{E}U_i^j = \int (Bz)^j \frac{z^{-\alpha}}{BZ}\, d\mu_i(z) = \frac{B^{j-1}}{Z} \int z^{j-\alpha}\, d\mu_i(z).$$

Subtract and factor the tensor integral to obtain (G.7). At $j = \alpha$, the integral is just the total mass of μ_i, proving (G.8). No cancellation of overlapping atoms after pushforward is needed. □

Which moments actually cancel? The next statement is the algebraic reason for choosing the two fractional exponents γ and $1 - \gamma$. The factor below α handles degrees $1, \ldots, \lfloor \alpha \rfloor$; the other factor handles the larger degrees. Only the degrees adjacent to α can fail to match, and turning off the appropriate perturbation eliminates either exception.

Corollary G.5 (All noninteger moment cancellations)**.** *Let $\alpha = m_0 + \gamma > 1$ be noninteger, where $m_0 \geq 1$ is an integer and $0 < \gamma < 1$. Choose an integer $L \geq m_0$. Use Lemma G.3 with exponent γ and degree $m_0 - 1$ for λ_-, and with exponent $1 - \gamma$ and degree $L - m_0$ for λ_+. Let their perturbations be ρ_-, ρ_+. Suppose the mass assumptions in Lemma G.4 hold. If $m_0 = 1$, require $\rho_- = 0$. Then the resulting priors have equal means. Their integer moments through degree $L + 1$ all agree except possibly moments m_0 and $m_0 + 1$; moment m_0 also agrees if $\rho_- = 0$, and moment $m_0 + 1$ also agrees if $\rho_+ = 0$.*

Proof. The two factors in (G.7), apart from positive normalizers and the power of a_-, are

$$\int u^{m_0 - j}\, d\nu_{\rho_-}(u), \qquad \int w^{j - m_0 - 1}\, d\nu_{\rho_+}(w).$$

The first vanishes for $1 \leq j < m_0$ and also for $j = m_0$ if $\rho_- = 0$. The second vanishes for $m_0 + 1 < j \leq L + 1$ and also for $j = m_0 + 1$ if $\rho_+ = 0$. In particular $j = 1$ always vanishes under the stated restriction. This proves mean matching before the normalization in Lemma G.4 is invoked. □

Substitution at the order-one endpoint. For $0 < \beta \leq 1/2$, use $\alpha = 1 + \beta$ and

$$a_- = e^{-8}, \qquad \lambda_- = \delta_{a_-} - e^{-8\beta} \delta_1.$$

This is precisely the unperturbed degree-zero factor with exponent β. Choose the unperturbed complementary factor of exponent $1-\beta$ and degree $L-1$, with negative mass $R_+ \le e^{-16}$. Then all moments through $L+1$ agree, and

$$\frac{A_1}{A_0} = \frac{1+e^{-8\beta}R_+}{e^{-8\beta}+R_+} \ge e^{7\beta}.$$

Indeed this ratio decreases with R_+, and, for $r = e^{-16}$,

$$\frac{d}{dx}\left[-\log\frac{e^{-x}+r}{1+re^{-x}}\right] = \frac{1-r^2}{(1+re^{x})(1+re^{-x})} \ge \frac{7}{8} \quad (0 \le x \le 8).$$

Integrate from 0 to 8β. This is the same tensor construction; the description using convolution in logarithmic coordinates is a change of variables.

Substitution near or between higher integers. Write $\alpha = r + \sigma s$ with $r \ge 2$, $0 < s \le 1/2$, and $\sigma \in \{-1, 1\}$. Perturb the low factor when $\sigma = 1$, and the high factor when $\sigma = -1$. The only potentially unmatched integer moment is r. If their negative masses satisfy

$$R_s \le \kappa, \qquad R_c \le \eta\kappa, \qquad \kappa = e^{-2\varepsilon\beta - C_0\eta},$$

then the same exact mass formula gives

$$\log(A_1/A_0) \ge 2\varepsilon\beta + C_0\eta - \log(1+\eta) \ge 2\varepsilon\beta + (C_0-1)\eta.$$

Thus no replacement of ε by a smaller fixed accuracy is involved.

G.3 From the weight construction to an impossibility theorem

We next connect the moment calculation to the observations. A *fingerprint* records how many symbols appeared once, twice, and so on; it discards their names. Randomly relabeling the constructed distributions makes these names uninformative. Poissonizing the number of weights then makes the fingerprint coordinates independent, which allows an exact statistical-distance calculation.

For probability laws P, Q, write $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$. Under equal prior probabilities, every test of P versus Q has average error at least $(1 - \mathrm{TV}(P,Q))/2$. Thus total variation at most $1/4$ forces testing error at least $3/8$. We will contradict this using a hypothetical entropy estimator with error probability at most $1/3$.

Lemma G.6 (Fingerprint separation with exact accuracy)**.** *Let $U_0, U_1 \in [0, B]$ have mean one and let $A_i = \mathbb{E}U_i^{\alpha}$. Fix $k \ge 2$, $b = k/2$, $\theta > 0$, and $0 < m \le 1$. Suppose*

$$\log(A_1/A_0) \ge 2\varepsilon(\alpha - 1) + m.$$

For $i = 0, 1$, let ζ_i be the right side of (G.2) in the compound-Poisson variant. Put

$$q_{ij} = \mathbb{E}\frac{e^{-\theta U_i}(\theta U_i)^j}{j!}, \qquad D = \sum_{j \ge 1} \frac{(q_{1j} - q_{0j})^2}{q_{1j} + q_{0j}},$$

with $0/0 = 0$. If

$$bD \le \frac{1}{16}, \qquad \max_i \zeta_i \le \frac{1}{64}, \qquad (e/4)^{k/2} \le \frac{1}{128}, \qquad e^{-\theta b/16} \le \frac{1}{128},$$

no estimator based on $n = \lfloor \theta b/4 \rfloor \ge 1$ iid samples can have additive entropy error at most ε with probability at least $2/3$, uniformly on the k-symbol simplex.

Proof. **Step 1: compute the fingerprint law.** Under hypothesis i, draw $Q \sim \mathrm{Poi}(b)$ weights and conditionally independent counts with means θU_i. Independent Poisson marking gives independent fingerprint entries $F_j \sim \mathrm{Poi}(bq_{ij})$, $j \geq 1$. Their affinity is

$$\exp\left\{-\frac{b}{2}\sum_{j\geq 1}(\sqrt{q_{1j}}-\sqrt{q_{0j}})^2\right\}.$$

Since $(\sqrt{a}-\sqrt{c})^2 \leq (a-c)^2/(a+c)$, the total variation of the two fingerprint laws is at most $\sqrt{bD} \leq 1/4$. Every equal-prior test consequently has average error at least $3/8$.

Step 2: turn the fingerprint into ordinary labeled data. Randomly assign distinct observed symbols to distinct labels in $[k]$ and randomly order their counts. If more than k distinct symbols are observed, output an arbitrary fixed record. This is one common randomized kernel; the latter event is contained in $\{Q > k\}$. On $Q \leq k$, it agrees in distribution with first assigning a uniform injection of all Q coordinates into $[k]$ and then retaining only the observed labels. The exceptional event has probability at most $(e/4)^{k/2}$. Conditional on the weights, the total count is $\mathrm{Poi}(\theta bS)$; its uniformly ordered labels are iid from the normalized weights, conditional on its length. On the normalization event, $S \geq 3/4$, and the probability of fewer than n labels is at most $e^{-\theta b/16}$ by the Poisson lower tail.

Step 3: an accurate estimate would produce an accurate test. Lemma G.1 places the two entropies within $m/[4(\alpha-1)]$ of centers separated by at least $2\varepsilon + m/(\alpha-1)$. Hence an accurate entropy estimate followed by the nearer-center decision is a correct test whenever the normalization, support, and availability events hold. A hypothetical estimator as in the statement would give testing error at most

$$\frac{1}{3}+\frac{1}{64}+\frac{1}{128}+\frac{1}{128}=\frac{35}{96}<\frac{3}{8},$$

a contradiction. □

Quantitative evaluation. The lemmas above give one positive-prior construction, one normalization identity, and one testing transfer for every noninteger order. Section G.4 supplies the interpolation nodes, perturbations, moment bounds, and likelihood estimates at the root in Lemma D.4, including the degree-zero factor at the order-one endpoint. Integer and bounded-root collision obstructions use the hidden-coordinate bound of Section C.2. These lower-bound arguments do not provide the matching noninteger upper approximation and noise bound.

G.4 Choosing the parameters at the target sample scale

The previous lemmas reduce the lower bound to three checks:

1. the two entropy centers differ by more than 2ε, with enough margin to absorb normalization;
2. the fingerprint distance is small;
3. normalization, the alphabet-size constraint, and sample availability fail with sufficiently small probabilities.

This subsection performs these checks with one parameter rule. The calculation is longer than the preceding constructions because its constants must remain uniform when α approaches one, approaches an integer, or grows with k. For fixed α, many of these uniformity estimates would be unnecessary.

Write $S_\alpha(k,\varepsilon,\zeta)$ for the least fixed number of iid observations that estimates H_α within ε with failure probability at most ζ, uniformly on Δ_k. We reserve δ below for the distance to the next integer. The notation $K = \log k$ is only an abbreviation within this lower-bound calculation; it does not denote another alphabet size. Likewise Q^2 below is a deterministic scalar in the root equation, unrelated to the random number of weights in the normalization lemma. Fix $0 < \varepsilon < 1/2$. For $K = \log k$, noninteger $1 < \alpha < K$, and $\beta = \alpha - 1$, write $\gamma = \alpha - \lfloor \alpha \rfloor$, $\delta = 1 - \gamma$, and

$$h = \gamma\delta, \qquad E = 2\varepsilon\beta, \qquad \Psi = \gamma^{1+1/\delta}\delta^{1+1/\gamma}, \qquad Q^2 = \frac{\Psi}{\alpha^2\beta^2}e^{K/\alpha - E/h}.$$

Define the regularized root and sample scale by

$$X = \inf\left\{ y \geq 1 : \frac{\alpha h}{\beta} y^\alpha \left[1 + \log \frac{y(1+\log y)}{Q^2} \right]_+ \geq 1 \right\}, \qquad R = \alpha k^{1-1/\alpha} X.$$

Theorem G.7 (Calibrated common lower bound)**.** *There are $c_\varepsilon > 0$ and k_ε such that, for all $k \geq k_\varepsilon$ and all noninteger $1 < \alpha < \log k$,*

$$S_\alpha(k, \varepsilon, 1/3) \geq c_\varepsilon R.$$

The construction uses the tensor priors of Lemma G.4 and the normalization and testing lemmas of Section G.1. The bounded-root case uses the common hidden-coordinate likelihood calculation. The interpolation parameters follow one rule for every noninteger order. The accuracy in the displayed formula is exactly ε.

We first give the count estimate and the interpolation bounds needed to verify this theorem. These arguments are included to make the calibration self-contained at the level of the interpolation and count estimates.

Why matching moments makes counts hard to distinguish. The probability of seeing a weight exactly j times is $\mathbb{E}[e^{-\theta U}(\theta U)^j/j!]$. Expanding the exponential expresses this probability through integer moments of U. Matching these moments removes the low-degree terms. The next lemma quantifies what remains if one moment is allowed to differ.

Lemma G.8 (One unmatched moment)**.** *Let mean-one laws $U_i \in [0, B]$ have matching integer moments through $L+1$, except possibly moment $r \leq L+1$. Suppose $\alpha \leq L+2$, $A_i = \mathbb{E}U_i^\alpha$, and, for some $D_0 > 0$, $\rho \geq 0$, $c_0 > 0$,*

$$|\mathbb{E}U_1^r - \mathbb{E}U_0^r| \leq \rho D_0, \qquad q_{ij} \geq c_0 \theta^r D_0/j! \quad (1 \leq j \leq r),$$

where $q_{ij} = \mathbb{E}[e^{-\theta U_i}(\theta U_i)^j/j!]$. Then

$$\sum_{j \geq 1} \frac{(q_{1j} - q_{0j})^2}{q_{1j} + q_{0j}} \leq \frac{C 2^r}{c_0 r!}\theta^r \rho^2 D_0 + C(A_0 + A_1) B^{-\alpha} \left(\frac{2e\theta B}{L+2} \right)^{L+2}.$$

If all the moments match, the first term is omitted and the denominator hypothesis is unnecessary.

Proof. For $j \leq L+1$, write

$$q_{ij} = \sum_{\ell=0}^{L+1-j} \frac{(-1)^\ell \theta^{j+\ell}}{j!\,\ell!} \mathbb{E}U_i^{j+\ell} + \text{Taylor remainder}.$$

Subtracting the two expressions cancels all matched moments. Equivalently, expand e^{-z} through degree $L+1-j$. The only uncancelled polynomial term is

$$a_j = \frac{(-1)^{r-j}\theta^r \Delta_r}{j!(r-j)!} \quad (1 \le j \le r), \qquad a_j = 0 \quad (j > r).$$

The integral Taylor remainder gives

$$|R_j| \le \frac{\theta^{L+2}(\mathbb{E}U_0^{L+2} + \mathbb{E}U_1^{L+2})}{j!(L+2-j)!}.$$

The tail above $L+1$ has the same factorial-moment bound. Since $\mathbb{E}U_i^{L+2} \le B^{L+2-\alpha}A_i$, summing yields

$$\sum_{j \ge 1} |R_j| \le C(A_0 + A_1)B^{-\alpha}(2e\theta B/(L+2))^{L+2},$$

where for tail indices $R_j = q_{1j} - q_{0j}$. If $q = q_{1j} + q_{0j}$, then $|a_j + R_j| \le q$ implies $(a_j + R_j)^2/q \le 2a_j^2/q + 2|R_j|$: for $|R_j| \le q$ use $2R_j^2/q \le 2|R_j|$, and otherwise use $(a_j + R_j)^2/q \le q$. Finally

$$\sum_{j=1}^{r} \frac{1}{j!(r-j)!^2} \le \frac{2^r}{r!}.$$

These estimates prove the assertion, including the case of exact matching. □

Why Chebyshev nodes are used. Extrapolating an interpolating polynomial outside its node interval can make its coefficients very large. The nodes below keep that increase controlled while leaving one small principal point a outside the interval. The endpoint coefficient also gives a nonzero atom in each hypothesis; this will provide the denominator in the count-distance bound. Here T_n and U_{n-1} denote the Chebyshev polynomials of the first and second kinds, characterized by $T_n(\cos t) = \cos(nt)$ and $U_{n-1}(\cos t) = \sin(nt)/\sin t$, with their polynomial extensions outside $[-1, 1]$.

Lemma G.9 (Interpolation coefficients used in the calibration)**.** *For degree $n \ge 1$, use the Chebyshev extrema on $[b, 1]$, and let ℓ_j be their Lagrange basis. If $b = c_b\eta/(n+1)^2$, $0 < \eta \le 3/5$, and $0 < a \le b/2$, then, for an absolute sufficiently small $c_b > 0$,*

$$\sum_j |\ell_j(a)| \le 1 + C\eta, \quad \ell_b(a) \ge 1, \quad \sum_{u_j \ne b} |\ell_j(a)| \le C\eta, \quad |\ell_1(a)| \ge b/2.$$

For $1/2 \le \xi < 1$, instead set $b = \xi/(n+1)^2$. If $a = \exp(-5)bz^{1/\xi}$ with $0 < z \le 1$, the signed measure $u^{-\xi}(\delta_a - \sum_j \ell_j(a)\delta_{u_j})$, divided by its positive Jordan mass, has negative mass at most $\exp(-\xi)z$ and principal positive mass at least $1/2$. The latter conclusion also holds for $n = 0$, using the single node 1 and $b = \xi$.

Proof. Outside the node interval, the signs of the Lagrange coefficients alternate, so their absolute sum is

$$|T_n(t)|, \qquad t = (2a - 1 - b)/(1 - b) < -1.$$

Writing $t = -\cosh u$ gives $u \le 2\,\mathrm{arctanh}\sqrt{b}$. Thus $|T_n(t)| = \cosh(nu)$. For the first choice of b, $n^2u^2 \le C\eta$, whence $\cosh(nu) \le 1 + C\eta$. Every factor in the product formula for $\ell_b(a)$ is at least one; subtracting this coefficient proves the third inequality. The exact endpoint formula is

$$|\ell_1(a)| = \frac{b-a}{1-b}\frac{|U_{n-1}(t)|}{n} \ge b/2,$$

since $|U_{n-1}(t)| \geq n$. For the second choice, $b \leq 1/4$ and $n \operatorname{arctanh}(\sqrt{\xi}/(n+1)) \leq \sqrt{\xi}$. (The function $v \mapsto \operatorname{arctanh} v/v$ is increasing, and $n \operatorname{arctanh}(1/(n+1)) \leq 1$.) Therefore $\cosh(nu) \leq \cosh(2\sqrt{\xi}) \leq \exp(2\xi) \leq \exp(4\xi)$, using $\cosh t \leq e^{t^2/2}$. The total weighted nodal mass divided by the principal mass $a^{-\xi}$ is at most $\exp(4\xi)(a/b)^{\xi} = \exp(-\xi)z$. Positive normalization only reduces the negative mass, and its principal mass is at least $1/(1+\exp(-\xi)z) \geq 1/2$. The degree-zero assertion follows directly. □

Proof of Theorem G.7. **Step 1: choose the sample intensity and interpolation degrees.** The target lower-bound scale is $R = \alpha k^{1-1/\alpha} X$. We therefore choose θ proportional to R/k; the eventual fixed sample size will be $\lfloor \theta k/8 \rfloor$. The interpolation degree L controls the Taylor remainder, while the support bound B is chosen so that θB is a small multiple of L. Put $x = X$. If $x \leq X_0$, for a constant X_0 chosen below, the common hidden-coordinate lower bound proves the claim. This case cannot occur for $1 < \alpha \leq 3/2$ once k is large: here $Q^2 \asymp_\varepsilon k^{1/\alpha}$, so the root's bracket is negative at every fixed X_0. The root comparison used in this observation is proved at the end of the argument.

Choose a nearest integer $r \geq 1$ and write $\alpha = r + \sigma s$ with $0 < s \leq 1/2$, $\sigma \in \{-1, 1\}$; at ties take the lower integer, so $r = 1$ throughout $1 < \alpha \leq 3/2$. Set

$$T = \alpha \log x, \quad v = \frac{\beta}{\alpha} e^{-T}, \quad w = h + v, \quad \zeta = \frac{c_* \, w}{1 - e^{-E}}, \quad \kappa = e^{-E - C_0 w},$$

$$L = \lceil C_L(T + \alpha) \rceil, \qquad \theta = c_n \alpha k^{-1/\alpha} x, \qquad B = c_m L/\theta.$$

Here $c_* = c_*(\varepsilon) > 0$ is chosen so that $\zeta \leq 1/8$ for all orders: since $h \leq \min\{\beta, 1/4\}$ and $v \leq \min\{\beta, 1\}$, it suffices to take a sufficiently small multiple of ε. Increasing X_0 ensures $v \leq E/4$ and $0 < w < 1/2$. The active factor has exponent s and degree n_s; the complementary factor has exponent $\xi = 1 - s$ and degree n_c:

$$(n_s, n_c) = \begin{cases} (r-1, L-r), & \sigma = 1, \\ (L-r+1, r-2), & \sigma = -1. \end{cases}$$

Define, for every order, the same parameters

$$b_s = \frac{c_b \zeta}{(n_s+1)^2}, \qquad b_c = \frac{\xi}{(n_c+1)^2}, \qquad a_c = \exp(-5) b_c (\zeta\kappa)^{1/\xi}, \qquad a_s = (B a_c)^{-1},$$

$$t = (a_s/b_s)^s/\kappa, \qquad \rho = \left[t \exp\left\{ C_1 \zeta \frac{n_s}{n_s+1} \right\} - 1 \right]_+ .$$

Use $u^{-s}\nu_\rho$ in the active position and $u^{-\xi}\nu_0$ in the complementary position, followed by positive normalization. When a degree is zero its only node is 1; the associated b is only a scale parameter. The low and high positions are exactly those of Corollary G.5.

Step 2: use the root equation to control the perturbation. For $x > X_0 > 1$, the defining inequality of the root is an equality. Because $x^\alpha = e^T$ and $v = (\beta/\alpha)e^{-T}$, it gives

$$1 + \log \frac{x(1 + \log x)}{Q^2} = \frac{v}{h}.$$

Taking logarithms in the definition of Q^2 and rearranging yields the identity below. This is the point where the lower-bound construction uses exactly the same scalar root as the upper-bound proof. The regularized root equation is

$$\frac{K}{\alpha} = \frac{E - v}{s(1-s)} + \log x + \log(1 + \log x) + 1 - \log \Psi + 2 \log \alpha + 2 \log \beta.$$

Writing $d_s = n_s + 1$, $d_c = n_c + 1$, substitution gives the exact identity

$$\log t = \frac{v + C_0 w}{\xi} + s(1 + 1/\xi)\log(s/\zeta) + \log\xi + s\log c_n + 2s\log(\alpha/\beta) + s\left[4 - \log(c_m c_b) + \log\frac{(d_s d_c)^2}{L\alpha^3(1+\log x)}\right].$$

For $C_L \geq 10$, $d_s d_c \asymp \alpha L$ and $L \asymp C_L\alpha(1+\log x)$ for every order, including $r = 1$. Thus the second line is $O_{C_L, c_m, c_b}(s)$.

If $n_s \geq 1$, then $\beta \geq 1/2$ and $w \asymp s + v$, $\zeta \asymp_\varepsilon s + v$. In particular the positive part of $s\log(s/\zeta)$ is $O_\varepsilon(s)$. Choosing c_n sufficiently small gives

$$\log t + C_1\zeta\frac{n_s}{n_s+1} \leq C_\varepsilon v - 2s.$$

If $n_s = 0$, then $r = 1$, $s = \beta$, $w \asymp s$, and ζ lies between two positive constants depending only on ε. The potentially unbounded logarithms cancel exactly:

$$s(1+1/\xi)\log s + 2s\log(\alpha/\beta) = \frac{s^2}{\xi}\log s + 2s\log\alpha.$$

All remaining terms in $\log t - s\log c_n$ are $O_\varepsilon(s)$. Thus the same choice of c_n makes $\log t \leq -2s$ and, automatically, $\rho = 0$. In particular the mean-matching restriction at degree zero is satisfied by the common parameter rule, without modifying the prior. For all orders, after increasing X_0,

$$0 \leq \rho \leq C_\varepsilon v \leq 1, \qquad \rho > 0 \Longrightarrow (r \geq 2 \text{ and } v \geq c_\varepsilon s).$$

The identity also gives $|\log t - s\log c_n| \leq C_\varepsilon$. To bound the principal point uniformly when s is very small, retain its individual terms before dividing by s:

$$\log(a_s/b_s) = \log c_n - E/s + v/(s\xi) + C_0 w/\xi + (1+1/\xi)\log(s/\zeta) + (\log\xi)/s + 2\log(\alpha/\beta) + O(1).$$

Since $v \leq E/4$ and $\xi \geq 1/2$, the two terms containing $1/s$ and E, v sum to at most $-E/(2s)$. At positive active degree, $\zeta \geq c_\varepsilon s$ and $\alpha/\beta \leq 3$ bound the remaining terms above uniformly. At degree zero, $s = \beta$ and $\zeta \asymp_\varepsilon 1$; the only unbounded logarithms cancel as $(1+1/\xi)\log s - 2\log s = (s/\xi)\log s \leq 0$. It follows that $a_s/b_s \leq C_\varepsilon c_n \exp\{-E/(2s)\} \leq 1/2$. For $n_s = 0$ this inequality simply verifies that the principal point lies below its only node. Fix C_1 before c_n. Since $\rho \leq C_\varepsilon v$ and $v/\zeta \leq C_\varepsilon$, a further decrease of c_n ensures $\rho a_s/b_s \leq \zeta$ with coefficient one; thus absorbing the perturbation into the nodal estimate does not require a circular choice of C_1.

Step 3: bound the positive and negative interpolation masses. Here "raw" means before dividing the signed measure by its positive mass. Let R_s, R_c be the negative masses after this division, and let P_s, P_c be the raw positive masses. For $n_s \geq 1$ the total active nodal mass is at most $b_s^{-s}(1+C\zeta)(1+\rho a_s/b_s)$, while its principal mass is $(1+\rho)a_s^{-s}$. Since $n_s/(n_s+1) \geq 1/2$, the definition of ρ gives $R_s \leq \kappa$ when C_1 is sufficiently large. The node at b_s is negative and has coefficient at least one, whereas the other nodes have total coefficient $O(\zeta)$. Hence

$$P_s a_s^s = 1 + \rho + O(\zeta\kappa), \qquad c_{\varepsilon, c_n}\kappa \leq R_s \leq \kappa.$$

For $n_s = 0$, exactly $P_s a_s^s = 1$ and $R_s = a_s^s = \kappa t b_s^s$. The same bounds hold because b_s is bounded above and below by positive ε-dependent constants in this case. Complementary interpolation gives $R_c \leq \zeta\kappa$ and $1 \leq P_c a_c^\xi \leq 2$.

Step 4: verify the entropy separation at the original accuracy. The exact ratio $(1 + R_sR_c)/(R_s + R_c)$ decreases in each negative mass when both masses lie in $[0, 1)$. Its logarithm therefore satisfies

$$\begin{aligned}\log(A_1/A_0) &\geq E + C_0w + \log(1 + \zeta\kappa^2) - \log(1 + \zeta)\\ &\geq E + C_0w - \zeta(1 - \kappa^2)\\ &\geq E + (C_0 - 2c_* - 2C_0\zeta)w \geq E + w,\end{aligned}$$

after choosing, for example, $C_0 \geq 4$ and c_* small. The penultimate inequality uses $1 - \kappa^2 \leq 2(1 - e^{-E}) + 2C_0w$. This is essential at order one, where both negative masses may not be small relative to the desired logarithmic margin. Only integer moment r can differ through degree $L + 1$; when $r = 1$ even that moment agrees because $\rho = 0$.

Step 5: bound the unmatched moment and keep the count probabilities positive. The moment defect must be small, but smallness alone is insufficient: the fingerprint distance divides its square by a count probability. We therefore identify actual atoms that provide lower bounds on those probabilities. First suppose $r \geq 2$. Let P_-, P_+ denote the raw positive masses in the low and high positions, and let Z be the tensor normalizer. Define

$$D_0 = \frac{B^{r-1}}{ZP_-P_+}\begin{cases} a_-^{-s}a_+^{-1}, & \sigma = 1,\\ a_-^{s-1}, & \sigma = -1.\end{cases}$$

The interpolation identities give the exact defect $\Delta_r = \rho C_-C_+D_0$, where $C_\pm = \prod_j(1 - a_\pm/u_{\pm,j}) \in (0, 1]$. For $\sigma = 1$, the pairs (a_-, a_+) and (b_s, a_+) belong to opposite parity measures and each contributes at least D_0 to its rth moment. For $\sigma = -1$, use (a_-, a_+) and (a_-, b_s). Their count means are at most $C\zeta/L < 1$: for the first sign use $\theta Ba_c = c_mLa_c \leq C\zeta/L$, and for the second use $\theta Bb_s = c_mLb_s \leq C\zeta/L$. Thus $(\theta U)^j \geq (\theta U)^r$ at these atoms for $j \leq r$, and $e^{-\theta U}$ is bounded below. Consequently

$$|\Delta_r| \leq \rho D_0, \qquad q_{ij} \geq c\theta^r D_0/j! \quad (1 \leq j \leq r).$$

Step 6: bound the power moments needed for normalization. The same atoms also control A_i and the second moment of U_i^α. For all $r \geq 1$, put $J = \int u^\beta d\lambda_-^+ = \int u^\beta d\lambda_-^-$. The equality follows from low-factor interpolation, including its possible perturbation. The exact factorization is

$$Z/B^\beta = J\int (a_+/w)^\beta d|\lambda_+|(w), \qquad c \leq \int (a_+/w)^\beta d|\lambda_+|(w) \leq 2.$$

The bounds on the last integral follow from the principal atom and $a_+/w \leq 1$. If $\sigma = 1$ and $r \geq 2$, its low exponent is s and $\beta - s = r - 1 \geq 1$. The near node contributes at most $b_s^{r-1} \leq C\zeta$, and all other nodes have total coefficient $O(\zeta)$. The endpoint 1 has coefficient at least $b_s/2$; either Jordan sign can be used because their β moments both equal J. Dividing by $P_s \asymp a_s^{-s}$ gives

$$ca_s^s\zeta/r^2 \leq J \leq Ca_s^s\zeta, \qquad D_0 \leq Cr^2/\zeta.$$

For $r = 1$, the low factor has degree zero and exactly $J = a_s^s$. Since $\zeta \asymp_\varepsilon 1$ here, the same two-sided bound on J holds with ε-dependent constants. No defect denominator is needed because all moments match. If $\sigma = -1$, the low factor is complementary and $\beta - \xi = r - 2 \geq 0$. Its absolute nodal coefficient sum is bounded, and the endpoint coefficient is at least $b_c/2$ (or one for degree zero). Since $a_c^\xi = \exp(-5\xi)b_c^\xi\zeta\kappa$,

$$e^{-C_\varepsilon r}\zeta\kappa \leq J \leq C\zeta\kappa.$$

The endpoint pair $(1, a_+)$ also gives

$$D_0 \le \frac{C}{a_-|\ell_1(a_-)|} \le \frac{e^{C_\varepsilon r}}{\zeta}.$$

Here $a_-^{-1} = \exp(5) b_c^{-1} (\zeta\kappa)^{-1/\xi}$ and $\zeta^{-s/\xi} \le C_\varepsilon s^{-s/\xi} \le 3C_\varepsilon$.

It follows in every degree that

$$D_0 \le e^{C_\varepsilon r}/\zeta \quad (r \ge 2), \qquad A_i \le c_n^{-s} e^{C_\varepsilon r}/\zeta.$$

For the second inequality when $\sigma = 1$ and $r \ge 2$, use $a_s^s = \kappa t b_s^s$, the lower bound on t, and $b_s^{-s} \le e^{Cr}$; the other sign follows directly from J. At degree zero use $J = a_s^s$ and the established bound on R_s.

All degrees satisfy $Z/B^\beta \le C_\varepsilon \zeta\kappa$ and

$$\int z^\alpha d(\mu_0 + \mu_1)(z) \le C_\varepsilon \zeta\kappa.$$

Indeed $z = a_- w/u \le w$. For a complementary high factor, its total nodal mass is $O(\zeta\kappa)$ and its principal atom satisfies $a_c^\alpha \le C\zeta\kappa$. For an active high factor, the coefficient bounds give $\int w^\alpha d|\lambda_+| \le C a_s^s \zeta \le C\zeta\kappa$; the low factor's total variation is at most two. Using $M_1 = 1 + R_s R_c$, $M_0 = R_s + R_c \ge c_{\varepsilon,c_n}\kappa$, the exact identity

$$\frac{\mathbb{E}U_i^{2\alpha}}{A_i^2} = B^\alpha \frac{Z}{B^\beta} \frac{\int z^\alpha d\mu_i}{M_i^2}$$

proves

$$\frac{\mathbb{E}U_i^{2\alpha}}{A_i^2} \le C_{\varepsilon,c_n} \zeta^2 B^\alpha.$$

Step 7: show that the count experiments are close. There are two contributions. The first is the controlled defect in moment r. The second is the remainder after matching all other moments through degree $L+1$. We bound them separately and then use the fingerprint-transfer lemma. Lemma G.8 bounds kD by a constant times

$$\frac{e^{C_\varepsilon r} k\theta^r \rho^2}{\zeta r!} + k(A_0 + A_1) B^{-\alpha} (2\mathrm{e}\theta B/(L+2))^{L+2},$$

with the first term omitted at $r = 1$. If $\rho > 0$, then $r \ge 2$, $\beta/\alpha \in [1/3, 1)$ and $v \ge c_\varepsilon s$. Thus $T \le \log(1/s) + C_\varepsilon$, and the root identity implies $sK/\alpha \le C_\varepsilon r$. Since $\rho \le C_\varepsilon v$,

$$\frac{e^{C_\varepsilon r} k\theta^r \rho^2}{\zeta r!} \le (C_\varepsilon c_n)^r \frac{v}{\zeta} \left(k e^{-T}\right)^{\sigma s/\alpha} \le (C'_\varepsilon c_n)^r.$$

Here v/ζ is uniformly bounded; the absolute logarithm of the last power is at most $(s/\alpha)(K+T) \le C_\varepsilon r$. For the remainder, use

$$B^\alpha/k = e^{-T} (c_m L/(c_n \alpha))^\alpha, \qquad \theta B = c_m L.$$

Uniformly in all orders $\zeta \ge c_\varepsilon e^{-T}$: indeed $1 - e^{-E} \le \min\{1, 2\varepsilon\beta\}$ and $v = (\beta/\alpha) e^{-T}$. Keeping the favorable factor $c_n^{\alpha - s} \le 1$, the prefactor has logarithm at most $2T + C_\varepsilon \alpha$. Thus, by taking c_m small and C_L large before decreasing c_n, the remainder is at most e^{-cL}. One then fixes c_n and increases X_0 to make kD as small as required by the common testing lemma.

Step 8: show that normalizing the weights does not destroy the gap. We now return to the residual $U_i^\alpha - A_i U_i$ from the first lemma. Controlling this residual, rather than U_i^α alone, preserves the factor $\alpha - 1$ needed near order one. The normalization margin is $m = w < 1/2$. We use the following elementary bound, which automatically removes the order-one singularity. Since $U_i \in \{0\} \cup [1, B]$ and $\mathbb{E}U_i = 1$, Jensen and $U_i^\alpha \le B^\beta U_i$ give $1 \le A_i \le B^\beta$. For $u \in [1, B]$, therefore,

$$|u^\beta - A_i| \le \max\{u^\beta, A_i\} \min\{1, \beta \log B\}.$$

The zero atom contributes no residual. Consequently, with $d = \min\{1, \beta \log B\}$,

$$\frac{\mathbb{E}(U_i^\alpha - A_i U_i)^2}{A_i^2} \le d^2 \left(\frac{\mathbb{E}U_i^{2\alpha}}{A_i^2} + \mathbb{E}U_i^2 \right) \le C_{\varepsilon, c_n} d^2 (\zeta^2 B^\alpha + B).$$

The common normalization failure probability is therefore bounded by

$$C_{\varepsilon, c_n} \left[\frac{B}{k} \left(1 + \frac{\beta^2}{w^2} \right) + \frac{d^2}{k w^2} (\zeta^2 B^\alpha + B) \right]. \tag{G.9}$$

This is the same variance bound for every noninteger order.

Here are its uniform numerical bounds. The root equation and $\Psi/(\alpha^2 \beta^2) \le 1$ give

$$K/\alpha \ge (E - v)/h \ge \varepsilon \beta / h, \qquad \beta / w \le K/(\varepsilon \alpha).$$

Also, after fixing c_n and increasing X_0,

$$\log(B^\alpha / k) = -T + \alpha \log(c_m L / (c_n \alpha)) \le -T/2.$$

At positive active degree, $\alpha \ge 3/2$, $\beta \ge 1/2$ and $\zeta / w \asymp_\varepsilon 1$. Thus (G.9) is at most

$$C_{\varepsilon, c_n} (K^2 e^{-K/3} + e^{-T/2}),$$

which is as small as needed by the final choices of X_0 and k. At degree zero, the same root satisfies $x \asymp_\varepsilon k^{1/\alpha}/K$, so $T \asymp K$ and $B \asymp_{\varepsilon, c_n} K^2$. Also $w \asymp \beta$, $\zeta \asymp_\varepsilon 1$, and $d \le \beta \log B$. Inserting these inequalities into the same formula (G.9) bounds it by

$$C_{\varepsilon, c_n} (B^\alpha + B)(1 + \log B)^2 / k = o(1),$$

uniformly, since $\alpha \le 3/2$ at degree zero.

Step 9: complete the reduction to a fixed sample size. Support overflow and sample shortage vanish uniformly. For positive degree, $\theta k \ge c_n \alpha k^{1 - 1/\alpha} \to \infty$; at degree zero $\theta k \asymp_{\varepsilon, c_n} k/K$. The exact parity gap is at least $2\varepsilon\beta + w$. Thus the single testing transfer Lemma G.6 applies at $n = \lfloor \theta k / 8 \rfloor \asymp_\varepsilon R$.

Step 10: verify the root estimate used at the order-one endpoint. This final elementary estimate justifies the degree-zero assertions used earlier in the proof. Note that

$$\log(\Psi / h^2) = (\gamma/\delta) \log \gamma + (\delta/\gamma) \log \delta \in [-2, 0].$$

At degree zero $\alpha = 1 + \beta \le 3/2$, whence $Q^2 \asymp_\varepsilon k^{1/\alpha}$ and $\alpha h / \beta = 1 - \beta^2 \in [3/4, 1]$. Let $y(1 + \log y) = Q^2/\mathrm{e}$. Then $y \asymp_\varepsilon k^{1/\alpha}/K$, and the root criterion gives $y \le x \le 2y$ for large k. These statements also show that the bounded-root case occurs only at positive active degree. This completes the proof with constants depending only on the fixed accuracy. □

What has been proved. The lower bound is quantitatively evaluated by one regularized-root parameter rule, one tensor construction, and one statistical transfer. The degree-zero interpolation factor receives its exact coefficient bound within the same argument. This section concerns the lower bound. The matching upper approximation and noise theorem is proved in Section E.3.

G.5 A complete fixed-real-order specialization

This specialization allows arbitrary fixed accuracy and a fixed confidence level. Its upper bound is a consequence of the common estimator, while its lower proof records the larger fixed separation needed for the additive-function transfer.

Theorem G.10 (Matching bounds at every fixed noninteger order)**.** *Fix a noninteger $\alpha > 1$, $\varepsilon > 0$, and $0 < \delta < 1/2$. Then, as $k \to \infty$,*

$$S_\alpha(k, \varepsilon, \delta) = \Theta_{\alpha,\varepsilon,\delta}\left(\frac{k}{\log k}\right).$$

The upper bound follows from Theorem 2.1, including its separate near-one construction. The lower bound is a specialization of the common tensor prior and Poisson likelihood certificate, with no restriction that the fixed accuracy be smaller than $1/2$.

Proof. The upper bound follows from Theorem 2.1. For a fixed noninteger order, h, Ψ, α, β are fixed, so the root satisfies $X \asymp_{\alpha,\varepsilon} k^{1/\alpha}/\log k$. For arbitrary fixed accuracy use $\varepsilon' = \min\{\varepsilon, 1/4\}$; this is only a fixed-order specialization, and no such accuracy substitution is made in the uniform theorem. Independent repetitions and a median give any fixed failure probability. Thus the same estimator supplies the upper bound stated here.

For the lower bound first take failure probability $1/3$, and allow any fixed $\varepsilon > 0$. Write $\alpha = m+\gamma$, $m \geq 1$, $0 < \gamma < 1$, and put $\xi = 1 - \gamma$ and $E = 2\varepsilon(\alpha - 1)$. Let

$$\kappa = \tfrac{1}{2}e^{-E-2}, \qquad L = \lceil C_2 \log k \rceil, \qquad n_- = m - 1, \quad n_+ = L - m.$$

For an interpolation degree $d \geq 1$, use the Chebyshev extrema on $[b_d, 1]$, where $b_d = c_b/(d+1)^2$ and $c_b > 0$ is a sufficiently small absolute constant. The node lemma gives an absolute $M \geq 1$ such that $\sum_j |\ell_j(a)| \leq M$ whenever $0 < a \leq b_d/2$. At degree zero use the single node 1 and the scale $b_0 = c_b$; the same bound holds.

For exponent $e \in \{\gamma, \xi\}$ and its corresponding degree $d \in \{n_-, n_+\}$, set

$$a_e = b_d\left(\frac{\kappa}{2M}\right)^{1/e}.$$

Form the unperturbed weighted interpolation factor $u^{-e}(\delta_{a_e} - \sum_j \ell_j(a_e)\delta_{u_j})$ and normalize its positive Jordan mass to one. Denote the resulting low and high factors by λ_-, λ_+, with principal points a_-, a_+ and negative masses R_-, R_+. The total weighted nodal mass divided by the principal mass is at most

$$M(a_e/b_d)^e = \kappa/2.$$

This bound also holds at degree zero, since its only node is $1 \geq b_0$. In particular $0 < R_-, R_+ \leq \kappa$, and each normalized factor retains at least $1/(1 + \kappa/2)$ of its positive mass at the principal point.

Use the common tensor-prior construction with these two factors. Its integer moments through $L + 1$ all match, because both perturbation parameters are zero. Moreover

$$\log(A_1/A_0) = \log\frac{1 + R_-R_+}{R_- + R_+} \geq -\log(2\kappa) = E + 2.$$

Here $B = (a_- a_+)^{-1} = \Theta_{\alpha,\varepsilon}(L^2)$. We record why A_0, A_1 are bounded above independently of k, which is needed in the count estimate. The low factor has fixed degree, fixed nodes, and fixed principal point. Since it annihilates $u^{\alpha-1}$,

$$J := \int u^{\alpha-1}\, d\lambda_-^+(u) = \int u^{\alpha-1}\, d\lambda_-^-(u) > 0$$

is a fixed constant. Factoring the tensor normalization gives exactly

$$\frac{Z}{B^{\alpha-1}} = J \int (a_+/w)^{\alpha-1}\, d|\lambda_+|(w).$$

The last integral is between $1/(1+\kappa/2)$ and $1+\kappa$: use the principal atom for the lower bound and $a_+/w \le 1$ for the upper bound. The exact tensor moment formulas now give $1 \le A_i \le C_{\alpha,\varepsilon}$, where the lower bound also follows from the mean-one condition and Jensen's inequality.

Choose a small absolute $c_3 > 0$ and put $\theta = c_3 L/B = \Theta_{\alpha,\varepsilon}(1/\log k)$. The common moment-matching count bound, with its defect term absent, gives

$$kD \le C_{\alpha,\varepsilon} k B^{-\alpha} \left(\frac{2e\theta B}{L+2}\right)^{L+2} \le C_{\alpha,\varepsilon} k B^{-\alpha} (2ec_3)^{L+2} = o(1),$$

by first choosing c_3 small and then C_2 large. Normalization uses the same residual bound as before, with a fixed logarithmic margin of one. Since $U_i \in [0, B]$, $\mathbb{E}U_i = 1$, and $A_i \ge 1$,

$$\mathbb{E}U_i^2 \le B, \qquad \frac{\mathbb{E}(U_i^\alpha - A_i U_i)^2}{A_i^2} \le 2\left(\frac{\mathbb{E}U_i^{2\alpha}}{A_i^2} + \mathbb{E}U_i^2\right) \le 2(B^\alpha + B).$$

Thus its failure probability is $O_{\alpha,\varepsilon}((B^\alpha + B)/k) = o(1)$. Support overflow and sample shortage also vanish because $\theta k \asymp k/\log k \to \infty$. The common fingerprint transfer therefore rules out an estimator at $\lfloor \theta k/8 \rfloor = \Omega_{\alpha,\varepsilon}(k/\log k)$. Nothing in these choices restricts the fixed E, and hence the lower bound holds for every fixed positive accuracy.

For a general fixed $\delta < 1/2$, an estimator with success $1-\delta$ can be repeated a constant number of times on independent blocks and its median has failure at most $1/3$. The binomial Chernoff bound proves this for a number of blocks depending only on δ. The preceding $1/3$ lower bound consequently gives the asserted lower bound for every such δ. □

H Logarithmic additive functionals and exact experiment transfer

H.1 A real-order universality theorem for additive logarithmic targets

For a functional T on Δ_k, write $S_T(k, e, \delta)$ for the least fixed number of iid observations permitting error at most e with failure probability at most δ, uniformly over Δ_k. Write S_α for the Rényi target $H_\alpha(p) = -(\alpha-1)^{-1}\log\sum_i p_i^\alpha$. The following reduction is independent of any evaluation of S_α. Its proof uses only the local leading-power assumption and exact sampling reductions.

Lemma H.1 (Dimension-free additive estimation)**.** *Suppose $f : [0,1] \to \mathbb{R}$ is continuous, $f(0) = 0$, and $g(x) = f(x)/x$ extends continuously to zero. For every $d, \delta > 0$ there is $N = N(f, d, \delta) < \infty$, independent of k, and a statistic $\widetilde{F}$ based on N iid observations such that*

$$\sup_{p\in\Delta_k} \Pr_p\left\{|\widetilde{F} - \sum_i f(p_i)| > d\right\} \le \delta.$$

Proof. Choose a polynomial B satisfying $\|B - g\|_\infty \le d/2$ and put $Q(x) = xB(x) = \sum_{j=1}^L c_j x^j$. Then

$$\left|\sum_i Q(p_i) - \sum_i f(p_i)\right| \le \frac{d}{2}\sum_i p_i = \frac{d}{2}.$$

For a block $X_1, \dots, X_L$ of iid observations let

$$W = c_1 + \sum_{j=2}^{L} c_j \mathbf{1}\{X_1 = \cdots = X_j\}.$$

Its mean is $\sum_i Q(p_i)$ and $|W| \le C_Q := \sum_j |c_j|$. The average of b independent blocks has variance at most C_Q^2/b. Taking an integer $b \ge \max\{1, 4C_Q^2/(\delta d^2)\}$ and applying Chebyshev proves the claim with $N = bL$; if Q is linear its aggregate is already known. Rao–Blackwellization with respect to the counts replaces each collision indicator by $\sum_i (N_i)_j/(N)_j$, the same factorial-count statistic used elsewhere in the framework. Polynomial approximation therefore proves the claim without a dimension-dependent distribution estimate. □

Theorem H.2 (Transfer for every fixed real leading order)**.** *Let $\alpha > 1$, $\beta > 0$, and $f \in C([0,1])$ satisfy*

$$f(0) = 0, \qquad f(x) > 0 \quad (0 < x \le 1), \qquad \lim_{x\downarrow 0}\frac{f(x)}{ax^\alpha} = 1 \quad \text{for some } a > 0.$$

Define $H_{f,\beta}(p) = -\beta^{-1}\log F_f(p)$, $F_f(p) = \sum_i f(p_i)$. For every $\varepsilon > 0$, every $0 < \zeta < \beta\varepsilon$, and every $0 < \delta < 1/2$, there are integers $m, N < \infty$, independent of k, such that

$$S_\alpha\left(\lfloor k/m\rfloor, \frac{\beta\varepsilon + \zeta}{\alpha - 1}, \delta\right) \le S_{H_{f,\beta}}(k, \varepsilon, \delta), \tag{H.1}$$

$$S_{H_{f,\beta}}(k, \varepsilon, \delta) \le N + S_\alpha\left(k, \frac{\beta\varepsilon - \zeta}{\alpha - 1}, \delta/2\right). \tag{H.2}$$

The lower inequality is understood for $k \ge m$. The constants m, N may depend on $f, \alpha, \beta, \varepsilon, \zeta, \delta$. No uniformity in a changing order or changing function is asserted.

Proof. Choose $\tau \in (0,1)$ sufficiently small that

$$e^{-\zeta}ax^\alpha \le f(x) \le e^{\zeta}ax^\alpha \qquad (0 \le x \le \tau). \tag{H.3}$$

For the lower bound choose an integer $m \geq 1/\tau$ and put $d = \lfloor k/m \rfloor$. Given $p \in \Delta_d$, form its replicated distribution on md symbols,

$$q_{ij} = p_i/m, \qquad 1 \leq i \leq d, \quad 1 \leq j \leq m,$$

and pad with zeros to reach k symbols. Sampling from q requires one sample i from p and an independent uniform label j; the latter contains no information about p. Summing (H.3) gives

$$\left| \log F_f(q) - \log \left(a m^{1-\alpha} \sum_i p_i^\alpha \right) \right| \leq \zeta.$$

Thus any estimator $\widehat{H}_f$ with error ε yields

$$\widehat{H}_\alpha = \frac{\beta}{\alpha - 1} \widehat{H}_f + \frac{\log(a m^{1-\alpha})}{\alpha - 1}$$

with error at most $(\beta\varepsilon + \zeta)/(\alpha - 1)$ and the same failure probability. This proves (H.1). The construction consists of exactly normalized probability vectors; no random-weight normalization or concentration of a prior functional is required. Moreover, the same equivalence holds exactly in the Poisson experiment: independent counts of means np_i/m aggregate to independent counts of means np_i, and multinomial splitting reconstructs the finer counts. For a positive reference p^*, replication preserves the likelihood quadratic form itself:

$$\sum_{i,j} \frac{(q_{ij} - q^*_{ij})(\widetilde{q}_{ij} - q^*_{ij})}{q^*_{ij}} = \sum_i \frac{(p_i - p^*_i)(\widetilde{p}_i - p^*_i)}{p^*_i}.$$

Thus every Poisson-kernel testing certificate for the base power target is transported without statistical loss; only the deterministic functional approximation costs the displayed ζ.

For the upper bound let $b = \min_{x \in [\tau, 1]} f(x) > 0$ and set

$$d_0 = \frac{b}{4}(1 - e^{-\beta\varepsilon}).$$

Since $f(x)/x \to 0$, Lemma H.1 gives an N-sample statistic $\widetilde{F}$ with $|\widetilde{F} - F_f| \leq d_0$ except on an event of probability $\delta/2$, uniformly in k. Independently run a Rényi estimator with error $(\beta\varepsilon - \zeta)/(\alpha - 1)$ and failure $\delta/2$. Output

$$\widehat{H}_f = \begin{cases} -\beta^{-1} \log \widetilde{F}, & \widetilde{F} \geq b/2, \\ ((\alpha - 1)\widehat{H}_\alpha - \log a)/\beta, & \widetilde{F} < b/2. \end{cases}$$

On the simultaneous success event, in the first branch $F_f \geq b/2 - d_0 > b/4$, so

$$\left| \frac{\widetilde{F}}{F_f} - 1 \right| \leq 4d_0/b = 1 - e^{-\beta\varepsilon}.$$

The resulting ratio lies between $e^{-\beta\varepsilon}$ and $2 - e^{-\beta\varepsilon} \leq e^{\beta\varepsilon}$, proving the required logarithmic accuracy. In the second branch $F_f \leq b/2 + d_0 < 3b/4 < b$. No coordinate can then be at least τ, because such a coordinate alone contributes at least b. Consequently (H.3) applies to every coordinate and

$$|\log F_f - \log(a P_\alpha)| \leq \zeta.$$

The error of the displayed output is at most $(\beta\varepsilon - \zeta + \zeta)/\beta = \varepsilon$. A union bound proves (H.2). □

Corollary H.3 (Consequence of any fixed-order minimax evaluation)**.** *Suppose a rate $R_\alpha(k)$ satisfies $S_\alpha(k, e, \delta) = \Theta_{\alpha,e,\delta}(R_\alpha(k))$ for every fixed $e > 0$, $0 < \delta < 1/2$, and $R_\alpha(\lfloor k/m \rfloor) = \Theta_{\alpha,m}(R_\alpha(k))$ for every fixed positive integer m. If $R_\alpha(k) \to \infty$, then every f in Theorem H.2 satisfies*

$$S_{H_{f,\beta}}(k, \varepsilon, \delta) = \Theta_{f,\alpha,\beta,\varepsilon,\delta}(R_\alpha(k)).$$

In particular, the integer evaluation proved above gives $\Theta(k^{1-1/r})$ for any C^r function positive on $(0, 1]$ satisfying $f(0) = \cdots = f^{(r-1)}(0) = 0 < f^{(r)}(0)$, with $r \geq 2$. For the integer rate, arbitrary fixed accuracy follows from a smaller-accuracy upper bound and the collision lower bound at that accuracy. Constant-size median amplification compares any two fixed failure probabilities in $(0, 1/2)$ without changing the alphabet order. For every fixed noninteger $\alpha > 1$, the fixed-order consequence in Section G.5 also gives the unconditional additive rate $\Theta(k/\log k)$. These fixed-order constants are not uniform as the leading order moves toward an integer.

Corollary H.4 (Complete alphabet dependence for positive analytic targets)**.** *Suppose f is continuous on $[0, 1]$, analytic on a neighborhood of zero, and positive on $(0, 1]$. Let $r \geq 0$ be the first index with $f^{(r)}(0) \neq 0$, assuming $f(0) \geq 0$. The ambient alphabet size k is known; when $f(0) > 0$, zero coordinates are included in the sum. For fixed $\beta, \varepsilon > 0$ and failure probability $1/3$:*

1. *If $r = 0$, no observations are needed for all sufficiently large k.*
2. *If $r = 1$, $O_{f,\beta,\varepsilon}(1)$ observations suffice. For every k at which the range of $H_{f,\beta}$ exceeds 2ε, at least one observation is needed; otherwise its range midpoint is a zero-sample estimator.*
3. *If $r \geq 2$, the complexity is $\Theta_{f,\beta,\varepsilon}(k^{1-1/r})$.*

Proof. Positivity and analyticity imply that r is finite and its leading coefficient is positive. For $r \geq 2$ apply the preceding transfer and the proved integer rate.

For $r = 1$, $h(x) = f(x)/x$ extends to a positive continuous function on $[0, 1]$. Consequently $F_f(p) \geq \min h > 0$ uniformly in k. Lemma H.1, with a fixed sufficiently small absolute tolerance, therefore estimates its logarithm with the requested accuracy. A zero-sample randomized estimator cannot attain success $2/3$ at two targets farther than 2ε apart, because their acceptable output intervals are disjoint. A range midpoint suffices if the range has width at most 2ε.

For $r = 0$, write $a = f(0) > 0$ and $M = \max |f|$. For every $t > 0$, at most $1/t$ coordinates of a probability vector exceed t. If $\omega_f(t) = \sup_{0 \leq x \leq t} |f(x) - a|$, it follows that

$$\sup_{p \in \Delta_k} |F_f(p)/k - a| \leq \omega_f(t) + \frac{2M}{kt}.$$

First let $k \to \infty$ and then $t \downarrow 0$ to obtain uniform convergence to a. Thus the deterministic output $-\beta^{-1} \log(ka)$ eventually has error at most ε. □

H.1.1 Why arbitrary smooth convex functions need additional data

Proposition H.5 (Smooth convex unit-Lipschitz targets can be arbitrarily hard)**.** *Fix $s > 0$ and $A \geq (s + 1)/s$, and define*

$$f_s(0) = 0, \qquad f_s(x) = \frac{e^{-A/x^s}}{Ase^{-A}} \quad (0 < x \leq 1).$$

Then f_s is C^∞, convex and increasing on $[0, 1]$, $\|f_s'\|_\infty = 1$, and $f_s^{(j)}(0) = 0$ for every $j \geq 0$. Nevertheless, for every fixed $\beta, \varepsilon > 0$, estimating $-\beta^{-1} \log \sum_i f_s(p_i)$ with failure probability at most $1/3$ requires

$$\Omega_{s,A,\beta,\varepsilon}(k^{2s})$$

observations as $k \to \infty$.

Proof. Every derivative at a positive x is a finite sum of real powers of x times e^{-A/x^s}, so its limit at zero is zero. Direct differentiation gives

$$f_s''(x) = \frac{e^{-A/x^s}}{Ase^{-A}} Asx^{-s-2}\{Asx^{-s} - (s+1)\} \geq 0.$$

Thus the derivative is increasing, and the chosen normalization gives $f_s'(1) = 1$.

First take even k, let $p_i = 1/k$, and let q have $k/2$ entries $(1 + ck^{-s})/k$ and $k/2$ entries $(1 - ck^{-s})/k$, where $c > 0$ is fixed. For sufficiently large k, q is a probability vector. Taylor expansion of $(1 \pm ck^{-s})^{-s}$ at one gives

$$-Ak^s\{(1 \pm ck^{-s})^{-s} - 1\} = \pm Asc + O(k^{-s}).$$

Therefore

$$\frac{F_{f_s}(q)}{F_{f_s}(p)} \longrightarrow \cosh(Asc), \qquad \chi^2(q,p) = c^2k^{-2s}.$$

Choose c so that $\beta^{-1}\log\cosh(Asc) > 2\varepsilon$. The two targets are then separated by more than 2ε for all sufficiently large k. For n iid observations,

$$\mathrm{TV}(q^{\otimes n}, p^{\otimes n}) \leq \frac{1}{2}\sqrt{(1 + c^2k^{-2s})^n - 1}.$$

If $n \leq c_0k^{2s}$ with a sufficiently small constant c_0, this is less than $1/3$. Binary testing consequently has average error greater than $1/3$, ruling out the proposed estimator. For arbitrary k, use the largest even alphabet size below k and pad with zeros. Since $f_s(0) = 0$, the target is unchanged. □

Proposition H.6 (A single smooth convex target with exponential complexity)**.** *Set*

$$f(0) = 0, \qquad f(x) = \frac{\exp(-e^{2/x})}{2e^2\exp(-e^2)} \quad (0 < x \leq 1).$$

This function is positive on $(0,1]$, increasing, convex, C^∞ and 1-Lipschitz, with all derivatives zero at the origin. For every fixed $\beta, \varepsilon > 0$, the sample complexity of its logarithmic additive functional is $\Omega_{\beta,\varepsilon}(k^2e^{4k})$ along even alphabet sizes.

Proof. Flatness follows by repeated differentiation. For the unnormalized function $u(x) = \exp(-e^{2/x})$,

$$\frac{u''(x)}{u(x)} = \frac{2e^{2/x}}{x^4}\{2(e^{2/x} - 1) - 2x\} > 0 \qquad (0 < x \leq 1).$$

The normalization makes $f'(1) = 1$, proving the stated Lipschitz bound. For even k, compare uniform p with a vector having half of its coordinates $(1 + t_k)/k$ and half $(1 - t_k)/k$, where $t_k = ce^{-2k}/k$. Then

$$e^{2k/(1\pm t_k)} = e^{2k} \mp 2c + o(1), \qquad \frac{F_f(q)}{F_f(p)} \longrightarrow \cosh(2c), \qquad \chi^2(q,p) = c^2e^{-4k}/k^2.$$

Choose c with $\log\cosh(2c) > 2\beta\varepsilon$ and apply the identical two-point likelihood calculation above. □

The proposition supplies smooth examples, rather than a discontinuous support functional: convexity, a common Lipschitz constant, and even the entire zero Taylor series do not imply a common polynomial sample rate. Theorem H.2 gives a complete reduction when a finite leading power exists. It does not justify a minimax formula for arbitrary positive continuous f.

Corollary H.7 (Nonpower additive functionals with matching bounds)**.** *Fix an integer $r \geq 2$, a continuous function $h : [0,1] \to (0,\infty)$, and $\beta, \varepsilon > 0$. For $f(x) = x^r h(x)$, estimating $H_{f,\beta}(p) = -\beta^{-1} \log \sum_i f(p_i)$ over Δ_k with accuracy ε and success probability $2/3$ requires and suffices to use $\Theta_{r,h,\beta,\varepsilon}(k^{1-1/r})$ samples.*

Proof. Apply Theorem H.2 with $\alpha = r$ and $a = h(0) > 0$. The already established integer iid rate has order $k^{1-1/r}$ for every fixed accuracy and confidence, and remains of this order under a fixed replication of the alphabet. Both inequalities in that theorem therefore have the displayed order. □

This includes $x^2 + x^{5/2}$, $x^2 e^x$, and $e^x - 1 - x$. The constants need not be uniform in a changing function or order.

I Approximation duality and the limits of scalar reductions

I.1 Exact multiplicative approximation and its count dual

The following duality preserves the requested logarithmic accuracy exactly. It concerns approximation and likelihood overlap. A minimax conclusion requires the additional variance, target-concentration, and sample-reduction conditions stated below.

Theorem I.1 (Exact multiplicative approximation duality)**.** *Let Θ be a compact metric space, let $f \in C(\Theta)$ satisfy $f_* := \min_\Theta f > 0$, and let V be a finite-dimensional linear subspace of $C(\Theta)$ containing the constants. Define*

$$d(f, V) = \inf_{v \in V} \left\| \frac{f - v}{f} \right\|_\infty .$$

Then

$$d(f, V) = \max \left\{ \int f \, d\omega : \int v \, d\omega = 0 \ (v \in V), \quad \int f \, d|\omega| \le 1 \right\}. \tag{I.1}$$

Here ω ranges over finite signed Borel measures. The infimum and maximum are attained, and $0 \le d(f, V) < 1$. If $d(f, V) > 0$, an optimal ω can be chosen with $\int f \, d|\omega| = 1$. Its Jordan parts have a common positive mass m, and the probability laws $\mu_+ = \omega^+/m$ and $\mu_- = \omega^-/m$ satisfy

$$\int v \, d\mu_+ = \int v \, d\mu_- \quad (v \in V), \qquad \frac{\int f \, d\mu_+}{\int f \, d\mu_-} = \frac{1 + d(f, V)}{1 - d(f, V)}. \tag{I.2}$$

Proof. Equip $C(\Theta)$ with the equivalent norm $\|g\|_f = \|g/f\|_\infty$. Since V is finite-dimensional, it is closed, and distance from f to V is attained. Hahn–Banach applied to the quotient $C(\Theta)/V$ represents this distance as the supremum of $\Lambda(f)$ over norm-one-or-smaller continuous linear functionals Λ that annihilate V. The Riesz representation theorem identifies $\Lambda(g) = \int g \, d\omega$. Its dual norm is

$$\sup_{|g| \le f} \left| \int g \, d\omega \right| = \int f \, d|\omega|.$$

This proves (I.1). The feasible measures have ordinary total variation at most $1/f_*$; hence their set is weak-star compact and the objective is continuous, proving attainment. A sufficiently small positive constant approximant has relative error strictly less than one, so $d(f, V) < 1$. When $d(f, V) > 0$, homogeneity makes the weighted variation constraint active. Because $1 \in V$, $\omega(\Theta) = 0$, so $\omega^+(\Theta) = \omega^-(\Theta) = m > 0$. Writing $A_\pm = \int f \, d\omega^\pm$ gives $A_+ + A_- = 1$ and $A_+ - A_- = d(f, V)$. Their ratio and the annihilation conditions give (I.2). □

Corollary I.2 (The same exact logarithmic threshold)**.** *Fix $\beta, \varepsilon > 0$, and set $\eta = \tanh(\beta\varepsilon)$ and $\kappa = e^{-2\beta\varepsilon} = (1 - \eta)/(1 + \eta)$. There exists $v \in V$ satisfying*

$$(1 - \eta) f \le v \le (1 + \eta) f$$

if and only if there exists $q \in V$ satisfying

$$\kappa f \le q \le f.$$

These equivalent conditions hold exactly when $d(f, V) \le \eta$. If $d(f, V) > \eta$, the laws in (I.2) have matching V-moments and

$$\frac{1}{\beta} \log \frac{\int f \, d\mu_+}{\int f \, d\mu_-} > 2\varepsilon.$$

Proof. Use $q = v/(1+\eta)$ in the forward direction and $v = (1+\eta)q$ in the reverse direction. The separation follows from $\log((1+t)/(1-t)) = 2\operatorname{arctanh}(t)$. □

This is an exact correspondence between an approximation band and separated prior *expectations*. To obtain a lower bound for estimating a realized functional, one must additionally establish that the functional concentrates around those prior expectations. For product random-weight constructions, this is a separate moment calculation; for general priors on matrices, it is not automatic.

Theorem I.3 (A finite-dimensional regularized dual)**.** *Under the preceding assumptions, let $V_0 \subset C(\Theta)$ be finite-dimensional and contain the constants, let $\phi_1, \dots, \phi_D \in C(\Theta)$, and let $t > 0$. Put*

$$\mathcal{D}_t(f) = \inf_{v \in V_0,\, c \in \mathbb{R}^D} \left\{ \left\| \frac{f - v - \sum_{j=1}^D c_j \phi_j}{f} \right\|_\infty + t\|c\|_2 \right\}.$$

Then

$$\mathcal{D}_t(f) = \max_\omega \left\{ \int f \, d\omega : \int f \, d|\omega| \le 1, \quad \int v \, d\omega = 0 \ (v \in V_0), \quad \left\| \left(\int \phi_j \, d\omega \right)_{j=1}^D \right\|_2 \le t \right\}. \tag{I.3}$$

Proof. The weighted norm has the dual representation used above. Thus the primal objective is the infimum over (v, c) of the supremum over $\int f \, d|\omega| \le 1$ of

$$\int f \, d\omega - \int v \, d\omega - \sum_j c_j \int \phi_j \, d\omega + t\|c\|_2.$$

The measure ball is convex and weak-star compact; for fixed (v, c) the integrand is continuous and affine in ω, and for fixed ω it is continuous and convex in (v, c). Sion's minimax theorem [Sio58] interchanges infimum and supremum. The infimum over v is finite precisely when ω annihilates V_0. The infimum over c of $t\|c\|_2 - c \cdot a$ equals zero for $\|a\|_2 \le t$ and minus infinity otherwise. This proves the identity. Compactness also proves dual attainment. □

Unlike in the unregularized case, an optimal signed measure need not saturate its weighted variation bound: scaling may violate the moment-norm constraint. Nevertheless, if its objective is $z > 0$ and $s = \int f \, d|\omega| \le 1$, its Jordan ratio equals $(s+z)/(s-z) \ge (1+z)/(1-z)$. Normalizing Jordan parts changes the size of their moment discrepancy by the inverse of their common mass. That mass must be retained in any likelihood estimate.

I.1.1 The Poisson basis gives the same quadratic form on both sides

Let $\theta = (\theta_e)_{e=1}^q$ range over a compact subset $\Theta \subset [0, \infty)^q$, and observe independent $N_e \sim \mathrm{Poi}(\nu_e \theta_e)$, where $\nu_e > 0$. Fix a reference θ^* with every $\theta_e^* > 0$; the reference need not belong to Θ. Write $\lambda_e^* = \nu_e \theta_e^*$. For $\ell \ge 0$ define

$$C_\ell(N; \lambda) = \sum_{j=0}^{\ell} \binom{\ell}{j} (-\lambda)^{\ell - j} (N)_j, \qquad H_\ell(N; \lambda) = \frac{C_\ell(N; \lambda)}{\sqrt{\ell! \lambda^\ell}}.$$

For a multi-index $a \in \mathbb{N}_0^q$, put

$$H_a(N) = \prod_e H_{a_e}(N_e; \lambda_e^*), \qquad \phi_a(\theta) = \prod_e \frac{\nu_e^{a_e/2} (\theta_e - \theta_e^*)^{a_e}}{\sqrt{a_e! (\theta_e^*)^{a_e}}}.$$

Then

$$\mathbb{E}_\theta H_a(N) = \phi_a(\theta), \qquad \mathbb{E}_{\theta^*} H_a(N) H_b(N) = \mathbf{1}_{\{a=b\}}. \tag{I.4}$$

Indeed, the generating function

$$\sum_{\ell \geq 0} C_\ell(N;\lambda) \frac{z^\ell}{\ell!} = e^{-\lambda z}(1+z)^N$$

has mean $e^{(\mu-\lambda)z}$ under $\mathrm{Poi}(\mu)$. The expected product of two such generating functions under $\mathrm{Poi}(\lambda)$ is $e^{\lambda z w}$, proving both identities by coefficient comparison and then independence.

For a finite polynomial $Q(\theta) = \sum_a c_a \phi_a(\theta)$, the statistic $\widehat{Q} = \sum_a c_a H_a(N)$ is unbiased and

$$\mathrm{Var}_{\theta^*}(\widehat{Q}) = \sum_{a \neq 0} c_a^2. \tag{I.5}$$

If $L_\theta = dP_\theta / dP_{\theta^*}$ denotes the likelihood ratio, direct summation of the Poisson series gives

$$K(\theta, \vartheta) := \mathbb{E}_{\theta^*} L_\theta L_\vartheta \tag{I.6}$$

$$= \exp\left\{ \sum_e \frac{\nu_e (\theta_e - \theta_e^*)(\vartheta_e - \theta_e^*)}{\theta_e^*} \right\} = \sum_{a \in \mathbb{N}_0^q} \phi_a(\theta)\phi_a(\vartheta). \tag{I.7}$$

For every finite signed measure ω on Θ, uniform absolute convergence on the compact square gives

$$\mathbb{E}_{\theta^*} \left(\int L_\theta \, d\omega(\theta) \right)^2 = \iint K(\theta, \vartheta) \, d\omega(\theta) d\omega(\vartheta) = \sum_a \left(\int \phi_a \, d\omega \right)^2. \tag{I.8}$$

Thus the Euclidean norm of polynomial coefficients is exactly reference sampling noise, whereas its dual norm is exactly the corresponding truncated likelihood overlap. The two sides use the same feature map, not separately chosen transforms.

If only $|a| \leq L$ is retained, put

$$R = \max_{\theta \in \Theta} \sum_e \frac{\nu_e (\theta_e - \theta_e^*)^2}{\theta_e^*}.$$

Minkowski's inequality and the multinomial theorem imply

$$\sum_{|a|>L} \left(\int \phi_a \, d\omega \right)^2 \leq \|\omega\|_{\mathrm{TV}}^2 \sum_{j>L} \frac{R^j}{j!} \leq \|\omega\|_{\mathrm{TV}}^2 e^R \frac{R^{L+1}}{(L+1)!}. \tag{I.9}$$

In particular the finite-dimensional dual is an actual likelihood certificate only after this tail is bounded. Weighted variation control alone gives $\|\omega\|_{\mathrm{TV}} \leq 1/f_*$, which may be too costly for sharp entropy rates.

For arbitrary θ, factorial substitution has the exact variance formula

$$\mathrm{Var}_\theta(\widehat{Q}) = \sum_{a \neq 0} \frac{\theta^a}{a! \nu^a} \left(\partial^a Q(\theta) \right)^2. \tag{I.10}$$

To verify this identity for a multivariate polynomial, expand its factorial estimator into monomials, apply the overlap identity in Lemma A.1 independently in every coordinate, and collect terms with common multi-index a. The coefficient is $\theta^a/(a!\nu^a)$ times the square of $\partial^a Q(\theta)$; the $a = 0$ term is the square of the mean and is subtracted. Consequently a reference norm bound is not a uniform variance bound. One must either control (I.10) on the relevant parameter region or use a norm incorporating all required references. Removing a polynomial term from the random statistic is legitimate only if its aggregate value is known.

I.1.2 Scope of the approximation duality

Scalar powers and additive functions. For $I = [a, b]$ with $a > 0$ and $f(x) = x^\alpha$, the unregularized duality applies for every real $\alpha > 1$, with the same exact accuracy threshold $\tanh((\alpha - 1)\varepsilon)$. Integer powers lie in a finite polynomial space, so their deterministic approximation error vanishes; the remaining constraint is count noise. General positive continuous f uses the same theorem. For x^α on an interval including zero, however, $f_* = 0$, so this theorem cannot simply be invoked. Restricting to $[a, b]$ and letting $a \downarrow 0$ requires uniform weighted measure bounds, a controlled affine correction using known mass, and tightness. None follows from the displayed duality alone.

Proposition I.4 (Pointwise relative approximation fails at zero)**.** *Let $b > 0$ and let $\alpha > 0$ be noninteger. For every polynomial Q,*

$$\sup_{0<x\le b} \left| \frac{Q(x)}{x^\alpha} - 1 \right| \ge 1.$$

Proof. For a nonzero Q, its first nonzero monomial has integer degree $r \neq \alpha$. If $r < \alpha$, $|Q(x)/x^\alpha|$ is unbounded near zero. If $r > \alpha$, $Q(x)/x^\alpha \to 0$, so the displayed supremum is at least one. The zero polynomial has error exactly one. □

Thus known-mass cancellations and aggregate approximation cannot be discarded.

A valid global formulation instead takes $\Theta = \Delta_k$ and $f(p) = \sum_i p_i^\alpha \ge k^{1-\alpha} > 0$ for $\alpha > 1$. The exact same duality then applies to polynomials on the simplex, including additive polynomials with known affine aggregate. It likewise applies to any positive additive target $f(p) = \sum_i f_0(p_i)$ with a positive minimum over the chosen compact parameter class. A rate conclusion from this abstract formulation requires a quantitative evaluation of the constrained approximation problem. The constructive iid proof supplies the needed certificates for powers directly.

Symmetry does not force additivity. For the iid count model, permutation averaging of a polynomial statistic preserves its degree and does not increase its worst-case bias or variance over the simplex. Indeed, averages preserve each pointwise bias band for a symmetric target, and $\mathrm{Var}(m^{-1}\sum_j Z_j) \le m^{-1}\sum_j \mathrm{Var}(Z_j)$ by Cauchy–Schwarz. Relabeling counts identifies the latter variances with those at permuted parameters. Therefore a symmetric polynomial suffices for a symmetric global approximation/noise certificate. Such a polynomial need not be additive. For example, on Δ_4 the symmetric polynomial $(\sum_i p_i^2)^2$ is not of the form $c + \sum_i g(p_i)$ for a twice differentiable g. Writing $p_4 = y = 1 - p_1 - p_2 - p_3$, the mixed derivative $\partial_{p_1}\partial_{p_2}$ of an additive expression is $g''(y)$, whereas that of the displayed polynomial is

$$8(p_1 - y)(p_2 - y) + 4\sum_i p_i^2.$$

At fixed y and $p_1 = p_2 = t$, this equals $32t^2 - 16t + 12y^2 + 4(1 - y)^2$, which varies with t on the interior of the simplex. Hence permutation symmetrization does not justify restricting the global primal to additive polynomials.

AI methodology

OpenAI ChatGPT/Codex assisted with mathematical exposition, proof review, literature comparison, and LaTeX preparation.

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